A Mobile App to search for the food additives by computer vision

by

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Final report

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Declaration

I hereby declare that all the work done in this Final Year Project is of my independent effort. I also certify that I have never submitted the idea and product of this Final Year Project for academic or employment credits.



WONG HO FAI, Michael

Date: 08/04/2026

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1. Introduction

1.1 Project Background

Nowadays, people are more and more focus on the food safety, nutrition, and personal health. When buying food on thier daily life, users usually want to know more about the ingredients, nutrition facts, allergen information of the food and to see whether the food product is suitable for their own health condition. But, the information shown on food packaging is often too detailed, difficult to read, or not easy to understand quickly let to many people may feel confused when choosing food products, especially for the users that who have special allergies, diet , or health concerns.

And mobile apps and AI technology are becoming more common in our daily life and these technologies make the problems that i have mention above become possible to solve by providing the food information in a faster and smarter way. Instead of reading each line of the food information on a package one by one , the users can use a mobile application to scan a barcode or upload a food label image, and then receive clearer results about all the food information and also getting a analysis report for the user special condition. This can save time and help users make better decisions when selecting food products.

Therefore, this project aims to develop an AI-powered mobile food information system. The system is designed to help users identify food products, understand important food information, and receive personalized warnings more conveniently. In addition, the system also includes supporting features such as scan history, food alerts, blog/community functions, and VIP features for advanced services. Overall, the project hopes to make food information easier to access and easier to understand for everyday users.

1.2 Problem / Improvement Areas

hen consumers purchasing food they usually have to read the ingredients list and nutrition label themselves. However, this may bring many difficulties to them. for example, ingredients lists often contain technical terms, names of additives, codes and chemical terminology that may be difficult for consumer to understand, creating barriers for them to understanding the food information and making it hard for consumers to make the right choice quickly.

Furthermore, for those with specific allergies, health conditions or dietary preferences, existing food labelling offers limited personalised support. Even when packaging contains the information, users still need to compare

ingredients against their own health circumstances . for example, checking whether a product contains specific allergens, excessive sugar or ingredients that they should avoid, etc. This process is both complex and prone to omissions or misjudgements.

Although some may argue that such apps already exist on the market, but I think that they are still some defect on it. For example, most apps are primarily in English and this is bad for chinese user and the people that are not knowing english, also their functions are fragmented, requiring users to switch between different platforms to access information and most importantly, they fail to effectively integrate personal health data to generate personalised alerts, eventually user are still needed to compare the food information with thier own health circumstances.

Therefore, this project aims to address the aforementioned issues by establishing an integrated mobile platform, enabling users to perform operations such as food scanning, ingredient identification, health analysis, alert viewing and record management within a single application.

1.3 Proposed Solution

To solve those problems, a advanced ,functionable food additives mobile searching app is the best solution, aims to help users understand food additives, nutritional content and potential health risks more quickly and intuitively via their mobile devices.

On the app, there are two ways to scan the food product and get the information, one is to scan the barcode and the system will auto find it in the database, also the user can take photo of the food label or upload the food label photo in their phone to the system and the system will using the OCR text recognition scanning to extract text and using the ai to make a usefully, understandable analysis report , and that report will combine the user health status and the product information and provide personalized reminders and suggestions not just some simple product information analysis report, make the users are no needed to read the product information line by line, also handle the compare by the app which to reducing the misjudgment or omissions that cause by manual compare, avoid the risk to the user health.

Furthermore, the app are also adding so many additional related function, for example the food safety alerts form the CFS, scan history review, blog , quiz games and voucher rewards, VIP features, including advanced analytics,

calorie tracking and healthy recipe recommendations, further improve the practicality of the app by integrate different function in one platform.

2. Project Plan

2.1 Resource Implications

Development and testing tools

Hardware:

PC	
CPU	AMD Ryzen 5 3600 6-Core Processor 3.59 GHz
RAM	16 GB 2133 MHz DDR4
GPU	NVIDIA GeForce RTX 3070 (8 GB)
Hard Disk Storage	2.05TB
Operating System	Windows 10
Mobile(Android)	Redmi Note 13 Pro 5G
CPU	Snapdragon 7s Gen 2 Mobile Platform 8 core 2.4GHz
RAM	12+6GB

Software:

Coding language	react native expo(front-end), node.js(back-end, Express.js)
Database	MongoDB(Local Host)
IDE	VS Code
AI API	POE(gemini-3.1-pro)
OCR	POE(gemini-3.1-pro)

2.2 Development and Operating Costs

Beginning at Year	1	2	3	4	5	Total(5 Year)
Development Costs(HKD)						
Hardware	$0	$0	$0	$0	$0	$0
AI API	$1599	$1599	$1599	$1599	$1599	$7995
Server	$9600	$9600	$9600	$9600	$9600	$48000
Contingency Fees	$1000	$1000	$1000	$1000	$1000	$5000
Operating Costs(HKD)						
Maintenance	$500	$500	$500	$500	$500	$2500
Advertisement(Only in Hong Kong with low frequency, On Google and IOS)	$30000	$30000	$30000	$30000	$30000	$150000
Google play Listing Fees	$200	$0	$0	$0	$0	$200
Apple store Listing Fees	$792	$792	$792	$792	$792	$3960
Total						$217655

2.3 Tangible and Intangible Benefits

- Tangible Benefits:
  1. Users can quickly identify food additive information by scanning food labels or barcodes.

  2. The system can use OCR and AI analysis technologies to automatically generate food safety reports, reducing the time users need to search for information manually.

  3. The system can provide personalized risk warnings based on user profile setting, such as allergies, dietary preferences, and health conditions.

  4. The system supports both barcode scanning mode and text recognition mode, which improves its usability and application range.

  5. The system can store scan history, allowing users to review previous food analysis results conveniently.

  6. VIP features provided to the users with additional services, such as advanced analysis, recipe recommendations, and calorie calculation functions.

- Intangible Benefits:
  1. The app can improve users awareness of food safety and food additives.

  2. The app can increase consumer confidence in food information transparency.

  3. The app can help people who are sensitive to certain food ingredients, such as allergy sufferers, pregnant women, and parents of children, make more suitable food choices.

  4. The app can improve the efficiency of users purchasing decisions and make food selection faster and more convenient.

  5. The app can help promote healthy eating awareness and encourage users to pay more attention to the ingredients in their daily food.

2.4 Cost-benefits Analysis

Beginning at Year	1	2	3	4	5	Total(5 Year)
Development Costs(HKD)						
Hardware	$0	$0	$0	$0	$0	$0
AI API	$1599	$1599	$1599	$1599	$1599	$7995
Server	$9600	$9600	$9600	$9600	$9600	$48000
Contingency Fees	$1000	$1000	$1000	$1000	$1000	$5000
Operating Costs(HKD)						
Maintenance	$500	$500	$500	$500	$500	$2500
Advertisement(Only in Hong Kong with low frequency, On Google and IOS)	$30000	$30000	$30000	$30000	$30000	$150000
Google play Listing Fees	$200	$0	$0	$0	$0	$200
Apple store Listing Fees	$792	$792	$792	$792	$792	$3960
Total Cost	$43691	$43691	$43691	$43691	$43691	$217655
Benefits						
Advertisement	$5000	$6000	$7000	$9000	$10000	$25500
VIP(No limit scanning, no Advertisement, history record, cooking recipe recommendation)	$30000	$40000	$50000	$60000	$70000	$250000
Total Benefits	$35000	$46000	$57000	$69000	$80000	$287000
Net Profit	-$8619	$2509	$13509	$25509	$36509	$69345
Cumulative Net Profit	-$8619	-$6182	$7327	$32836	$69345	$69345

Return on Investment (ROI):

= (Total Benefit-Total Cost)/ Total Cost × 100%

=(287000-217655)/ 217655*100%

=0.318601(31.9%)

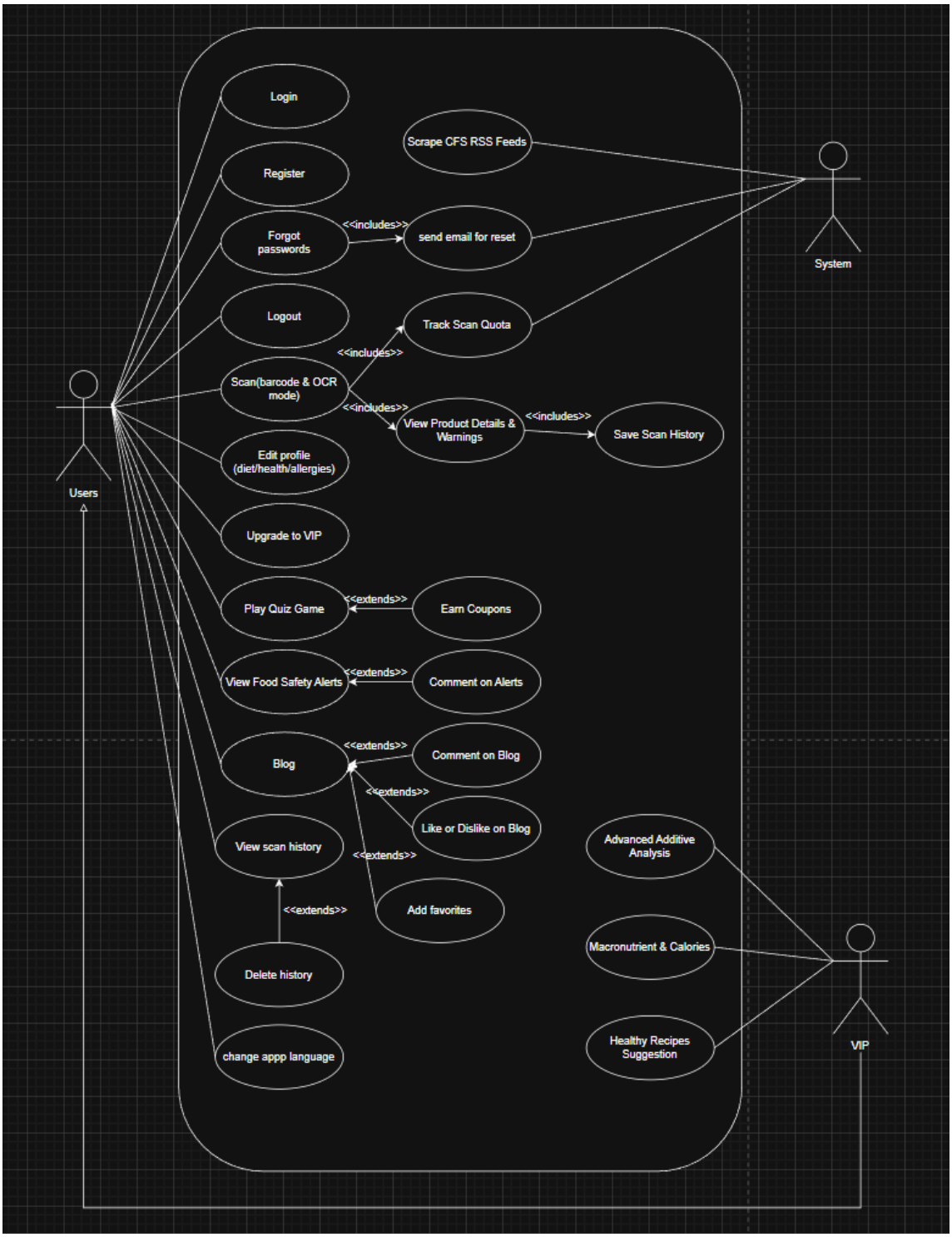
For the app the cost mainly on the AI API and also the server hold fee, other tool are all open source and human resources is only me so it is cost little for the development of this app, but on the benefits we have income in advertisement and VIP mode fee, Although the app can not have profit in the first year but as we can see along with the user increase the benefits will also increase and on the third year the cumulative net profit become positive and keep increasing year by year, and totally in five year we have 31.9% ROI, which means that this project are financially feasible in the long term. Therefore, the project benefits are outweigh the costs and it is considered worthwhile to develop.

2.5 Development Schedule

Phase	Name	Start Date	End Date	Description
1.1	System Design(interface sketch)	01/10/2025	7/10/2025	Finalize the interface sketch.
1.2	system flow chart	8/10/2025	14/10/2025	Finalize the system flow chart.
1.3	architecture design	15/10/2025	21/10/2025	Finalize the architecture design.
2.1	Interface Development	22/10/2025	28/10/2025	Interface Development (login page)
2.2		29/10/2025	5/11/2025	register page
2.3		6/11/2025	11/11/2025	forgot password page
2.4		12/11/2025	18/11/2025	error report page
2.5		19/11/2025	25/11/2025	user profile page
2.6		26/11/2025	2/12/2025	main page(food scanning)
2.7		3/12/2025	9/12/2025	food report page
2.8		10/12/2025	16/12/2025	info page (food safety news) and history page
3.1	Image Preprocessing	17/12/2025	24/12/2025	Image Preprocessing
3.2	Database Access	25/12/2025	11/1/2026	Database Access
3.3	OCR	12/1/2026	27/1/2026	OCR
4.1	Function Development	28/1/2026	4/2/2026	Function Development (language setting)
4.2		5/2/2026	12/2/2026	food safety report
4.3		13/2/2026	20/2/2026	user discussion platform
4.4		21/2/2026	26/2/2026	food product additives error report
4.5		27/2/2026	6/3/2026	VIP
4.6		7/3/2026	11/3/2026	cooking recipe recommendation
4.7		12/3/2026	17/3/2026	Calorie Counting
5.1	Testing Phase	18/3/2026	28/3/2026	Functional testing, text recognition accuracy testing.
5.2	update	29/3/2026	14/4/2026	update and debug.
6	Deployment Phase	15/4/2026	23/4/2026	Release a preliminary usable version.

3. Systems Analysis

3.1 Use Case Model (Use Case Diagram & Description)



Description:

Use Case ID	UC01	
Use Case Name	Login	
Overview	User login to the app	
Actor	User	
Precondition	Registered the account	
Post-condition	Start using the app	
Flow of Event	Actor	System
	1. Input email and password. 2. Click the Login button.	1. validate of the email and password is in the system or not.

Items	Descriptions	
Use Case ID	UC02	
Use Case Name	Register	
Overview	User register a account for the app	
Actor	User	
Precondition	Having a available email	
Post-condition	Login to start using the app	
Flow of Event	Actor	System
	1. Input email, username, age, weight , height, password, allergies and diet preferences, health status. 2. Click the register button	1. Validate all the info. has been inputted by user and verify the password is fit to the condition. 2. Save the data to the database. 3. Route the user back to login page.

Items	Descriptions	
Use Case ID	UC03	
Use Case Name	Forgot passwords	
Overview	If user forgot passwords, user can use it to reset it.	
Actor	User	
Precondition	Registered the account	
Post-condition	Can be using the new password to login to the app	
Flow of Event	Actor	System
	1. input email. 2. Click the send button. 3. Input 6-digit OTP to the page 4. Input the new password	1. validate the email 2. if the user is in the database generate a random 6-digit OTP code and send it to the user

		email. 3. Route the user to the OTP code input page wait for input. 4. Validate the OTP code, by compare the input and database OTP code 5. Route the user to the new password input page, and when user has inputted the new password validate it to see it is match to the password condition or not. 6. If yes, route the user back to the login page for login.

Items	Descriptions	
Use Case ID	UC04	
Use Case Name	Scan(barcode & OCR mode) and view Product Details, analysis report & Warnings	
Overview	User can Scan for the food label or the barcode to get the food additives and ingredients info.	
Actor	User	
Precondition	Device are having a available camera, and usable account of the app	
Post-condition	To view the personalize food analysis report and food additives info.	
Flow of Event	Actor	System
	Barcode: 1. Click the Scan icon button to the Scan page. 2. Click the Scan mode button 3. Scan the food barcode 4. View the personalize report and warnings(if have) OCR: 1. Click the Scan icon button to the Scan page. 2. Click the text mode button 3. User can choose to take a photo	Barcode: 1. The system will compare the barcode with the database and return the data then connect and send the analysis request , user data and food data to the AI API. 2. Route the user to the product detail page to view the detail and analysis report. OCR: 1. After received the photo form the user the system will connect and send the OCR request to the AI API.

	to the food label or upload one. 4. Check the OCR result, input the name of the product and scan the real barcode of the product then click the save button. 5. View the personalize report and warnings(if have)	2. After received the OCR result form the AI API route the user to the edit page to view the OCR result, input product name and scan the barcode to the database. 3. After the click the save button in the edit page then connect and send the user data ,food data(OCR result) and the analysis report request to the AI API and route the user to the product detail page wait for the result return. 4. When the result return show it on the bottom of the page.

Use Case ID	UC05	
Use Case Name	Logout	
Overview	User logout to the app	
Actor	User	
Precondition	Registered the account and has login to the app	
Post-condition	Login with the other account/register a new account	
Flow of Event	Actor	System
	1. Go to the profile page and roll down 2. Click the Logout button.	1. Route the user to the login page.

Use Case ID	UC06	
Use Case Name	Scan history save	
Overview	Save the scan history to the database	
Actor	User	
Precondition	After the user scanned a product	
Post-condition	Can view the report in next time	
Flow of Event	Actor	System
	1. Click the save button in the product detail page	1. Save all the things in to database including the OCR result, food name , food barcode, AI

		analysis report etc. 2. Route the user back to the main page.

Items	Descriptions	
Use Case ID	UC07	
Use Case Name	View CFS alerts and Scrape CFS RSS Feeds	
Overview	User can view the food safety alerts form the CFS	
Actor	User	
Precondition	User has logged in	
Post-condition	View the latest 3 CFS alerts	
Flow of Event	Actor	System
	1. View alerts on the main page. 2. Click in to the alerts user are interested in.	1. Check the alerts JSON and put the latest 3 CFS alerts to the alerts page(the alerts JSON will renew every day at 03:00 am) 2. Check if the alerts is within 7 days, if yes show the “New!!” next to the alerts title.

Items	Descriptions
Use Case ID	UC08
Use Case Name	View CFS alerts comment
Overview	User can comment under the food safety alerts form the CFS
Actor	User
Precondition	User has logged in and has click in to one of the alert
Post-condition	Comment under the CFS alerts
Flow of Event	Actor 1. Comment under the alerts the user was interested in.

Items	Descriptions
Use Case ID	UC09
Use Case Name	View scan history
Overview	User can view the scan history
Actor	User
Precondition	User have scan a product before
Post-condition	/

Flow of Event	Actor
	1. Click the history icon button 2. Click in and view the history user are interested in.

Items	Descriptions
Use Case ID	UC10
Use Case Name	Delete scan history
Overview	User can delete the scan history
Actor	User
Precondition	User have scan a product before
Post-condition	/
Flow of Event	Actor
	1. Click the history icon button 2. Click the delete button next to the scan history

Items	Descriptions
Use Case ID	UC11
Use Case Name	Blog
Overview	User can share their own view or reply to other user on the blog.
Actor	User
Precondition	User have register a account
Post-condition	Using the blog function
Flow of Event	Actor
	1. Click the blog icon button 2. Click add to start a new blog.

Items	Descriptions	
Use Case ID	UC12	
Use Case Name	Blog comment	
Overview	User can comment under the other user blog .	
Actor	User	
Precondition	User have register a account	
Post-condition	/	
Flow of Event	Actor	System
	1. Click the blog icon button 2. Click any discussion that user are interested to comment.	1. Load all the comment for the blog that user has click in. 2. When comment add, save it to the database.

Items	Descriptions	
Use Case ID	UC13	
Use Case Name	Like or Dislike on Blog	
Overview	User can like or dislike other user on the blog.	
Actor	User	
Precondition	User have register a account	
Post-condition	/	
Flow of Event	Actor	System
	1. Click the blog icon button. 2. Click any blog that user are interested in. 3. Click the like or dislike button to the blog.	1. Check the like and dislike number in the database when user click in to the blog 2. Show the correct like and dislike number on the blog 3. When user like or dislike the blog , system will edit the like or dislike number in the database.

Items	Descriptions	
Use Case ID	UC14	
Use Case Name	Add favorites	
Overview	User the blog that they want to favorites for easy access	
Actor	User	
Precondition	User have register a account	
Post-condition	/	
Flow of Event	Actor	System
	1. Click the blog icon button 2. Click in to any blog user want and click the add to favorites button.	1. Set the favorites to True for that blog port of the current user.

Items	Descriptions	
Use Case ID	UC15	
Use Case Name	Recommend Recipe	
Overview	User can view the recommend recipe on the food safety report after they scan the food label or barcode, if they are subscribed the VIP mode.	
Actor	User(VIP mode subscribed)	
Precondition	User have subscribed the VIP mode	
Post-condition	Have a Recommend Recipe in the food safety report.	
Flow of Event	Actor	System
	1. Click VIP dashboard on the main	1. validate of the VIP

	page. 2. Take a photo or upload a photo of food ingredients 3. View the Recommend Recipe	identity of the user. 2. When received the photo input form the user, and it will send the photo and the recommend recipe request to the AI API 3. Get back the result(the recipe and the recipe analysis) and put on the Recommend Recipe page.

Items	Descriptions	
Use Case ID	UC16	
Use Case Name	Calorie calculate	
Overview	User can set their own calorie goal and then when user has scan the food that they have ate today, they can add the food in to the calorie calculator for calorie calculate, if they are subscribed the VIP mode.	
Actor	User(VIP mode subscribed)	
Precondition	User have subscribed the VIP mode	
Post-condition	View the Calorie calculate.	
Flow of Event	Actor	System
	1. Click the VIP dashboard button on the main page. 2. Scan a food label or barcode. 3. Add the food product by gram(g) to the calorie calculator 4. View the Calorie calculate	1. validate of the VIP identity of the user. 2. Using user input and the calorie info. that the system get when scanning the food label to calculate the calorie, and according to the number calorie fill up the goal bar on the calorie calculate page.

Items	Descriptions
Use Case ID	UC17
Use Case Name	Advanced Analysis
Overview	User can view the more detail food safety report after they scan the food label or barcode, if they are subscribed the VIP mode.
Actor	User(VIP mode subscribed)
Precondition	User have subscribed the VIP mode
Post-condition	Viewing a upgrade version of food safety report.

Flow of Event	Actor	System
	1. Click the VIP dashboard button on the main page. 2. Scan a food label or barcode. 3. View the upgrade version of food safety report(including the NOVA Classification, Additive Classification etc.).	1. validate of the VIP identity of the user. 2. Connect to the AI API to analysis deeper to the scanned product.

Items	Descriptions	
Use Case ID	UC18	
Use Case Name	Discount game and earn coupons	
Overview	User can view a discount game on the main page and play it can get a coupons	
Actor	User	
Precondition	User need to login	
Post-condition	Get a partner shop Coupons	
Flow of Event	Actor	System
	1. Click the “play now” button on the main page(discount game crad). 2. Play the quiz game. 3. Get the coupons and view it in the my profile page.	1. connect to ai API and return the food additives quiz game.

Items	Descriptions
Use Case ID	UC19
Use Case Name	Upgrade to VIP
Overview	Normal user can upgrade to the VIP user for additional function and also unlimited token for the AI food report.
Actor	User
Precondition	User need to login
Post-condition	Be able to use the VIP additional function and the unlimited token for the AI food report.
Flow of Event	Actor 1. Click the Upgrade to VIP button on the main page or in the profile page. 2. Choose the plan that user want to upgrade 3. Choose the payment method and pay

Items	Descriptions	
Use Case ID	UC20	
Use Case Name	Edit profile(diet/health/allergies)	
Overview	User can edit their setting of diet/health/allergies in the profile setting page, incase the user will have any diet/health/allergies change.	
Actor	User	
Precondition	User need to login	
Post-condition	Change the diet/health/allergies setting.	
Flow of Event	Actor	System
	1. Click the profile icon button to get to the profile page. 2. Roll down on the profile page and click the setting button 3. Change the diet/health/allergies setting 4. Click the save button	1. when the user click save , save the new data to the database.

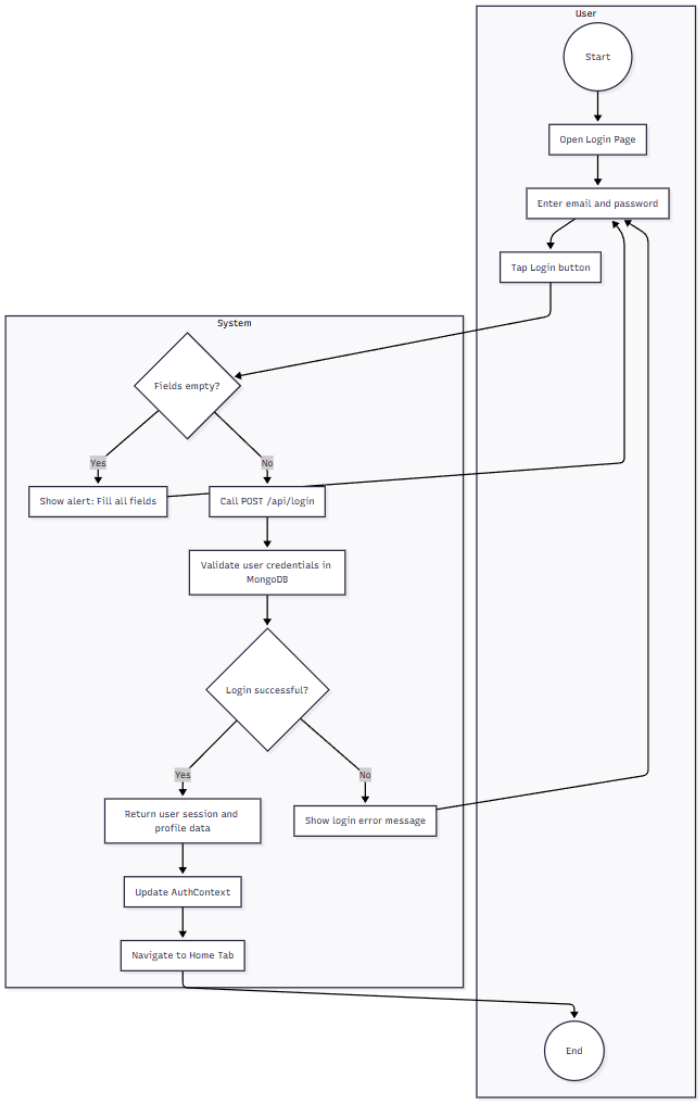
Use Case ID	UC21	
Use Case Name	Track Scan Quota	
Overview	The system will keep check the quota of the user to generate the food report, for normal user there 5 quota and for the VIP is unlimited.	
Actor	System	
Precondition	User are using the app	
Post-condition	/	
Flow of Event	Actor	System
	1. User use the scan function and generate the food report	1. check the quota when the user go to the scan icon button and also when the user has just generate the food report, the system will -1 quota of that user.

Use Case ID	UC22	
Use Case Name	Change app language	
Overview	The user can click the language button on the app top right corner to change the app language between Chinese and English.	
Actor	User	
Precondition	User are using the app	
Post-condition	/	
Flow of Event	Actor	System
	1. User click on the language button on	1. change the language of the

	the top right corner	app to other language, if Chinese change to English, if in English change to Chinese.

3.2 Activity Diagrams (on workflow) with Description

3.2.1 login

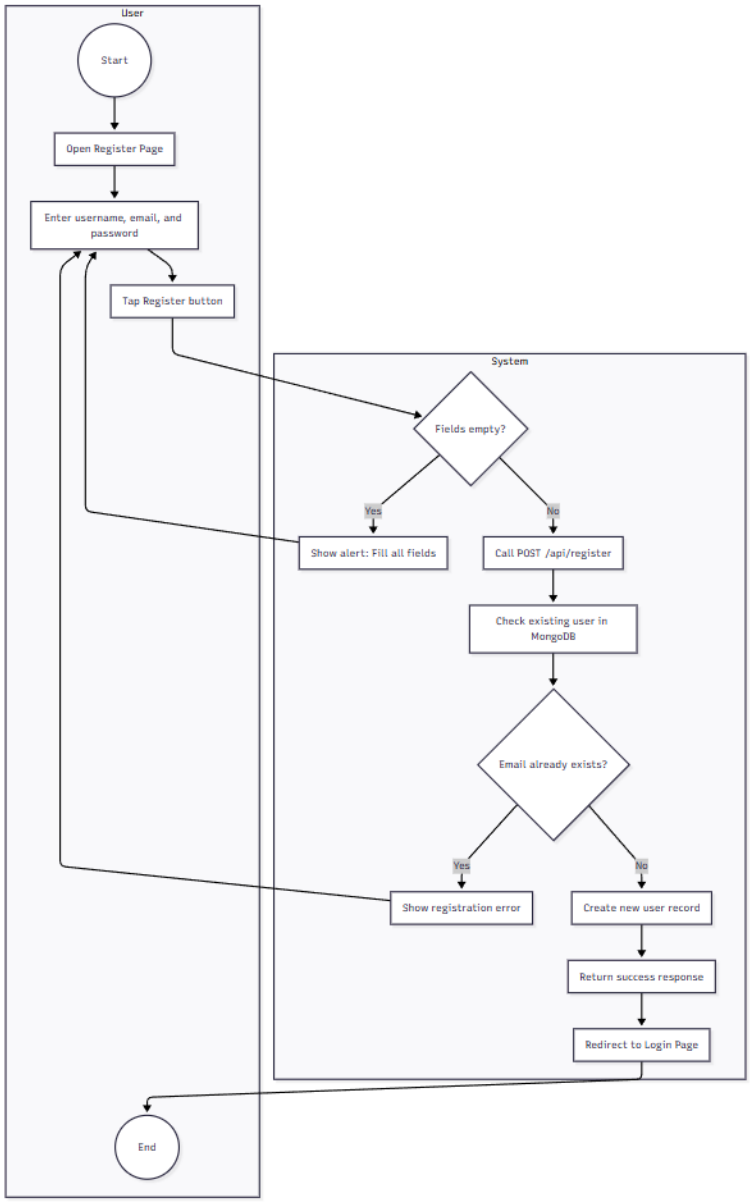


Descriptions:

When the user open the app it will first come to the login page and user need to type in the email and password and click the login button to login, and the system will verify the input ensure the input are filled and filled correctly, if any missing or incorrect input, it will prompt a reminding to user to correct. once the input are verified with the data in database, the system will get the user JWT token and data(user setting) and store

them in local storage then navigate the user to the application's main page. And if the user data(email and password) is not correct the system will pop up a error message and told the user to type the login data again.

3.2.2 User Registration Activity Diagram

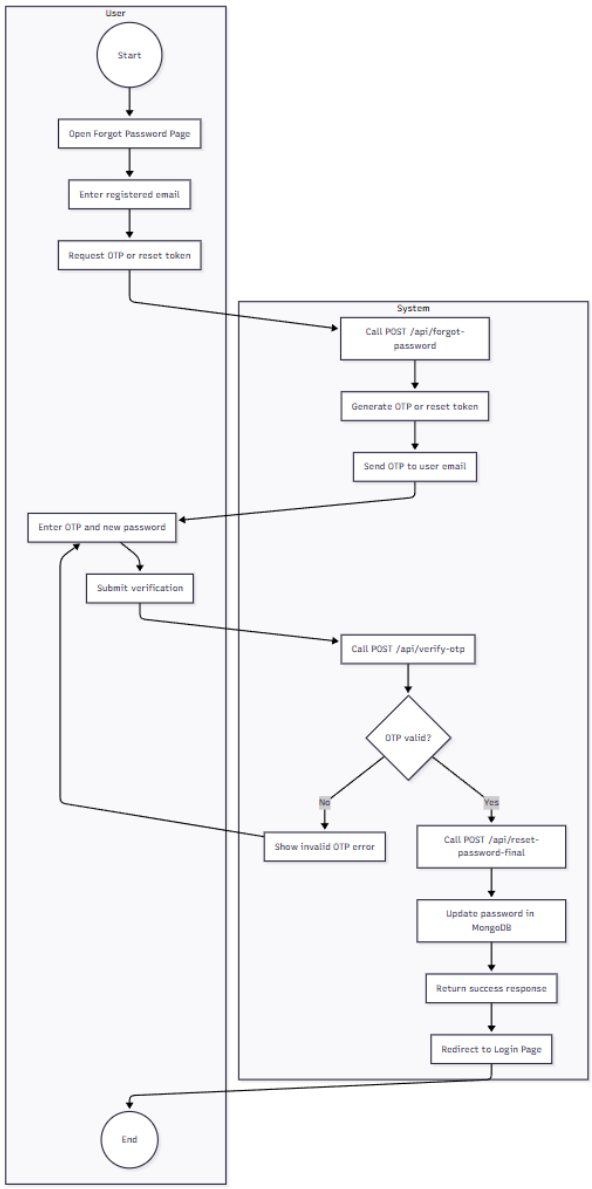


Descriptions:

when new user come to the app, they need to click the register button to register a new account, and on the register page the user need to enter suck data like email, username, ,age ,password, diet, food allergies, etc. and click the register button on the register page and the system will check is the email is used or not, and for the password system will confirm the password is match with the confirm password field or not and also check the password meet the password requirements(need to be 8 digit and with

one capital letter and one lowercase letter), once it is all good the system will alert account created and navigate the user to the login page for user to login. but if there are any requirements don't meet, it will alert the error message to the user and told them to correct it.

3.2.3 Forgot Password Activity Diagram

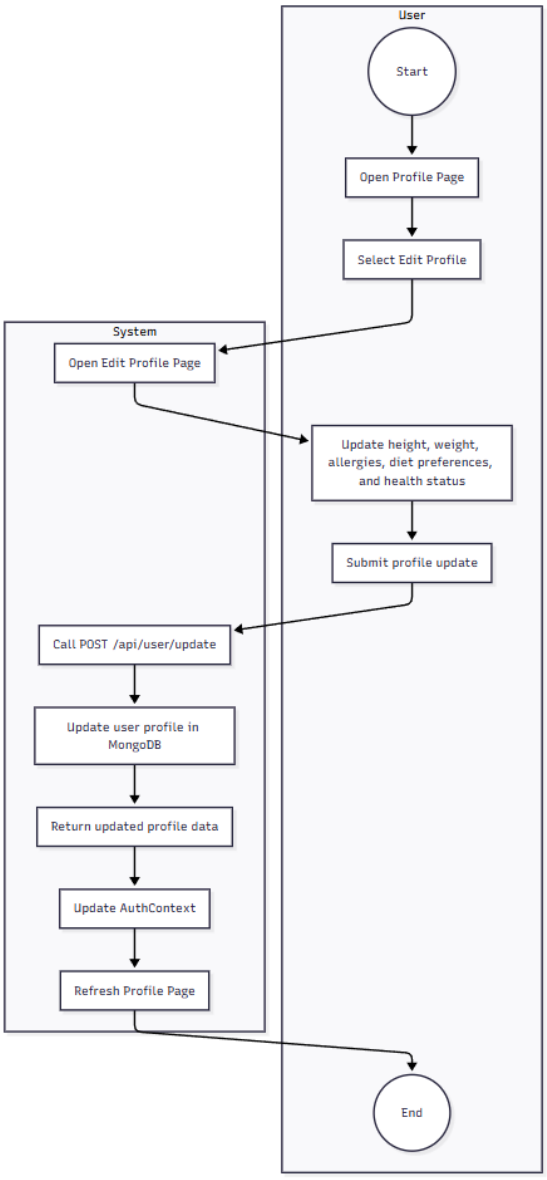


Descriptions:

When the user forgot the password, they can go to the forgot password page and enter the email and click the send button for the reset password email, and when the user inputted the email the system will check the dose the email exist with a user on the database and also check is the input field is empty or not, if all is good the system generate a random 6 digit OTP code in the database then send a reset password

email(contain a 6 digit OTP reset password code) to the user inputted email and alert the user with massage "reset link sent", then user need to go to the email and check for the OTP code and back to the app to enter the OTP code to the forgot password page, then the system will verify the inputted OTP code with the database OTP code, if match the system will let the user to type in the new password, then save and change the password in the database, last navigate the user to the login page for login.

3.2.4 Edit Profile Activity Diagram

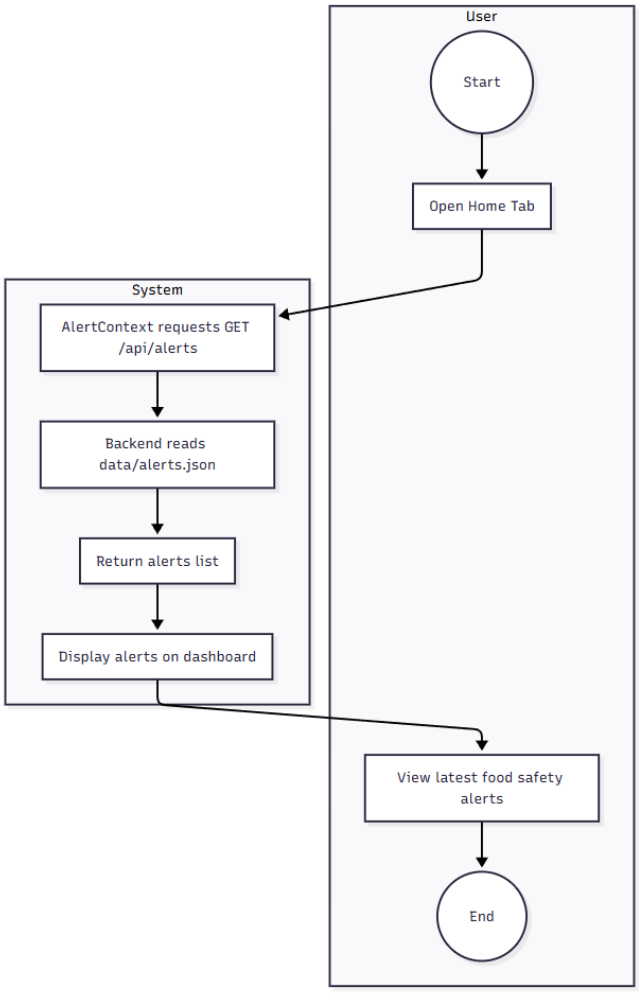


Descriptions:

The user can edit any user setting when they are logged in to the app and want to modify any user data(height, age, diet etc.), first they need to go to the profile page and roll down to look for the setting button then the system will navigate

the user to the edit profile page to let user edit any of the user data and when user are done with changing the user data they can click the submit button and the system will save it to the database and navigate the user back to the profile page.

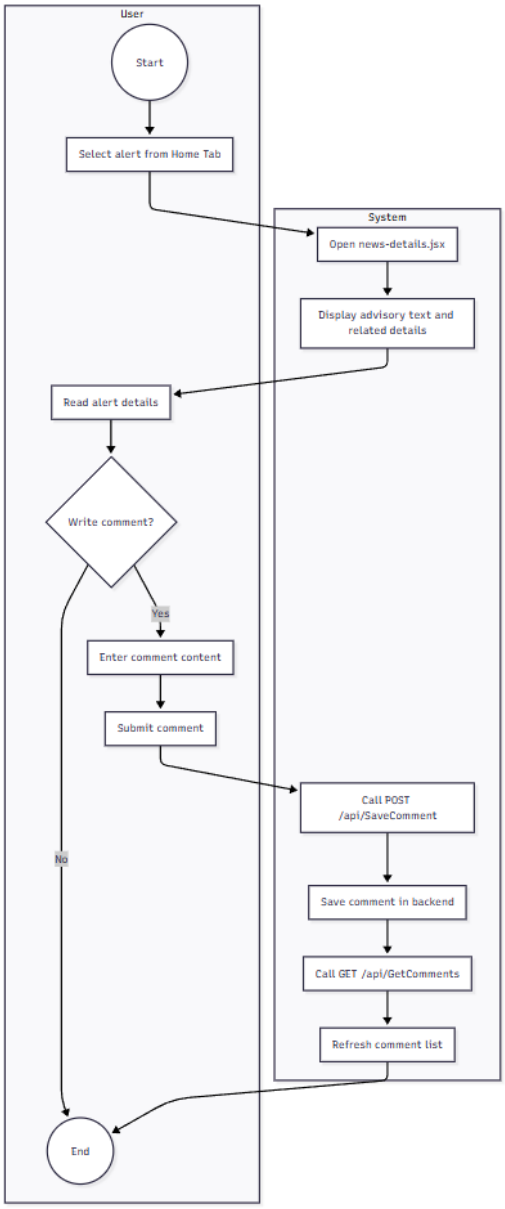
3.2.5 View Food Safety Alerts Activity Diagram



Descriptions:

The user can open the home page to view the food safety alerts, and the food safety alerts directly show on the home page, for the flow when user open the home page the system will call the backend to read for the alerts.json to get the alerts then display the alerts on the home page, the json will update every days 3 am by connecting the data.gov.hk API to get the latest alert. And for the alerts show rule, there will show 3 latest alerts, and when the alert ports days are within 7 days a “New!!” red color sign will show next to the alert title.

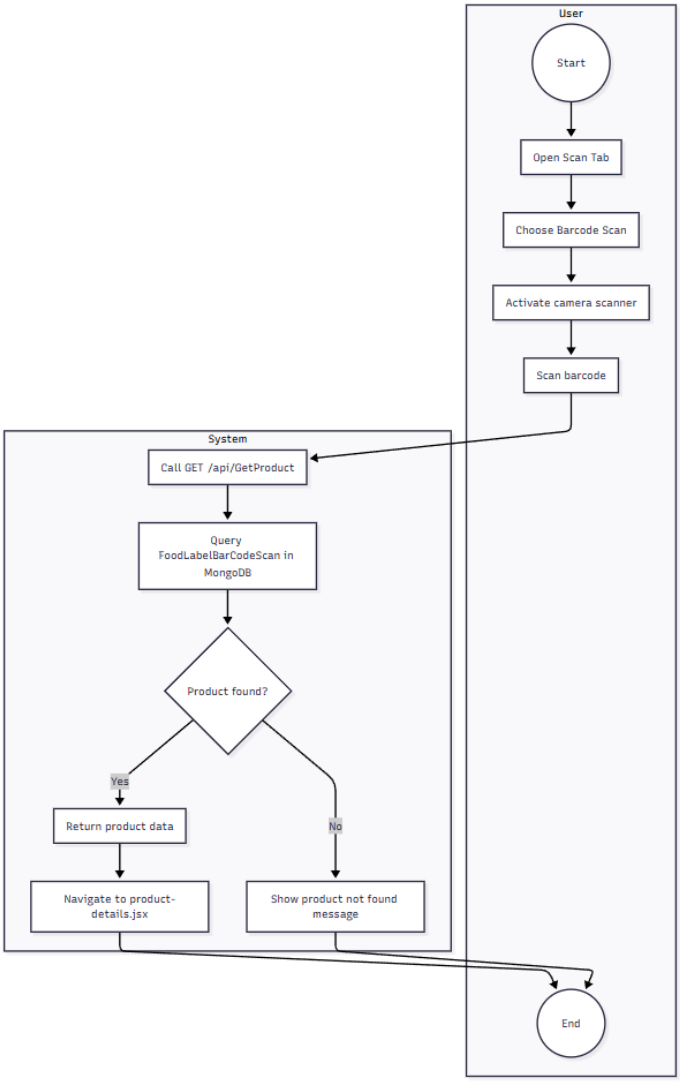
3.2.6 View Alert Details and Post News Comments Activity Diagram



Descriptions:

To view the alerts details, like such ports date, alert product information, alert source etc., user can first go to the home page and select the alert that they were interested then system will navigate the user to the alert details page for user to view the details, also user can roll down to view other user comment or comment the alert by their own, once the comment has added the system will save it to the database and refresh the comment list.

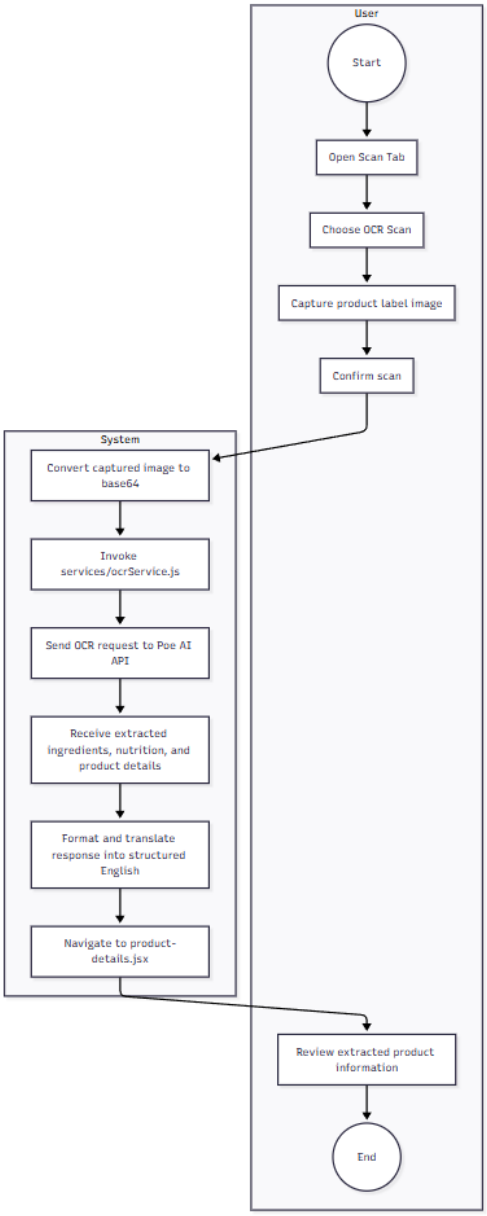
3.2.7 Scan Product by Barcode Activity Diagram



Descriptions:

By using the barcode mode, the user can go to the scan page and select the barcode mode then the page will let user to scan for the barcode, once the system has detected a barcode it will read for the data(the product code number) and find it in the database, if found the system will return the data to the fontend then navigate user to the product detail page and call for the AI API to generate a personalize report, if not found the system will show the “not found” error message to the user and let the user to scan again.

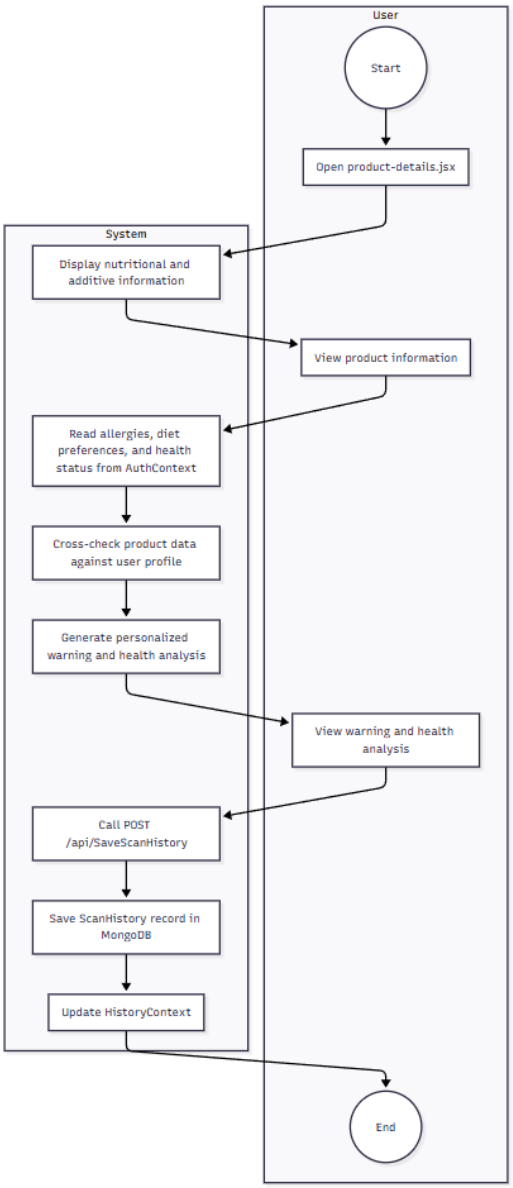
3.2.8 Scan Product by OCR Activity Diagram



Descriptions:

And for the text mode of the scan function, when the user are in the scan page they can select the text mode and the user can select whether to take a phot of the food label or upload a food label photo and when user are using either of one method to provided the photo of a food label the system will connect to the AI API and send it to the AI then get the returned OCR result of the food label and navigate the user to the edit page to scan for barcode, product name input and the OCR result edit then if all the above moves are done and submit button are clicked the scan part is done.

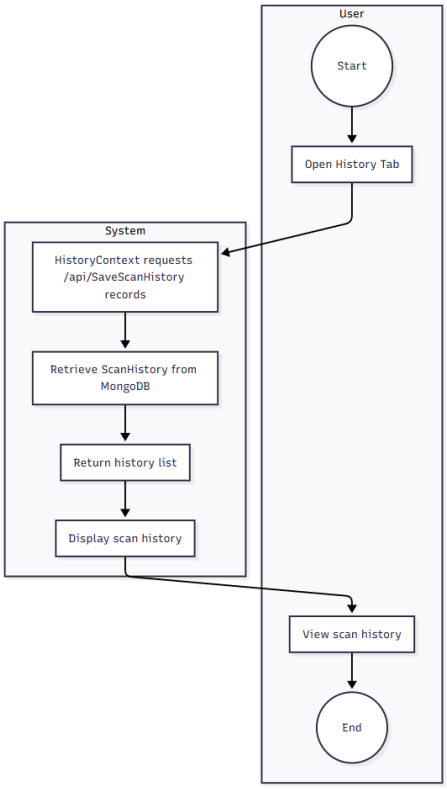
3.2.9 View Product Details and Save Scan History Activity Diagram



Descriptions:

And when the 3.2.10 scan part is done the system will send the user setting and also the OCR result to the AI and first navigate the user to the product detail page showing the product name, barcode number, OCR result and waiting for the food analysis report(AI return result), and once the AI has return the report it will show on the product detail page. And after the user are done with viewing the report and click the back to history page button(which on the bottom of the product detail page) the system will save all the data(name, barcode number, OCR result and the report) in the product detail page to the database.

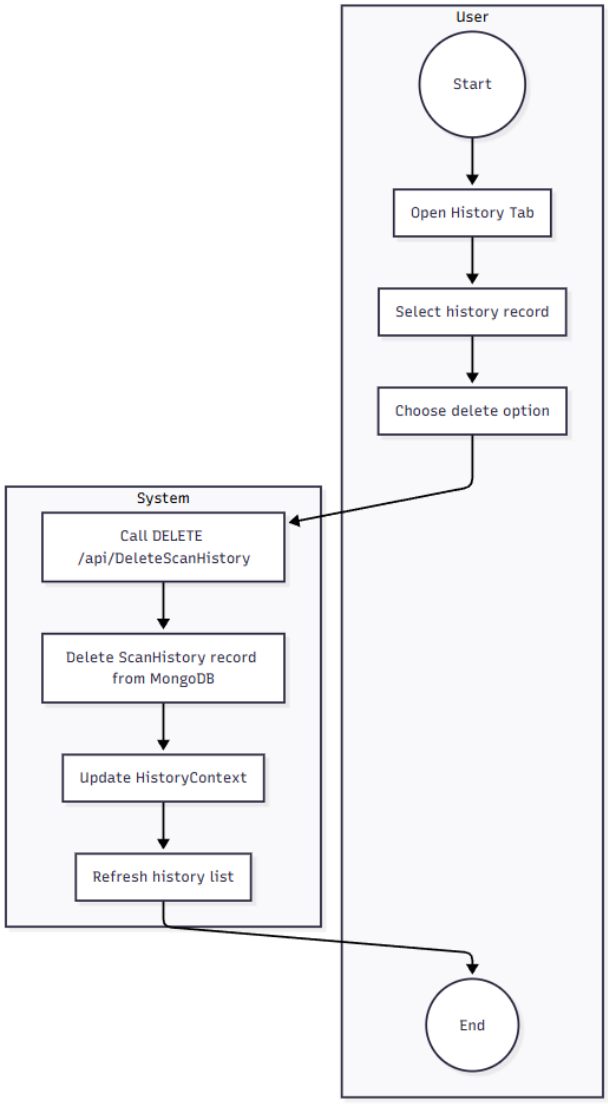
3.2.10 View Scan History Activity Diagram



Descriptions:

The user can open the history page to view all the scan history, but only with the barcode number and the product name, if the want to view the detail they need to click the view button that next to the product name, if the view button is clicked the system will navigate the user to the corresponding product detail page and return the historical detail data (name, barcode number, OCR result and the report) for user to view.

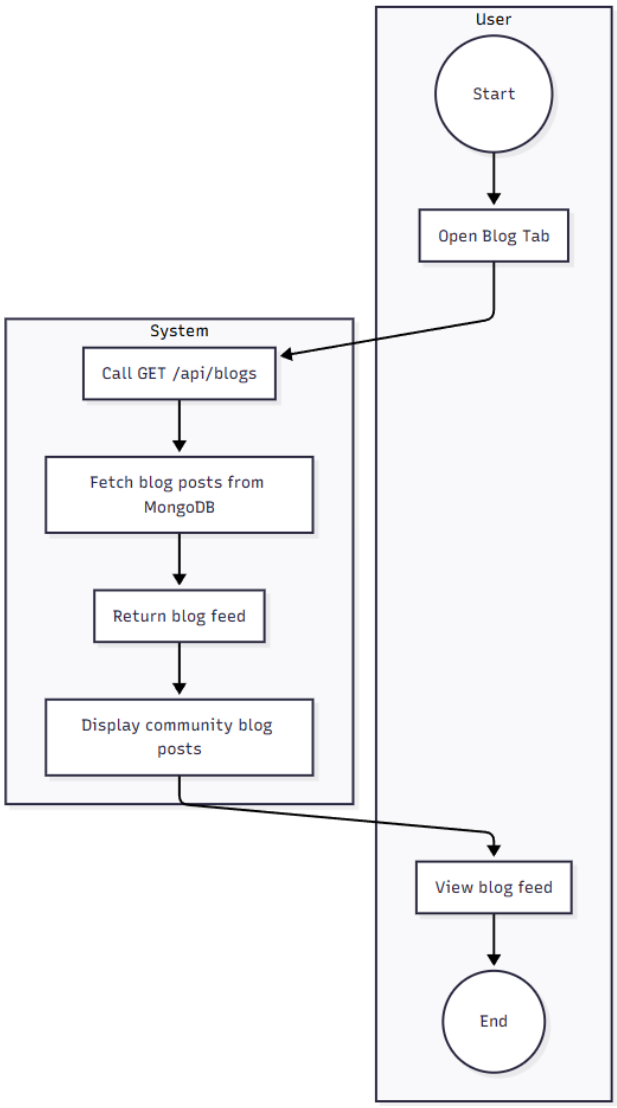
3.2.11 Delete Scan History Activity Diagram



Descriptions:

When user want to delete a scan history, they can go to the history page and click the delete button that next to the view button in each scan history and the system will delete the corresponding historical detail data.

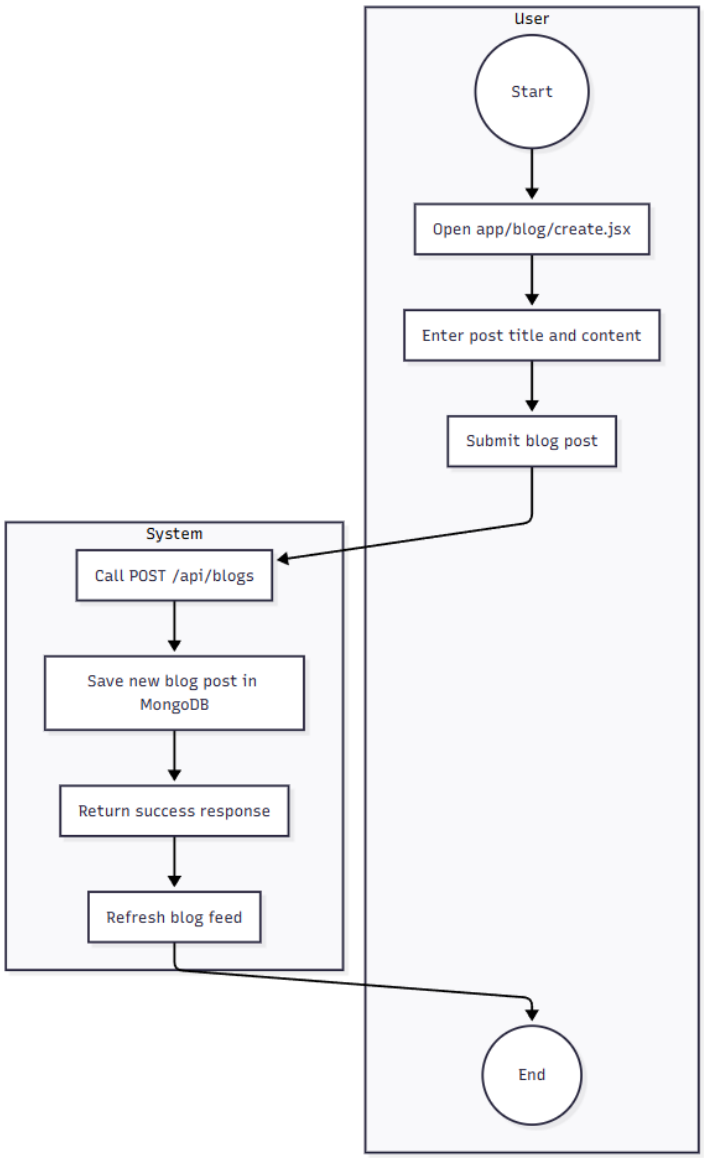
3.2.12 View Blog Feed Activity Diagram



Descriptions:

For the blog function, user can click the blog tab to navigate to the blog page and the system will fetch all the post that in the database(the post may created by the user themselves of other user), and show on the blog page.

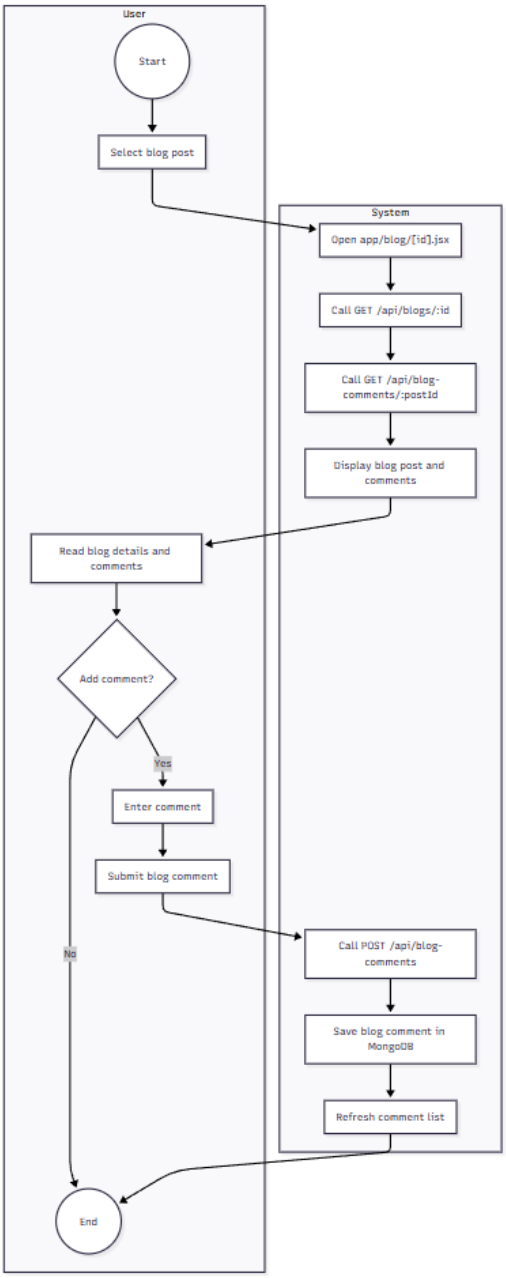
3.2.13 Create Blog Post Activity Diagram



Descriptions:

To create a blog post user can first go to the blog page and then click the “+” button on the top right corner of the blog page, and the system will navigate the user to the create blog page for user to enter the blog title and also the content then click the submit button, the system will save the blog to the database and refresh the blog list then navigate the user back to the blog page.

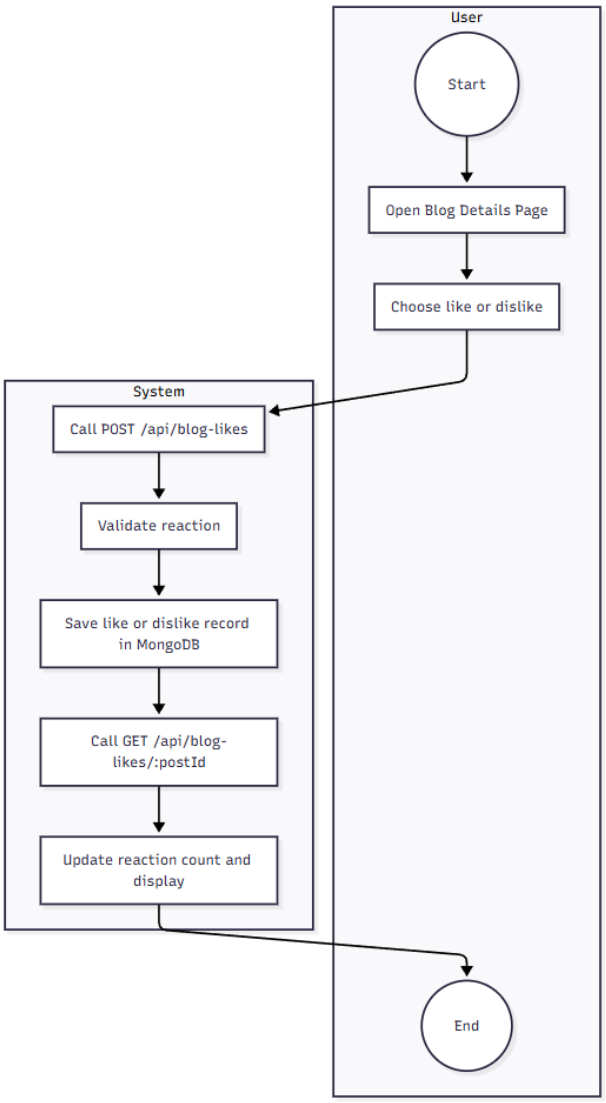
3.2.14 View Blog Details and Blog Comments Activity Diagram



Descriptions:

For the blog detail, user can first do the flow of 3.2.12 but on the blog page there are only showing the blog title and the user that who post it so to view the detail user need to select the post that they were interested and the system will navigate the user to the blog detail page to view the detail and comment of that post, and user are also comment the blog by roll down to the comment section and type in the comment box and click send button, the system will save the comment and refresh the comment section.

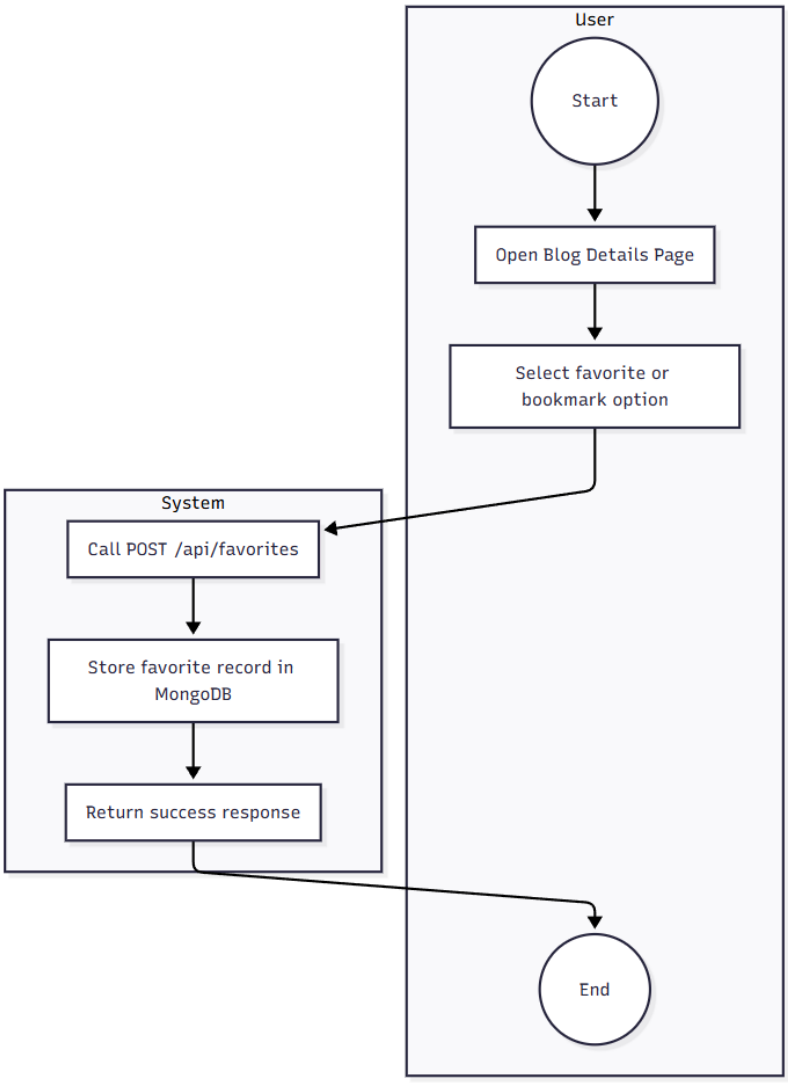
3.2.15 Like or Dislike Blog Post Activity Diagram



Descriptions:

This is also the function that include to the blog function, when the user has selected one of the blog and go in to the blog detail page, there will be two button above the comment section of the page, one is likes and the other is dislike, user can select either of one on each post or when user has selected one of it they can change it by click the other one, and no matter each moves the system will save the record to database and refresh the page.

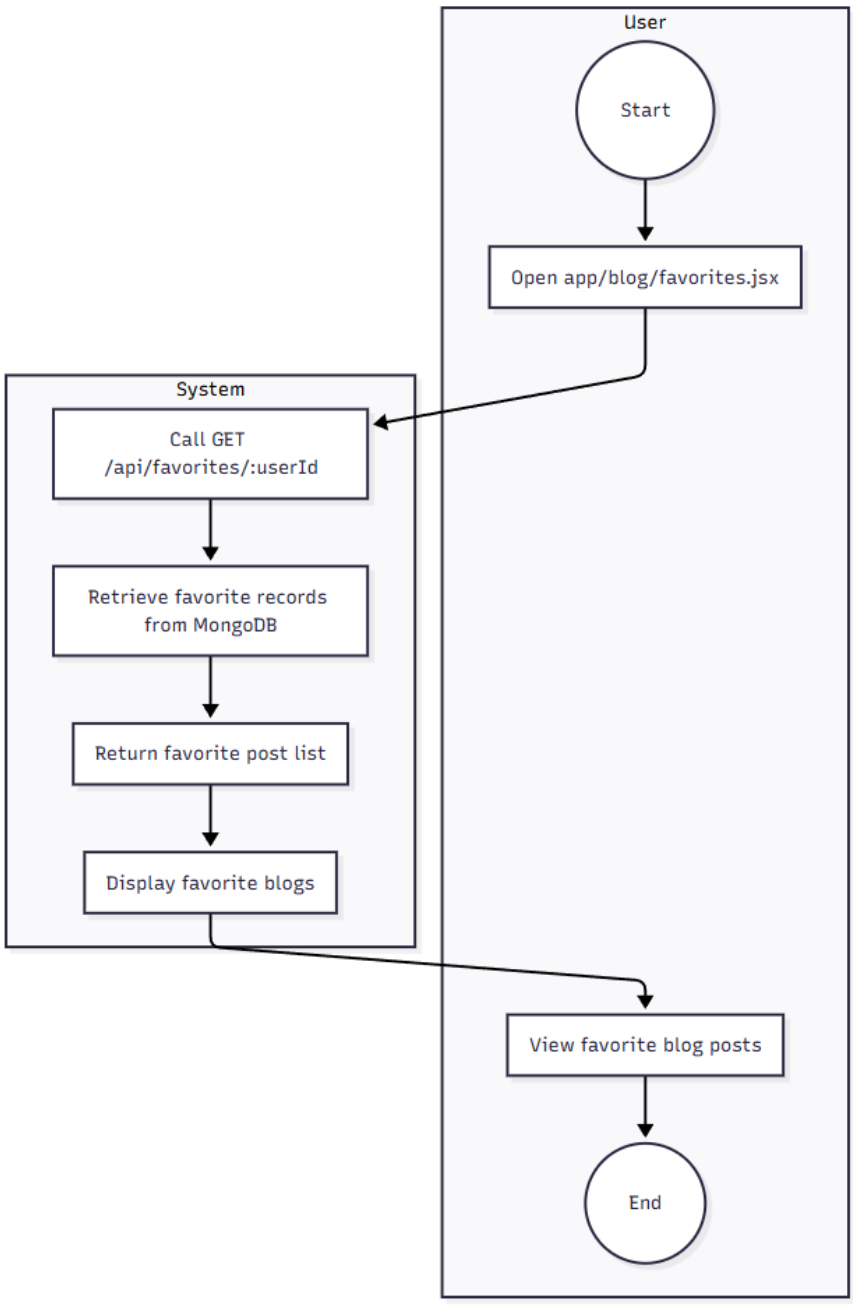
3.2.16 Add Blog Post to Favorites Activity Diagram



Descriptions:

User can add the blog post to favorites, when the user has selected one of the blog and go in to the blog detail page, there will be a add to favorites button on the left side of the like/dislike button, when the user click it the blog will added to the user favorites list on the database and when user go to the user favorites list they can see the favorited blog there help user to easier to find the blog.

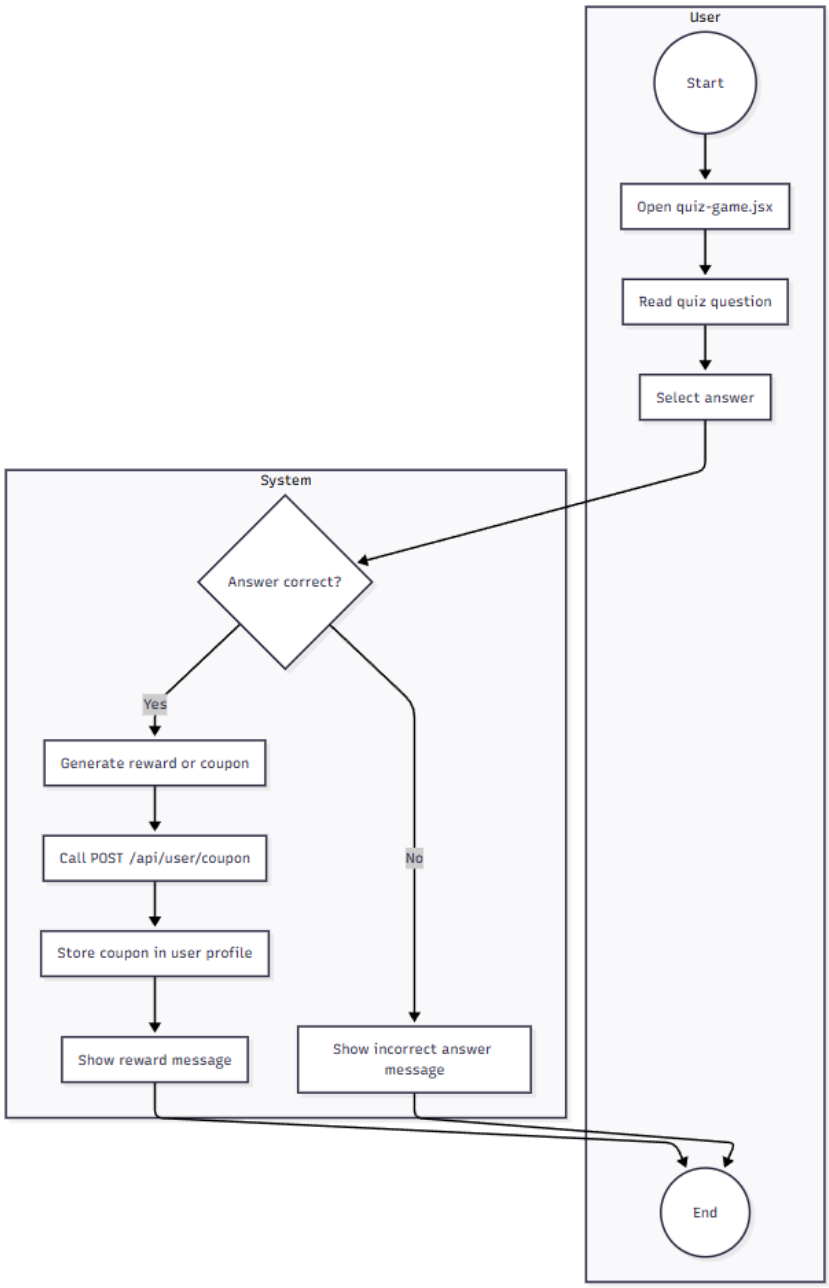
3.2.17 View Favorite Blog Posts Activity Diagram



Descriptions:

When user has done the moves in 3.2.16(which already added some blog to favorited), user can find a favorite button on the profile page and when user click in the system will fetch all the blog posts that are added to favorite, and display it as the format that same with the blog page (only title and author name), for user easier to find the blog that they want.

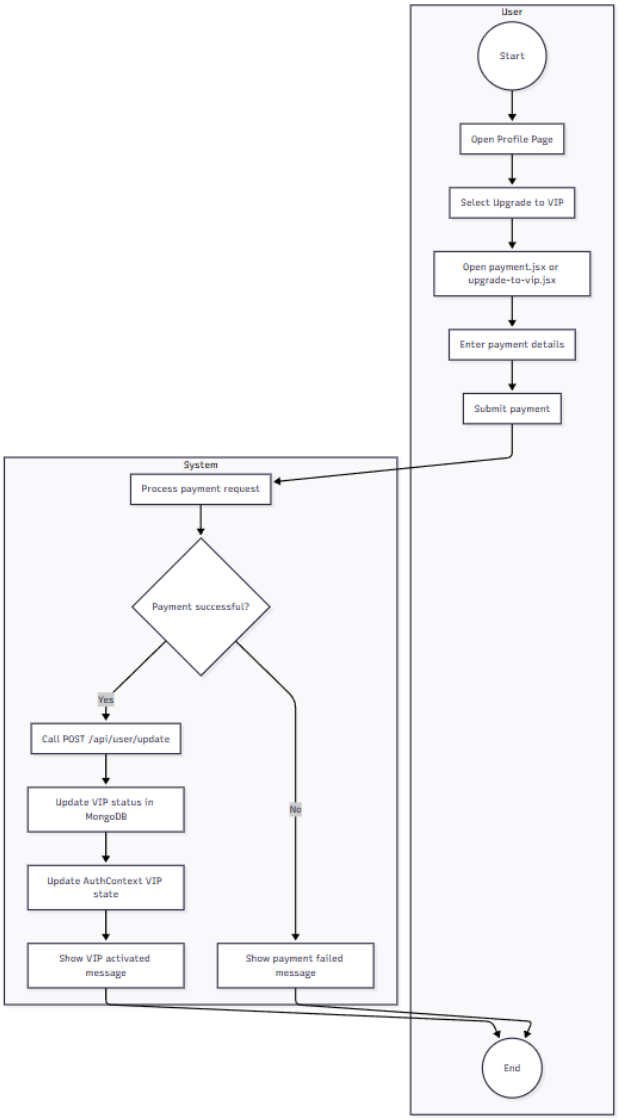
3.2.18 Play Quiz Game and Receive Coupon Activity Diagram



Descriptions:

The user can view the discount game on the main page and the user can click the play now button to play it and the system will use the AI API to get a set of question then the user answer the question the system check the answer if correct add a point if incorrect no point then go to next question until all the question are answered the system show game over and view the final score then a coupon will generate and let the user check it in the profile page.

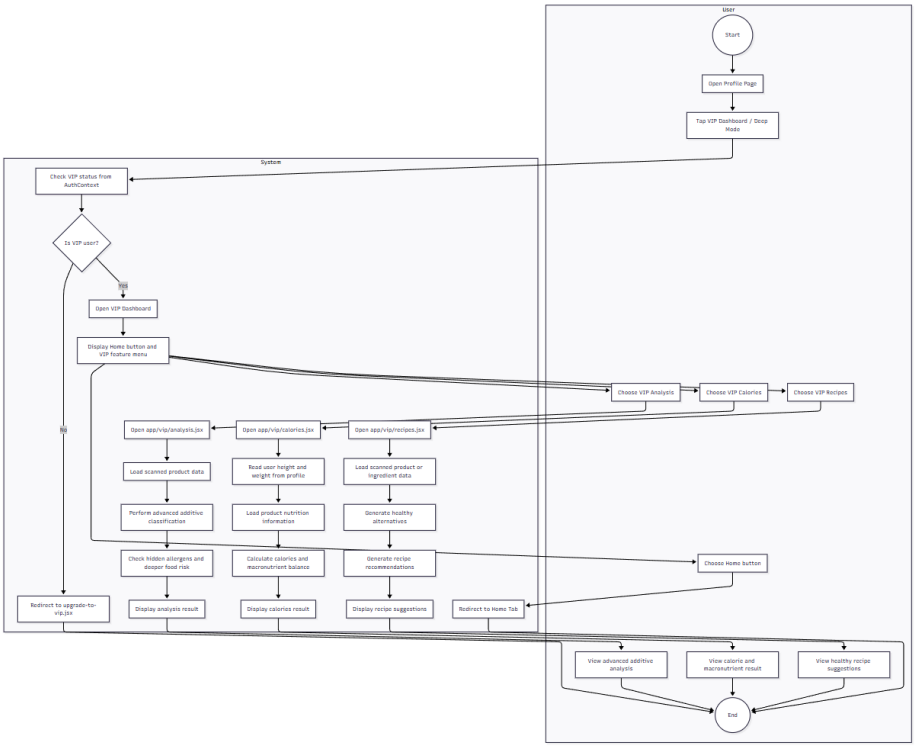
3.2.19 Upgrade to VIP Activity Diagram



Descriptions:

The user can go to the VIP upgrade page which is on the home page and profile page and on the VIP upgrade page the user will first see what is about the VIP mode, and if the user want to upgrade they can click the upgrade button and the system will navigate the user to the VIP plan select page for user to choose what type of VIP level they want and the system will take the user to the payment page let user choose the payment way then the system will alert to the user “VIP Activated” and change the user database “Isvip=true” and navigate the user back to the profile page. and if the payment is fail system will alert the user “payment fail” and let the user to do the payment process again.

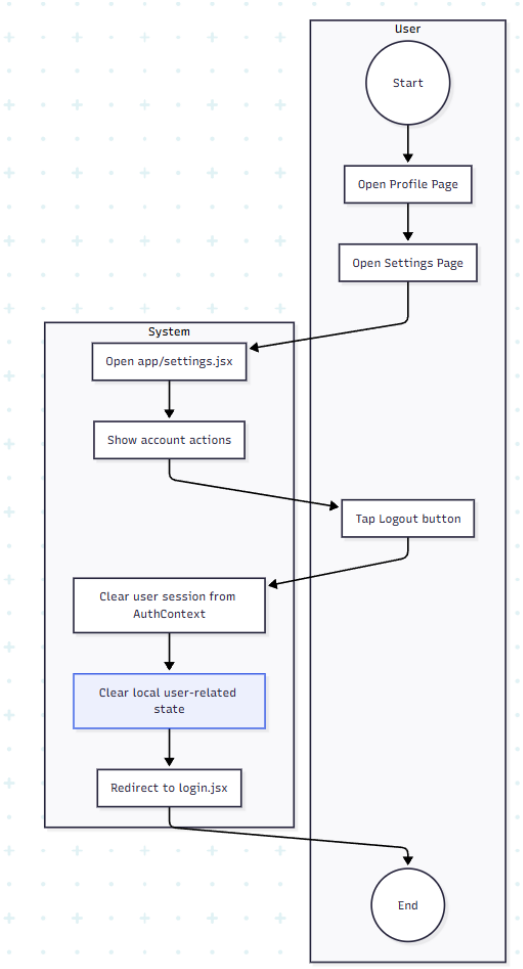
3.2.20 Access VIP Features Activity Diagram



Descriptions:

When the user has Upgraded to VIP, the system will show a VIP Dashboard on profile page and also the home page, then when user click it the system will check is the user is VIP or not then there will be 3 function to let the user choose to use Recipes, Calories & Analysis. For Recipes the system are letting the user to take a photo or upload a photo to the system and then the system will send the photo to the AI and get the result to show to the user, and the result are mainly recognize what ingredients are in the picture, analysis of the ingredients(too check whether the ingredients is harmful for the user) and a recommend recipe(if the ingredients in the photo are contain some harmful ingredients or additives the recipe will use other ingredients to replace it in the recipe), and for the calorie calculator the system can let the user to set the goal of the calorie then the system will first load all the item scan history of the user and let them select the scanned product and input how many they have eat(in gram) then count the calorie for the food and fill up the goal bar with corresponding calorie to let the user easier to see how many percent of calorie they has eat, and last for the Advanced analysis the system will also load all the scan history of the user and when user select one of the history product the system will send the data to the AI and get back a more detailed analysis report for the user, and on the report there will having a additive classification, NOVA level, hidden allergen detection and macronutrient balance and all this deep analysis are not in the normal scan analysis.

3.2.21 Logout

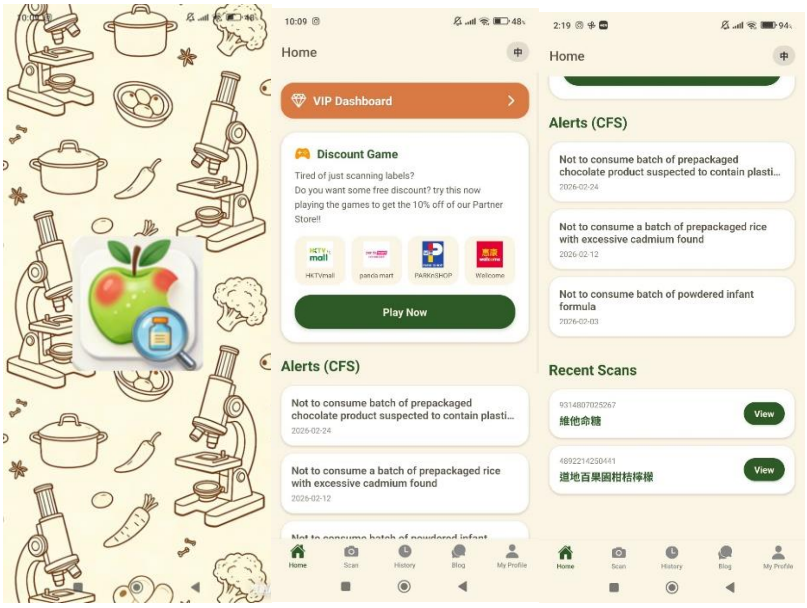


Descriptions:

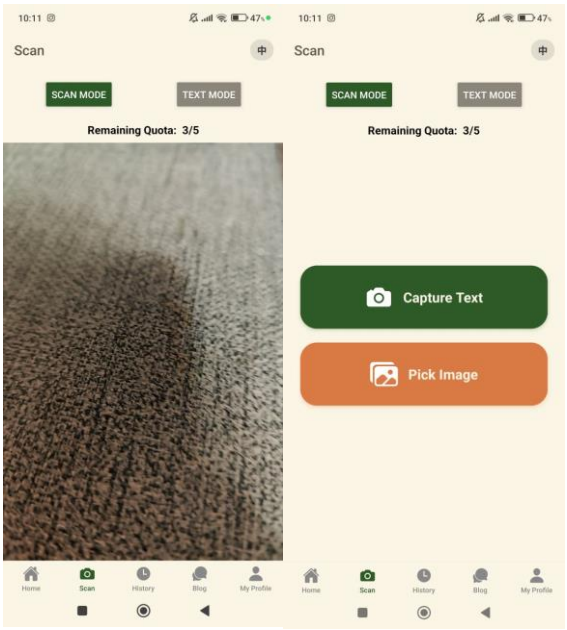
The user can go to the profile page and roll down to find the logout button and when the user click it the system will clear the local state and navigate the user to the login page.

3.3 User Interface Prototype

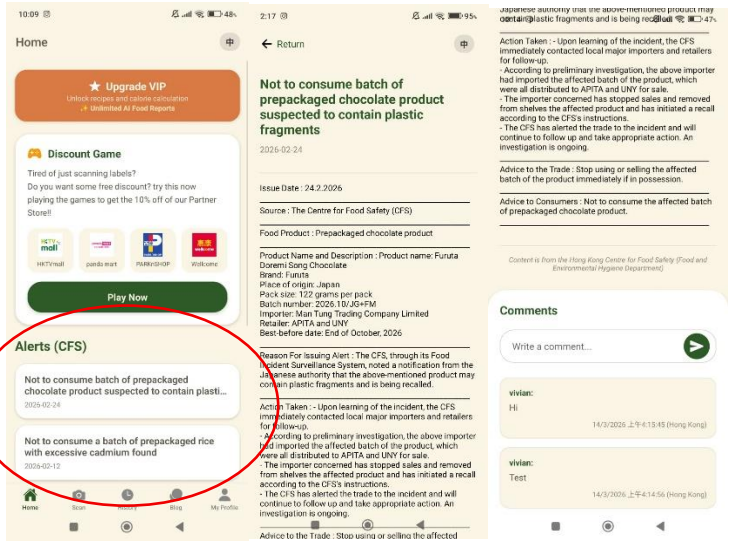
Landing page and home page:



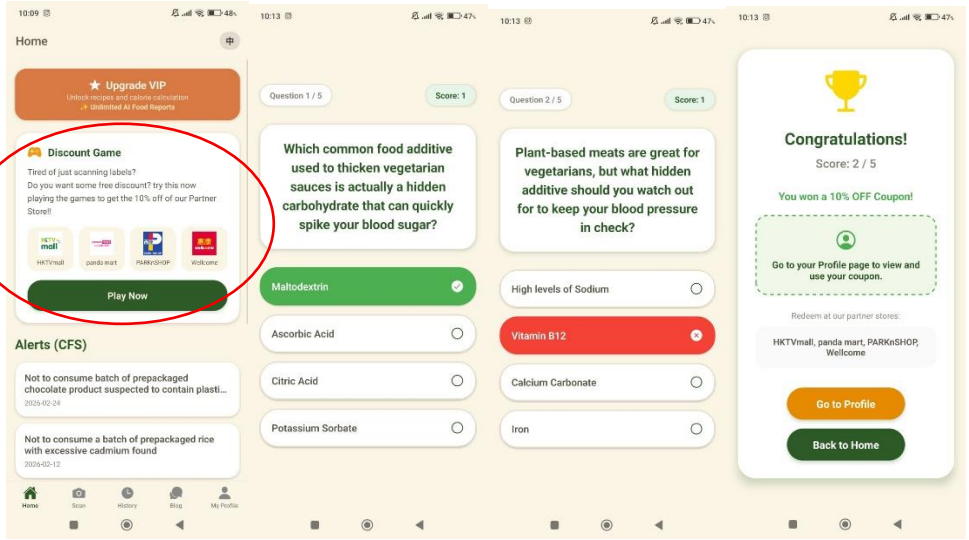
Scan page(bar code mode and Text mode(OCR)):



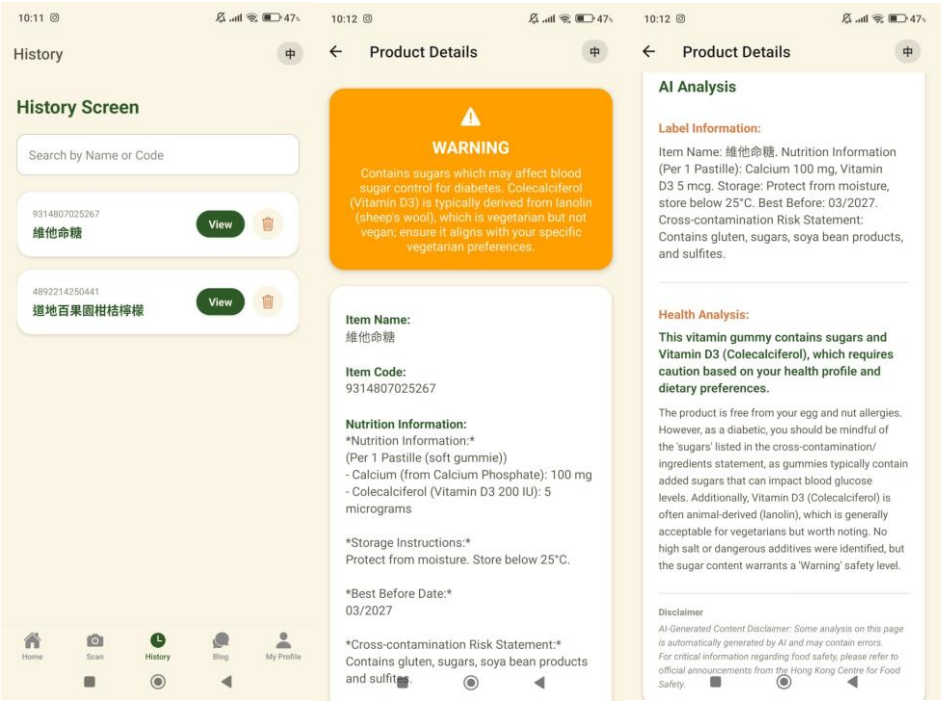
Alert news section and alert detail page:



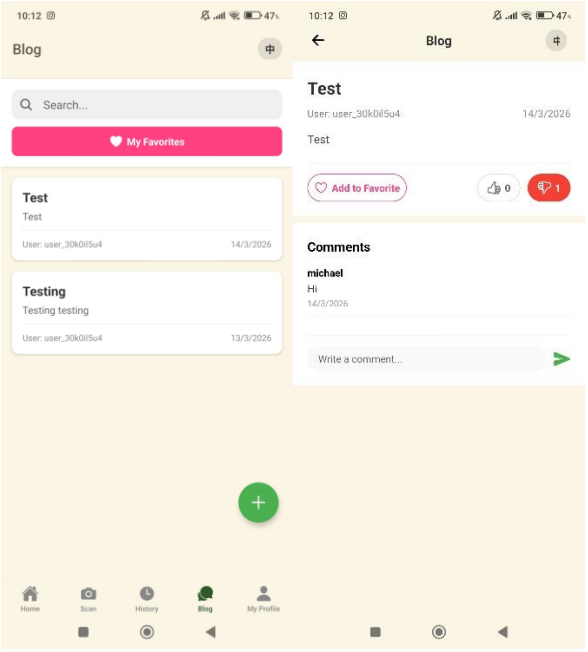
Quiz game section and page:



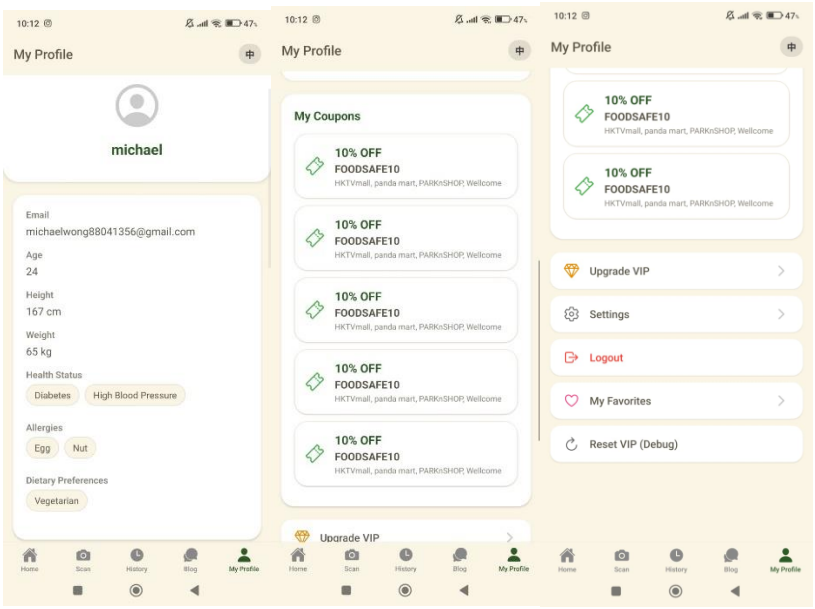
History and product detail page:

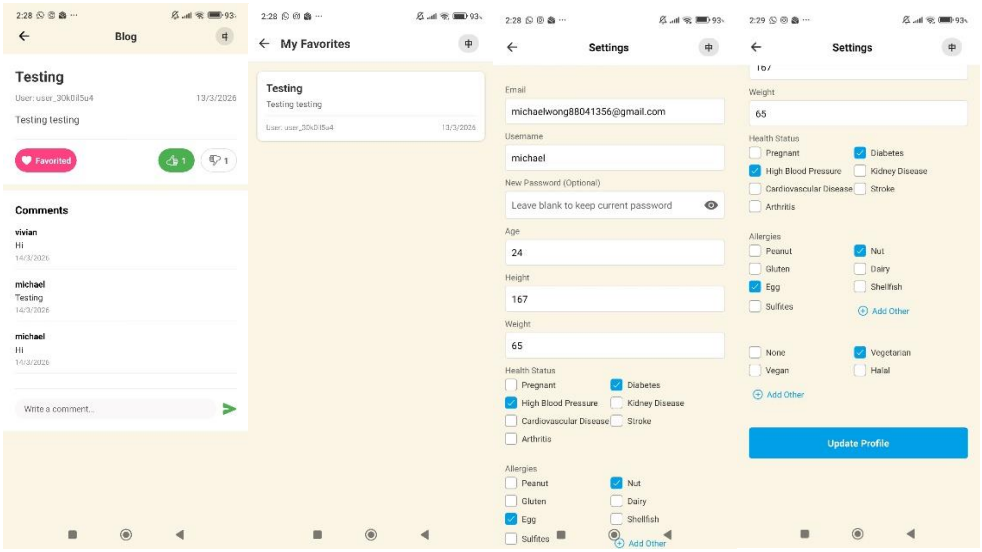


Blog and blog posts detail page:

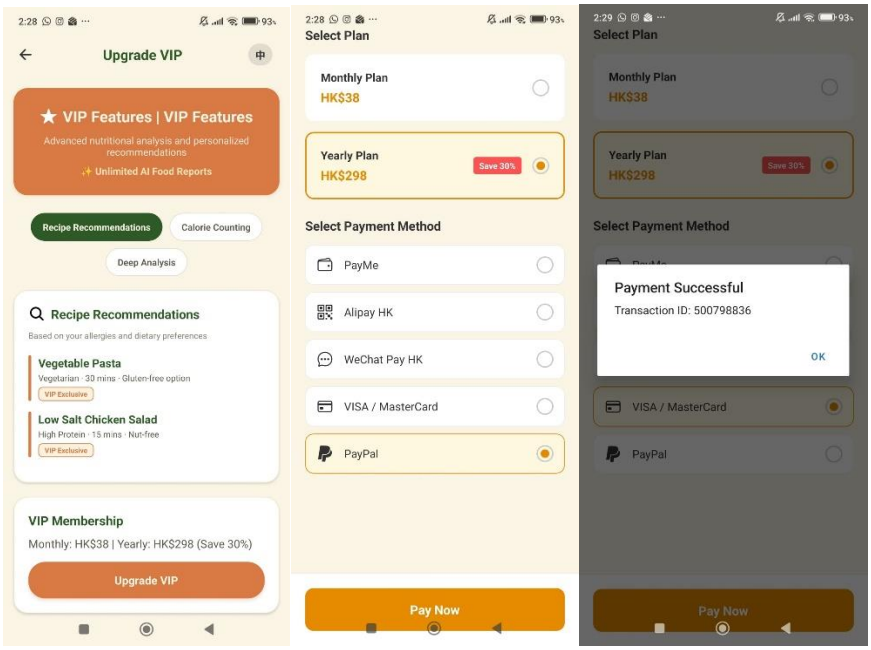


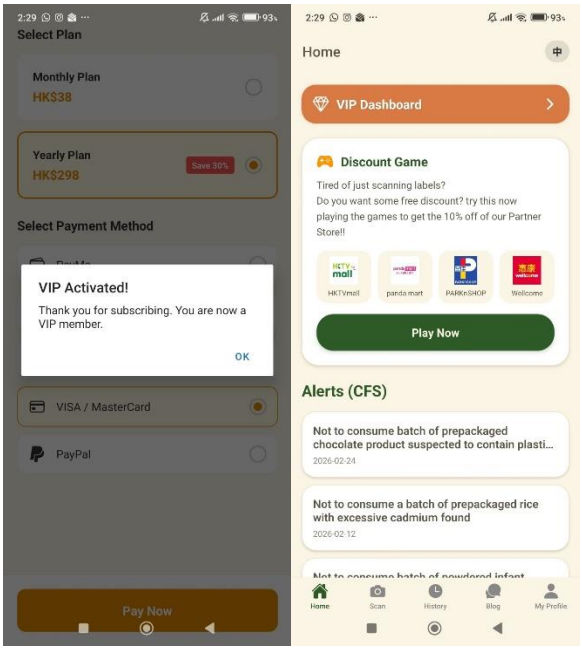
Profile , favorites and setting page:



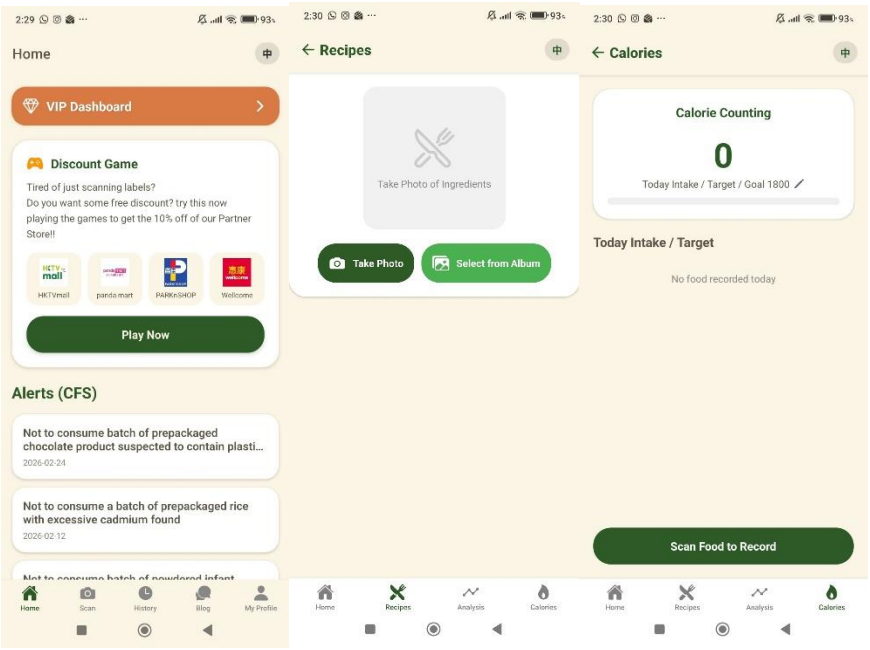


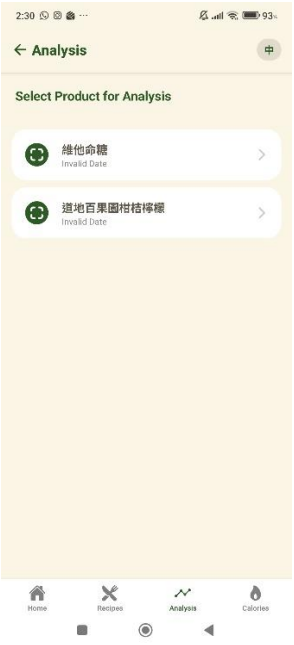
VIP upgrade and payment page:



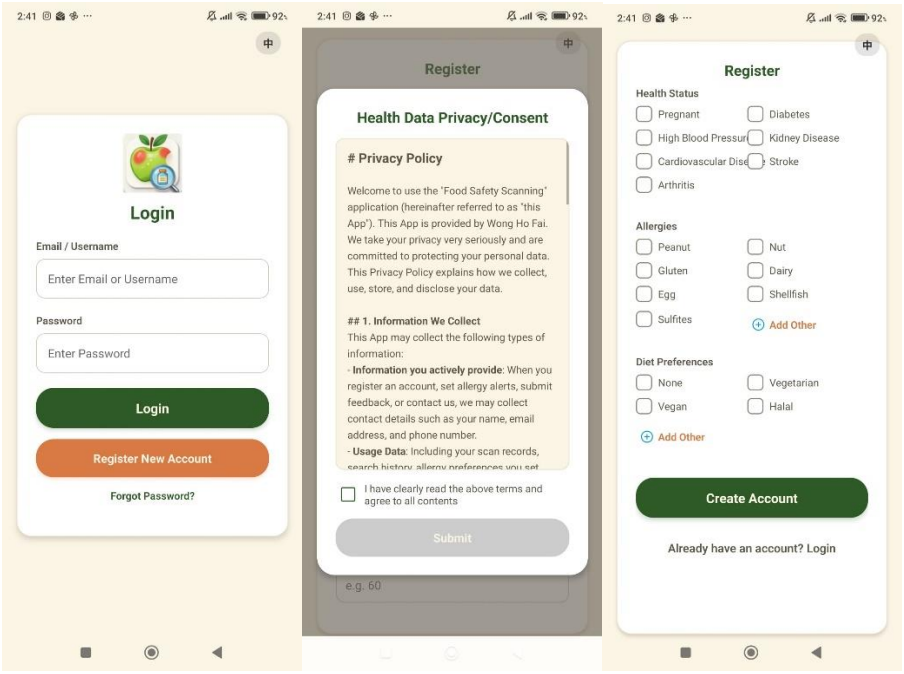


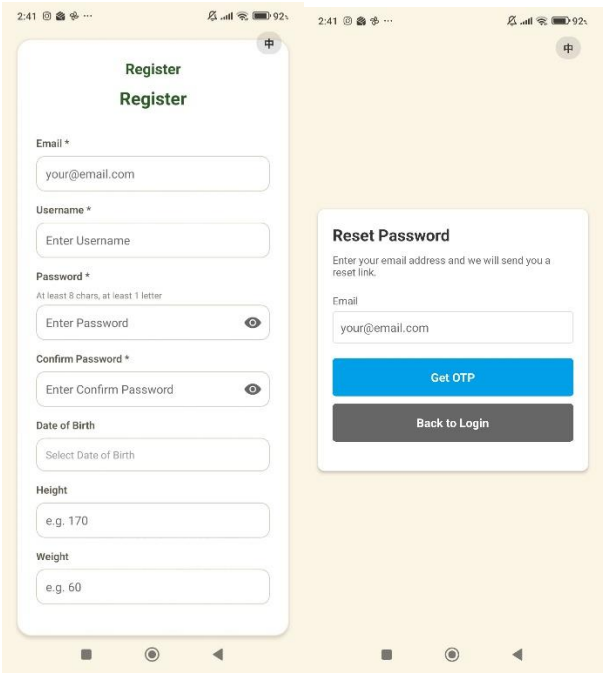
VIP feature page:





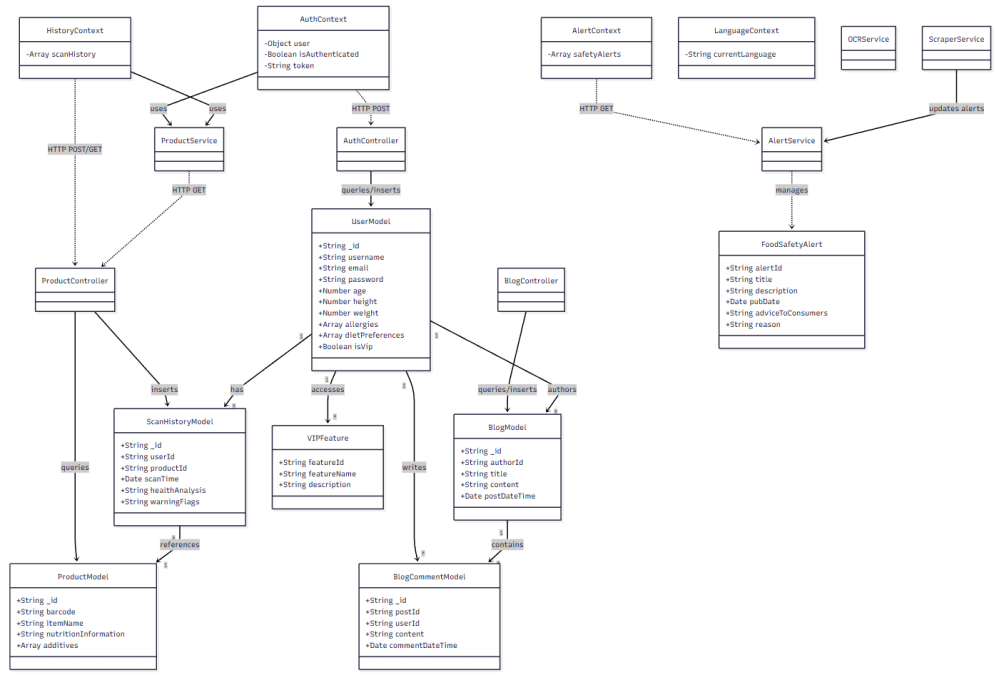
Login, Register and Forgot Password page:



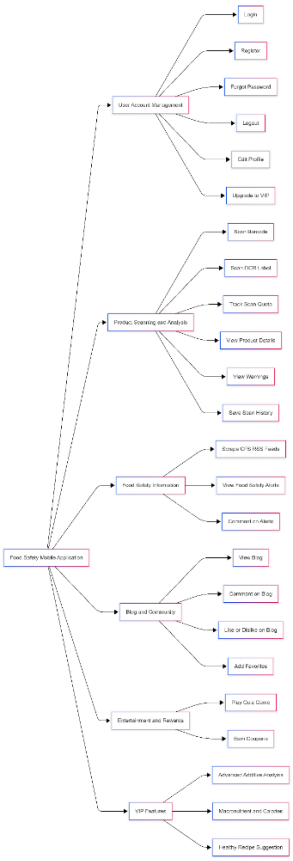


For detail screenshot of the system in A3.

3.4 Class Diagrams (with attributes only)

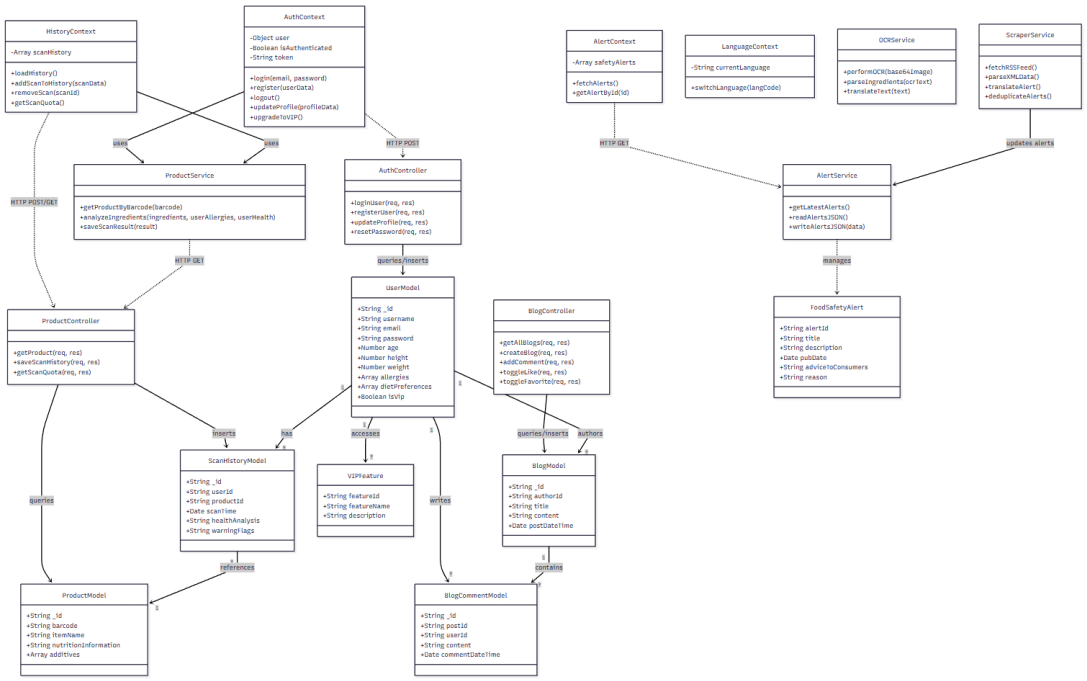


3.5 FDD Diagrams



4. Systems Design

4.1 Class Diagrams & Descriptions (with attributes & methods)



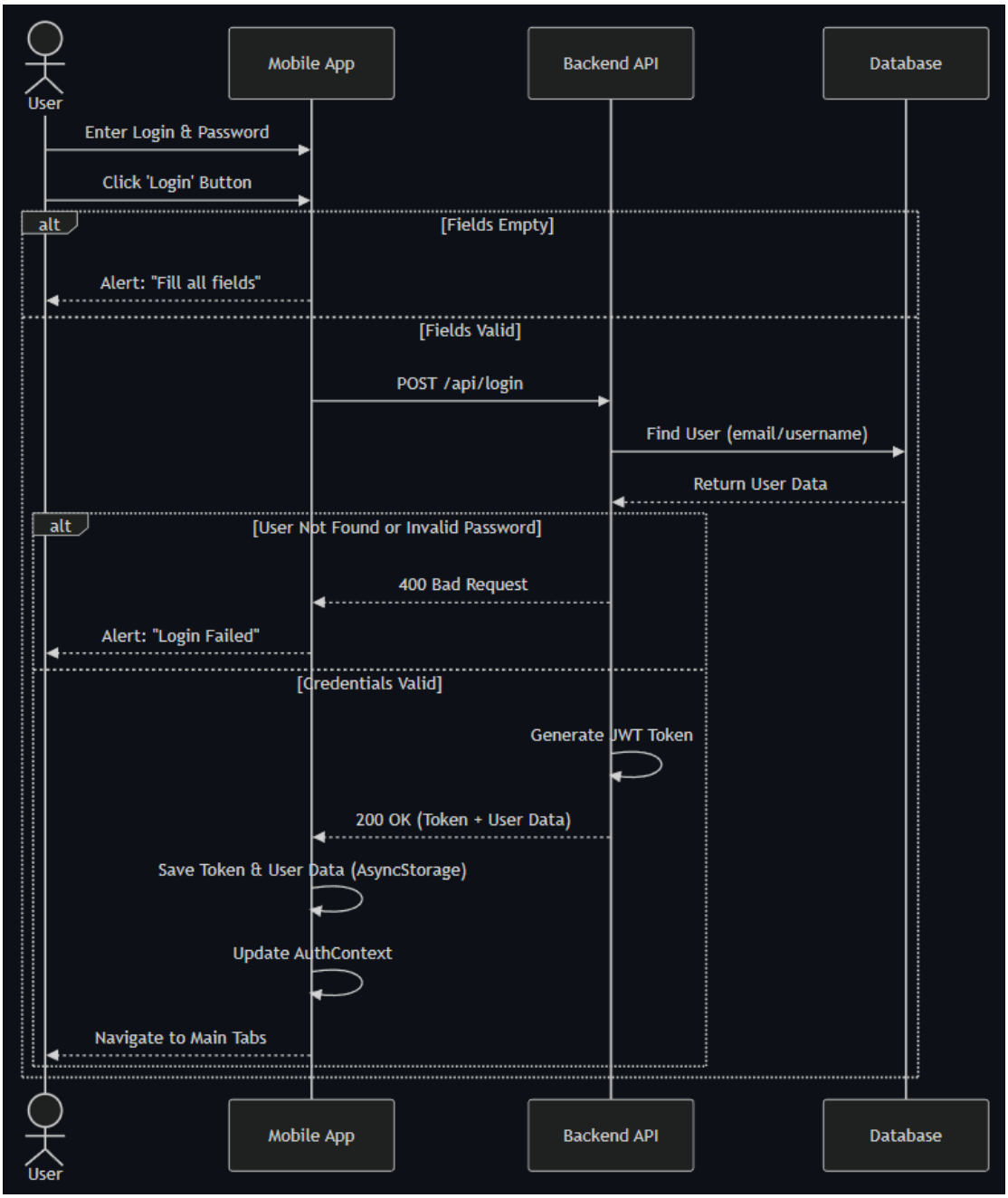
Class	Descriptions
AuthContext	This class is the class that manages the user authentication state in the frontend of the system. It stores the current user object, authentication status, and access token. It also includes methods for login, registration, logout, profile update, and VIP upgrade.
HistoryContext	This class is the class that manages the scan history data in the frontend of the system. It stores the list of scanned product records. It also includes methods for loading scan history, adding a new scan record, removing a scan record, and checking the scan quota of the user.
AlertContext	This class is the class that manages the food safety alert data in the frontend of the system. It stores the list of safety alerts retrieved from the backend. It also includes methods for fetching alert data and viewing a specific alert by its ID.
LanguageContext	This class is the class that manages the language setting of the application. It stores the current selected language and provides the method to switch the application language according to the user preference.
OCRService	This class is the service class that handles optical character recognition processing in the system. It includes methods to extract text from the uploaded image, parse ingredient

	text, and translate the text into the required language.
ProductService	This class is the service class that handles product-related operations in the frontend. It includes methods for retrieving product information by barcode, analyzing ingredients based on user allergies and health conditions, and saving scan results.
AuthController	This class is the controller class that handles user authentication operations in the backend. It includes methods for user login, user registration, profile update, and password reset. This class communicates with the user model for data retrieval and storage.
ProductController	This class is the controller class that handles product and scan history operations in the backend. It includes methods for retrieving product information, saving scan history records, and checking the scan quota of the user. This class communicates with the product model and scan history model.
BlogController	This class is the controller class that handles blog-related functions in the backend. It includes methods for retrieving all blog posts, creating a new blog post, adding comments, toggling likes, and toggling favorite actions. This class communicates with the blog model for blog data management.
AlertService	This class is the service class that handles food safety alert data in the backend. It includes methods for retrieving the latest alerts, reading local alert data from JSON files, and writing updated alert data into JSON files. It manages the food safety alert records used by the system.
ScraperService	This class is the service class that handles the scraping and processing of food safety alert data from external sources. It includes methods for fetching RSS feeds, parsing XML data, translating alert content, and removing duplicate alert entries before storing them in the system.
UserModel	This class is the model class that stores all user account information in the database. It includes username, email, password, age, height, weight, allergies, diet preferences, and VIP status. This class acts as the main entity for user-related data in the system.
ProductModel	This class is the model class that stores product information in the database. It includes barcode, item name, nutrition information, and additives. This class is used to retrieve product details when the user scans or searches for a product.

ScanHistoryModel	This class is the model class that stores the scan history records of users in the database. It includes the user ID, product ID, scan time, health analysis result, and warning flags. This class is used to keep track of the products scanned by each user.
FoodSafetyAlert	This class is the entity class that stores food safety alert information in the system. It includes the alert ID, title, description, publication date, advice to consumers, and reason for the alert. This class is used by the alert service to manage alert records.
BlogModel	This class is the model class that stores blog post information in the database. It includes the author ID, title, content, and post date and time. This class is used for storing and retrieving user-generated blog posts in the community feature.
BlogCommentModel	This class is the model class that stores blog comment information in the database. It includes the related post ID, user ID, comment content, and comment date and time. This class is used to store and retrieve comments under blog posts.
VIPFeature	This class is the class that stores the premium features available to VIP users in the system. It includes the feature ID, feature name, and description. This class is used to represent the exclusive functions that can be accessed by VIP users.

4.2 Sequence Diagrams with Description

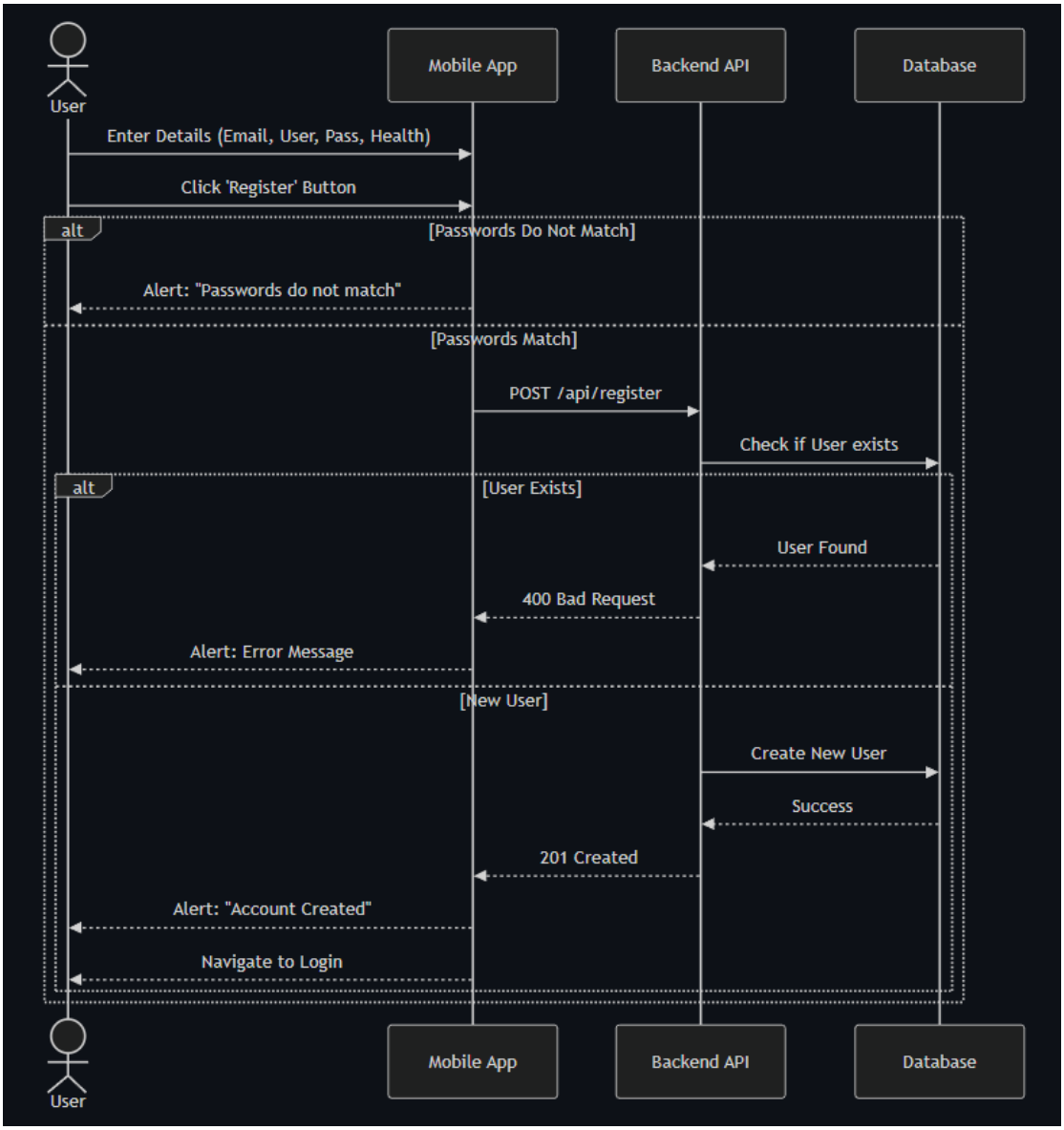
1. Login Workflow



Description:

The user enters their email and password and the system first validates if fields are empty. If valid, it sends a request to the server and then the server verifies the user against the database if success, a JWT token is returned and saved locally and navigate the user to the home page of the app.

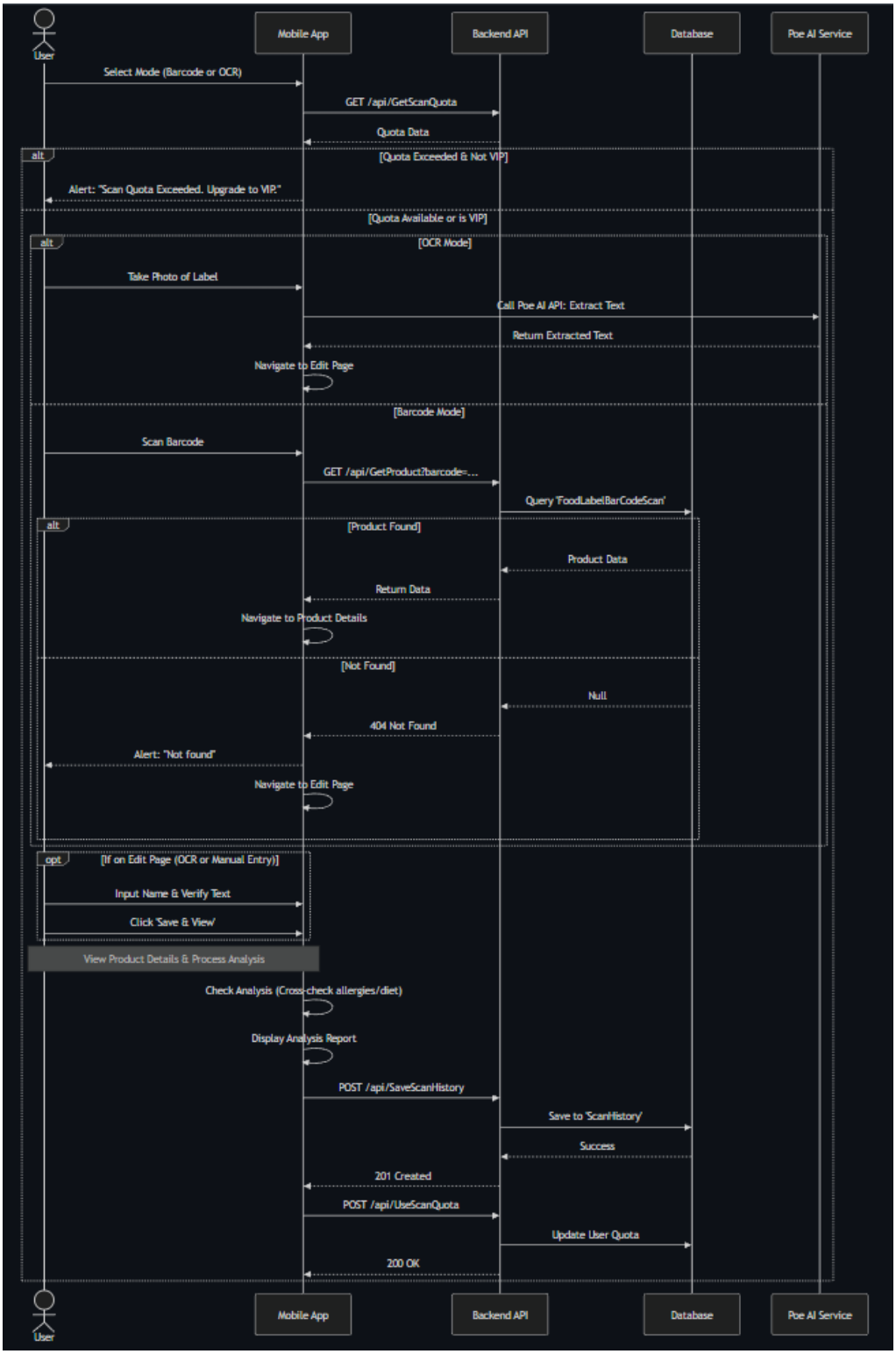
2. Register Workflow



Description:

The user enters registration details for example email, user name , diet etc. then the system will checks if passwords match if match the data will sent to the server and the sever will check for existing users before creating a new account and if all the checking is ok then system will create the new account for the user and alert a “account created” message then navigate the user to the login page of the app.

3. Scan Workflow (Barcode & OCR)

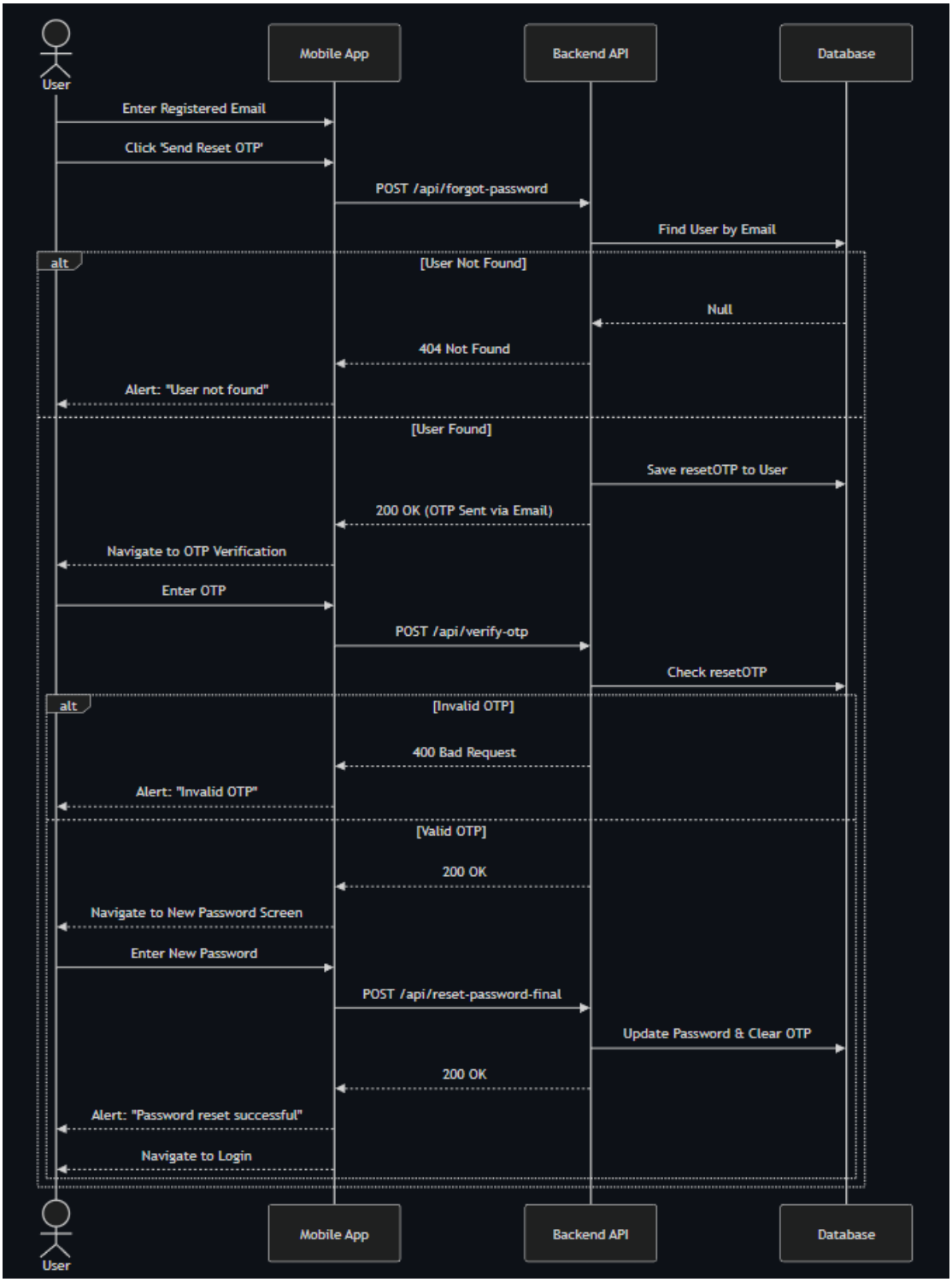


Description:

The system will first check the quota of the user(if the user is not a VIP user), if the user already use up all the quota(daily 5 quota for the normal user) the scan page will be lock and show the alert message “today quota use up, and the quota will renew at 12:00am”, and if the quota is not use up or it is a VIP

user it will let the user to select mode of scan, and there are 2 mode one is barcode scan mode and the other is text mode, and in text mode, the user is needed to take a photo of the food product label or upload a food product label photo then the system will call the AI API connect with it and send the request and photo to the AI for OCR process then when the result return the user will be navigate to the edit page to see the OCR result, the user can input the name of the food product, scan the barcode of the product and also edit the OCR result if there are any problem of it, then when all done user can click the save button to the next step and being navigate to the product detail page meanwhile the system will connect to the AI API again and send the user data, request and also the OCR result to the AI for analysis report,
And when the AI result is return it will show on the product detail page for user to view, after the viewing all the thing in the product detail page user can click the back to history page button to get back to the history page and meanwhile the system will save all the content in product detail page to the database (ScanHistory collection) then update the quota of the user. And on the Barcode Mode, it is only being different on the start, when the user selected the barcode scan mode the system will open the phone camera and let user to scan the barcode for finding the product, and when the barcode is detected the system will decode the barcode and find it on the database once the product is found the system load the food product data to the app and navigate the user to the product detail page and display the data, meanwhile will do the AI analysis I have mention above and the rest of the moves are also same with above.

4. Forgot Password Workflow

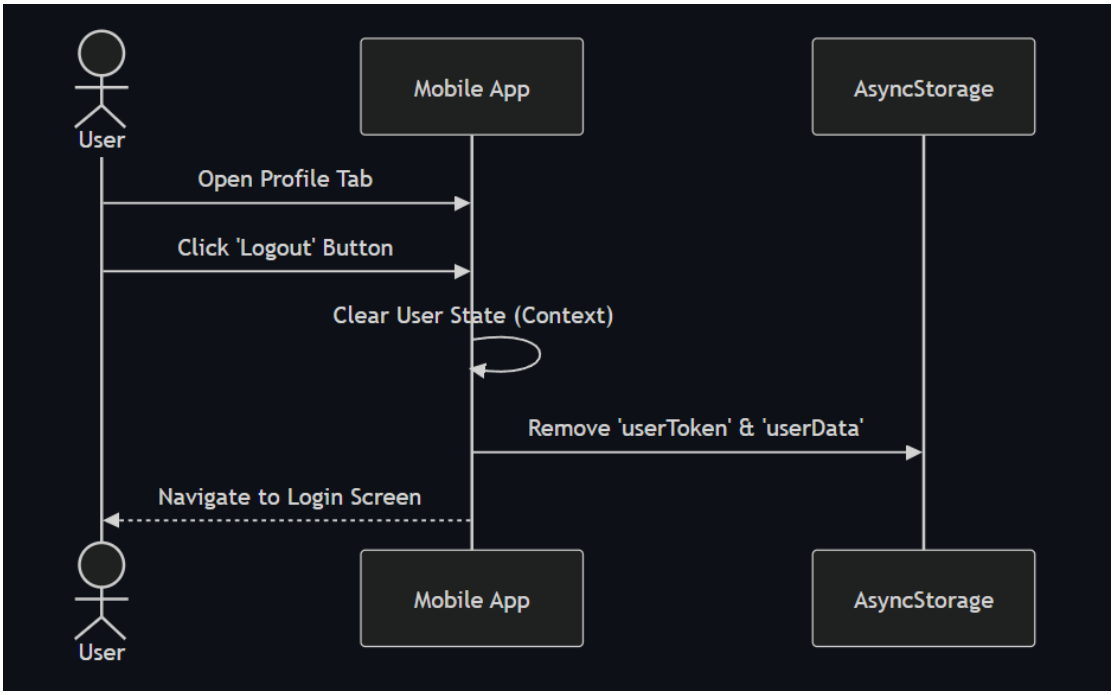


Description:

The user are requested a password reset by providing their email to the system and when the user have inputted the email and click the send OTP the sever will checks if the email exists in the database, if found the system will generate a OTP code and send it to the user email for password reset, and the user need to check the email they has input to find out the OTP code

then back to the app and input the OTP code to the system and the system will verify the OTP code and if the OTP code is valid the system will let the user to enter a new password and submit it and the system will change the password in the database, at last the system will navigate the user to the login page for login.

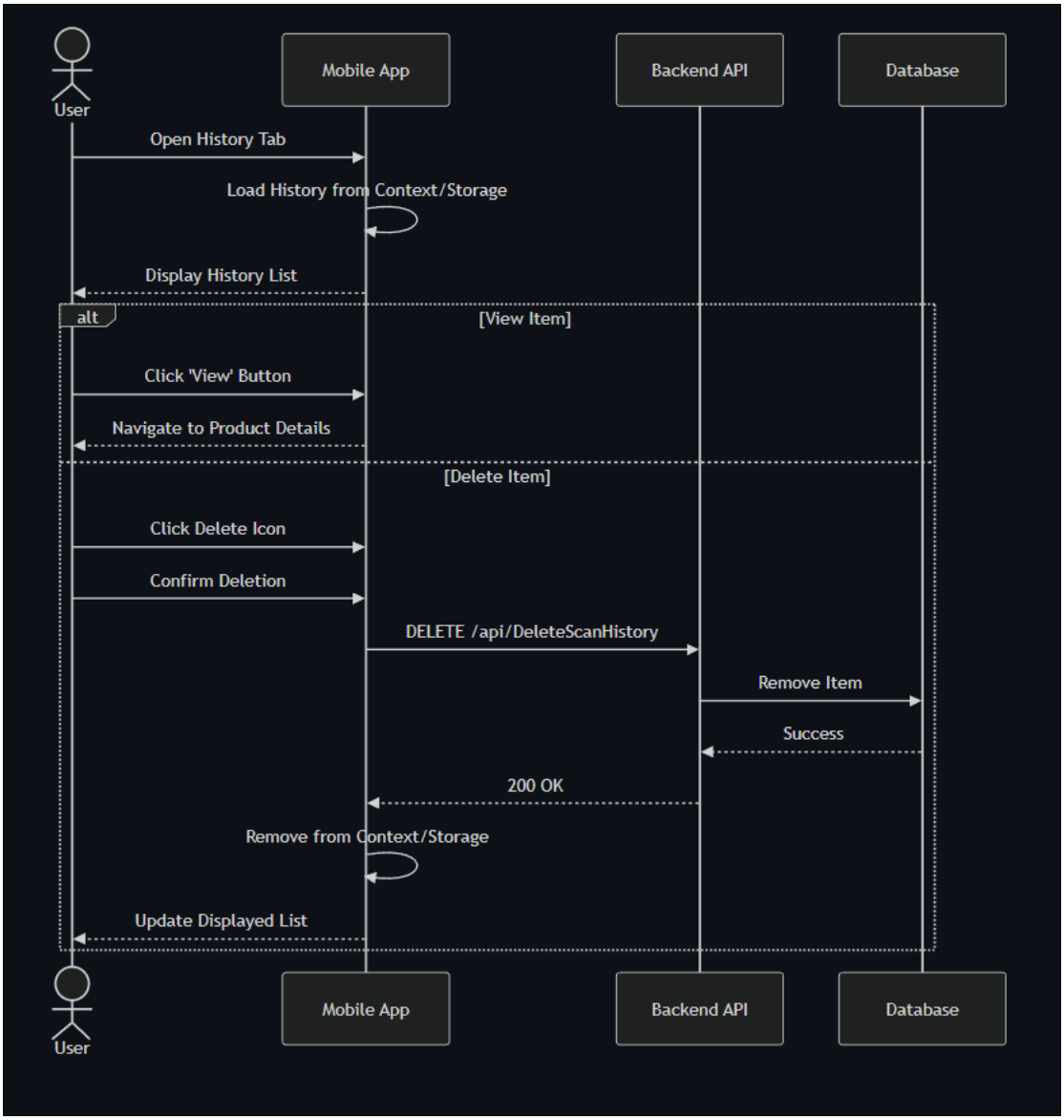
5. Logout Workflow



Description:

The user can logout from the profile tab, and the system will clears the local user state (Context) and removes the persistent authentication token from storage then navigate the user back to the login page.

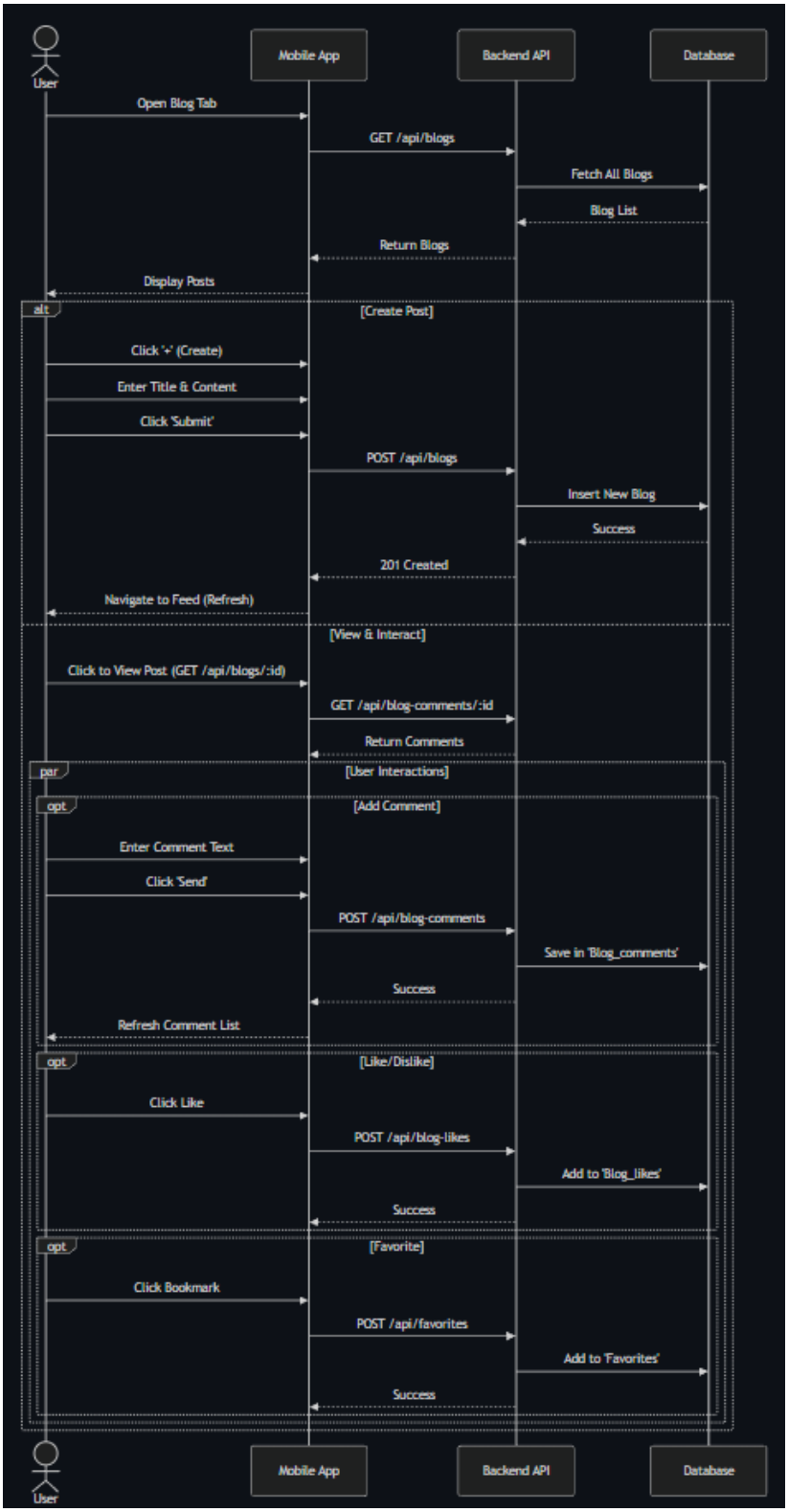
6. History Workflow



Description:

The user can use it to views their past scans, when the user open the history page the system will first loads the history data from database and display all the history by a list, and use can choose to view the history details or delete history, if user want to view the history details they can click the view button and the system will navigate them to the product detail page and load all the corresponding historical scan data from the database for viewing. And if the user want to delete the scan history they can click on to the delete button and the system will remove the history form the database then update the history page, and that all of the history.

7. Blog Workflow

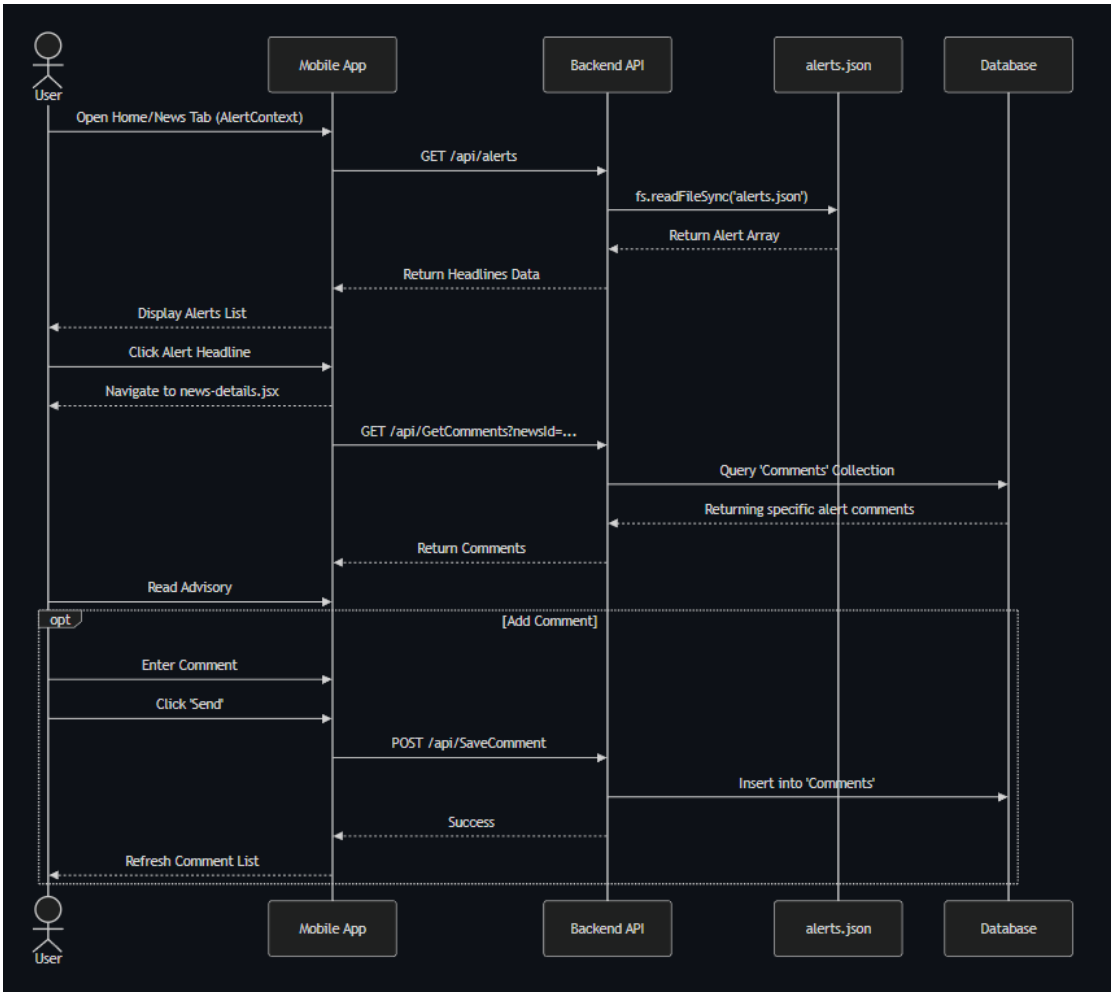


Description:

In the blog the users can see the community blog posts here, when the user go to the blog page the system will load all the blog the has been created and display them as a list for user to select, when user want to view the detail

of the one post they can simply click on that post and the system will navigate the user to the blog detail page and display the post title, content and also the comment that are belong to this blog post. And on this page user can roll down to the comment section and enter some comment on the comment box and send it then the system will save the comment to the database and refresh the comment section, furthermore user can also hit the like or dislike button here too and when the like or dislike button are clicked the system will record it and refresh the page, and lastly on this page user can also add the blog post to the favorite by clicking the favorite button then the system will add the post to the user favorite list. Beside of that user can click the “+” button on the blog feed page to create a new post when the button was clicked the system will navigate the user to the post edit page to input the post title and post content, after input the user can click submit to save it to the database and the system will save it , refresh the blog list and navigate the user to the blog page, and that all of the blog.

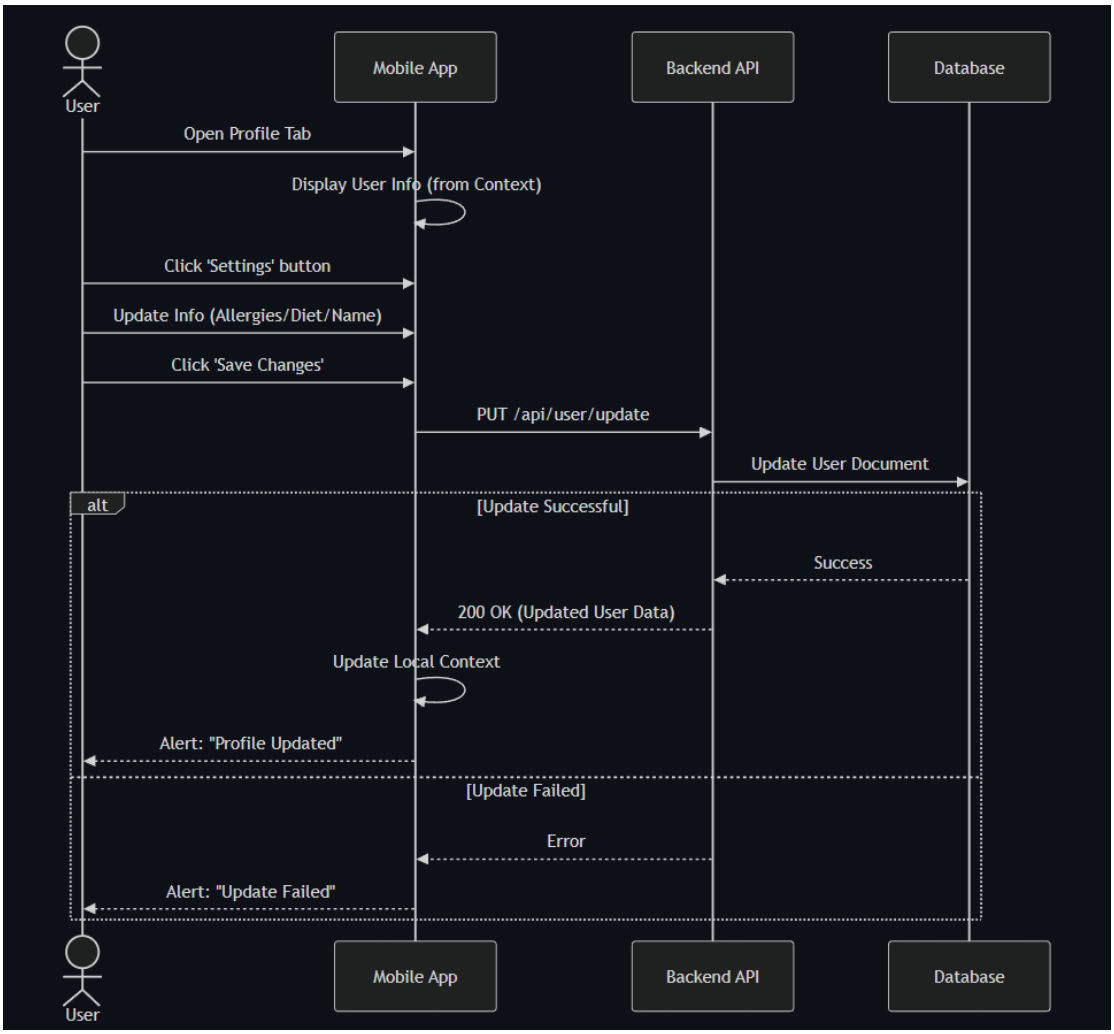
8. News Workflow



Description:

When the user go to the home page the system will fetch 3 latest alert to the alert news section and if the alert are within 7 days there will also having a “New!!” sign next to the alert new title, and when the fetch is done the alert will displayed on the home page new section, and if the user want to view the detail of the alert they can click on to the new they want and system will load the alert content and also the comment that are belong to the alert, beside of reading the alert the user can also enter a comment to the comment section of the alert, and the system will save it to the database and refresh the comment section, and it is all of the alert news part.

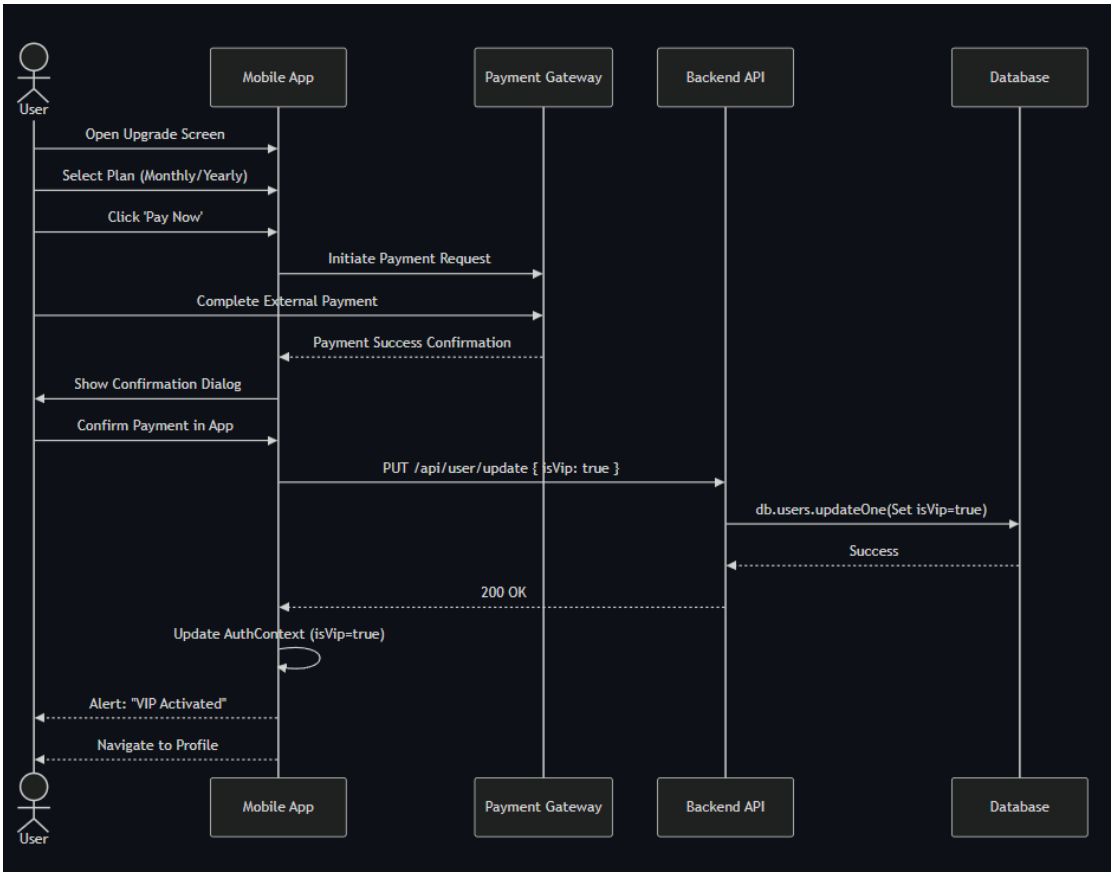
9. Profile Management Workflow



Description:

Users can view the personal information on the profile page and also they can click the setting button to update their personal information in the profile tag by clicking the setting button, and when the user done with the edit and save the info. the info. will send to the sever and refresh the page for update.

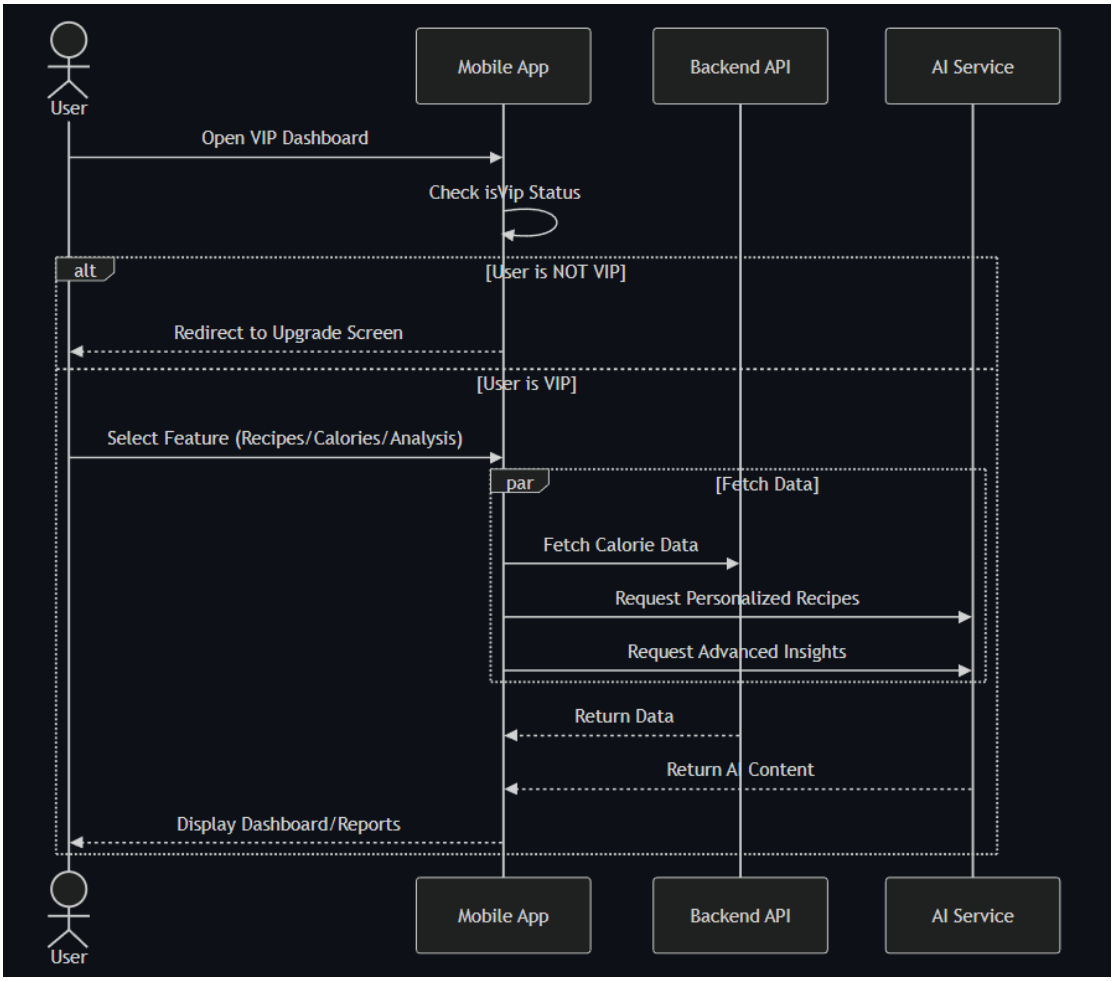
10.VIP Upgrade Workflow



Description:

Users can subscribe to a VIP plan by clicking the upgrade button on the home page or the profile page, and the user can view the upgrade detail for example what benefit will be get when being a VIP user, then if the user want to be the VIP user they can go to the next page to select which subscription plan they want then click pay now to go to the payment page and choose the payment method and pay , once the payment are done the system update the isVip status of that user to true representing it is a VIP user and last navigate the user back to the profile page, and that all for the VIP upgrade.

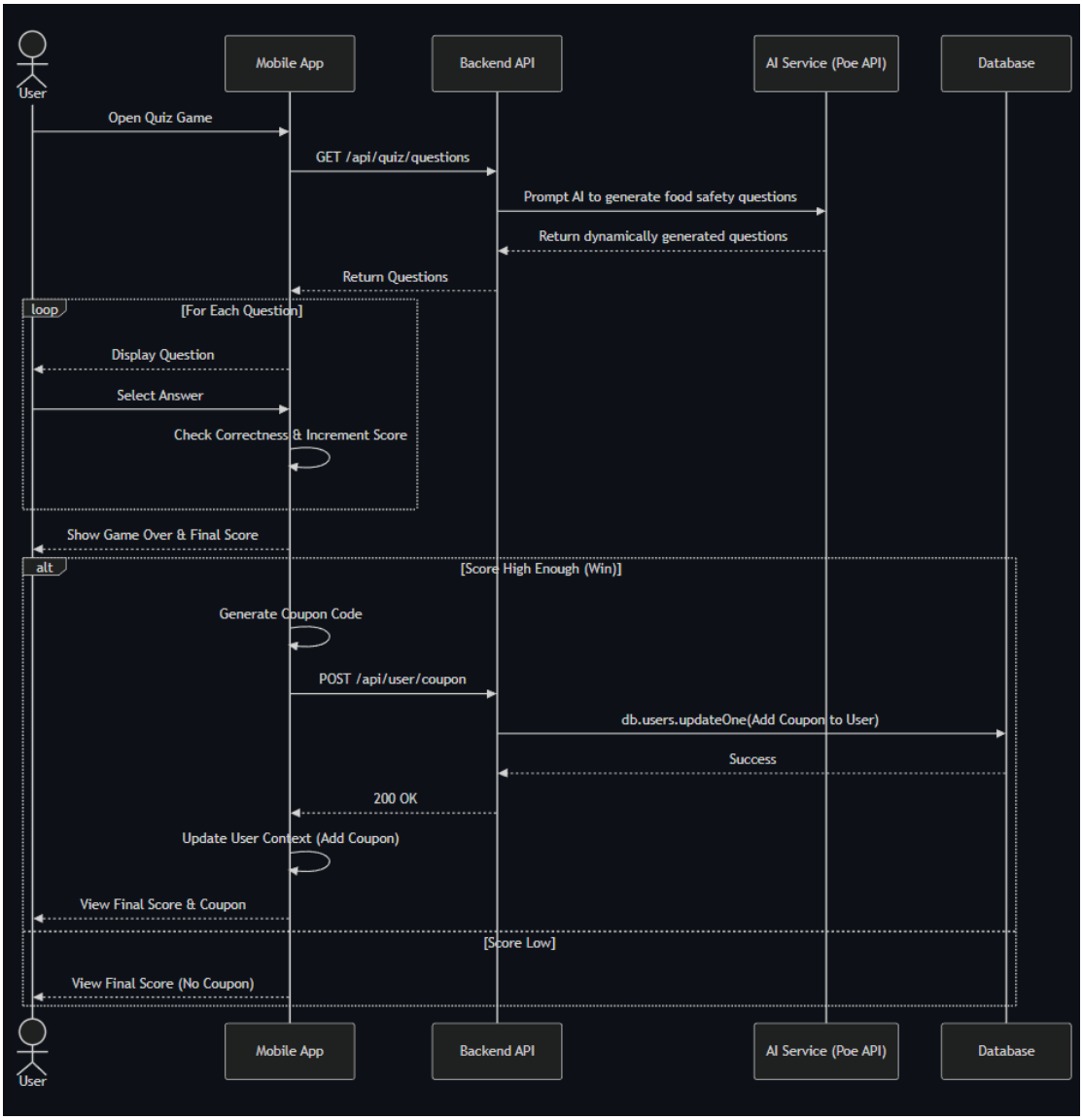
11.VIP Features Workflow



Description:

When the user has upgrade to VIP, they can access exclusive features, the VIP upgrade button will change to VIP dashboard and the user can click on it to view and use the exclusive features but when they click the VIP dashboard the system will check the user VIP status if the pass, the user can go ahead and use it, for calories count the system will fetch the data from database thought backend, and wait for the user to select the scanned food and input how many they have taken for counting the calories. And for the recipes the system will require a photo of the food ingredients, user can either take photo or upload photo and the system will connect to the AI service for the result, and for the deep analysis the system will load the history scan data too and when user selected one of the product it will send the data to the AI service and wait for the result, and if the user are done with the VIP features they can click the home button on the VIP dashboard anytime to get back to the home page, and that all for the VIP features.

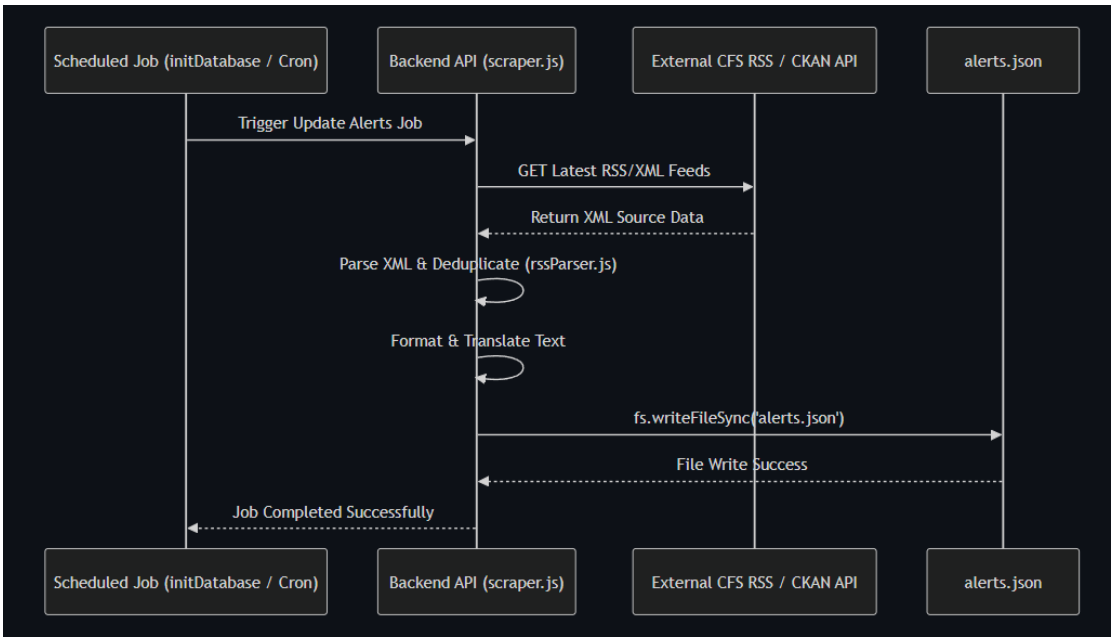
12.Discount Game Workflow



Description:

The user can play the game for coupon in the home page and when the user go in to the game the system will connect to the AI service to get the questions then let the user answer it, in total there are 5 questions, after 5 questions the system will show the game over screen and show the final score and also generate a coupon to the user when the user have get a coupon the system will add a coupon to the database and show it on the profile page, and it is all about the discount game.

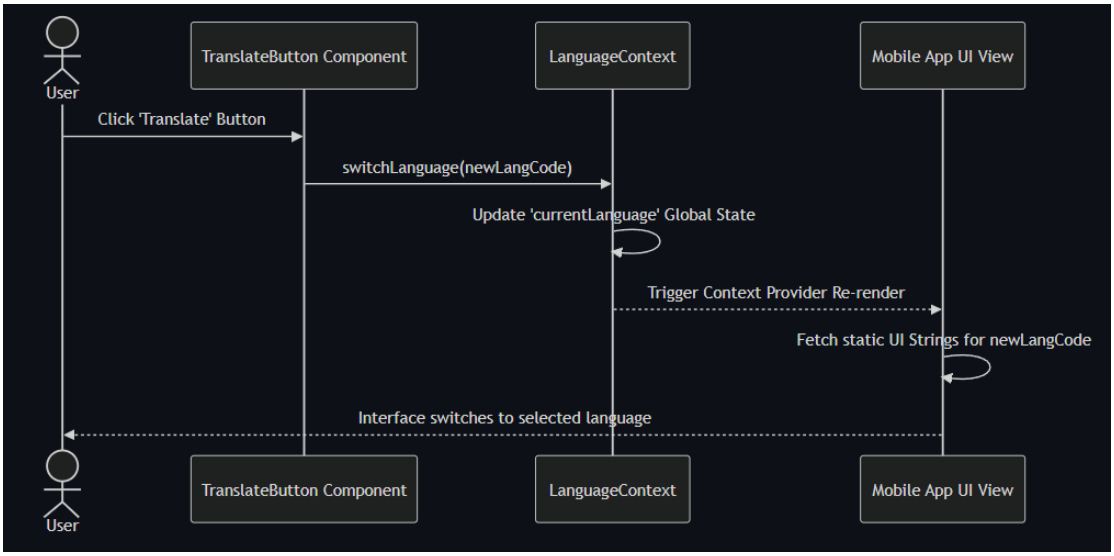
13.Backend Scraper & Alert Update Workflow



Description:

It is show the scheduled automated process where the Node.js backend will fetches the latest food safety alerts from the external government API (data.gov.hk) form everyday 3 am and parses the data locally to the alerts.json to prevent database read overload.

14.Multilingual Toggle Workflow

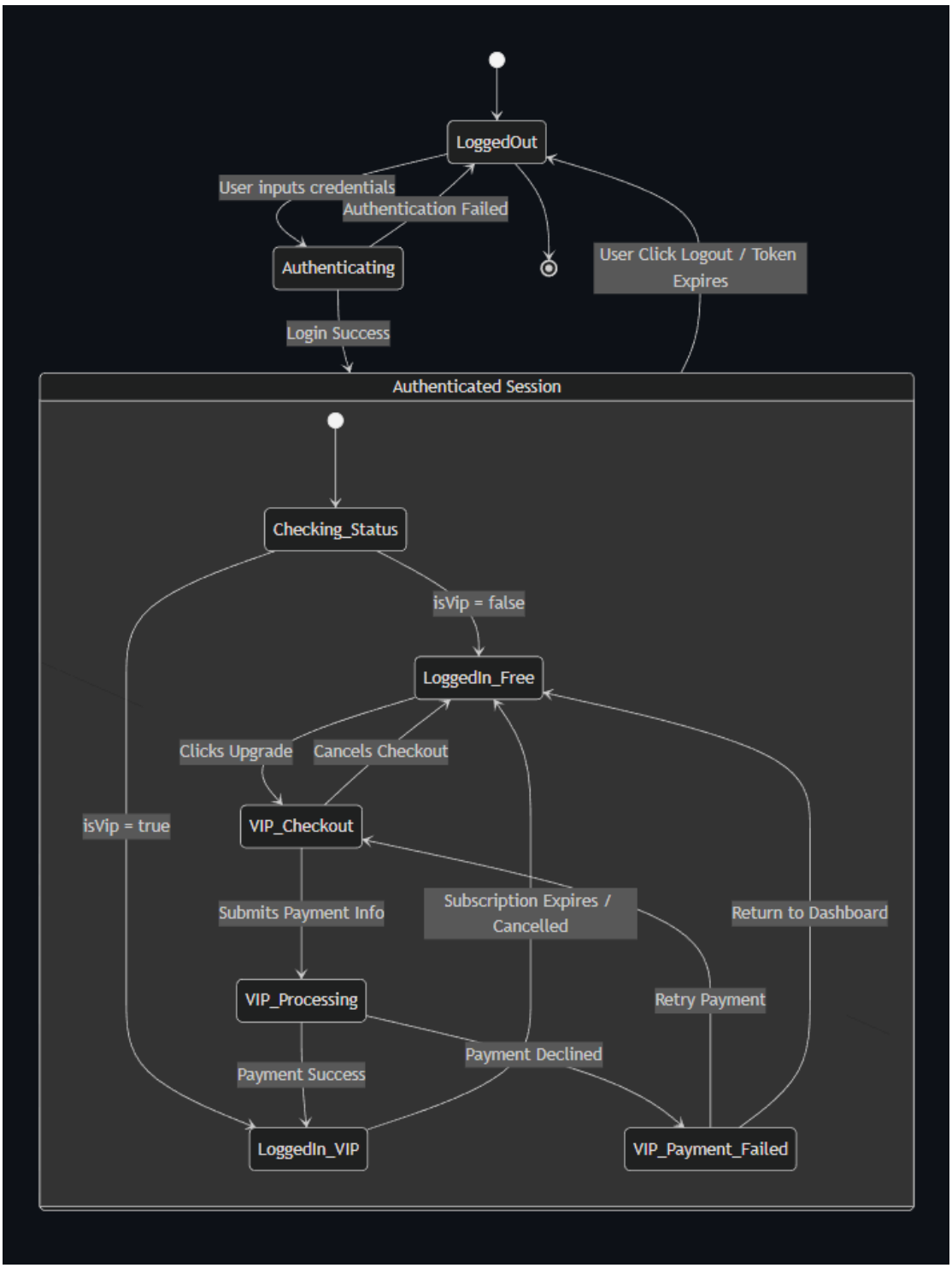


Description:

The user can switch the interface language for example English to Traditional Chinese or the opposite. The application updates the global state via LanguageContext, prompting all dependent react components to logically re-render in the selected language.

4.3 State Charts with Description

1. User Authentication & Session State



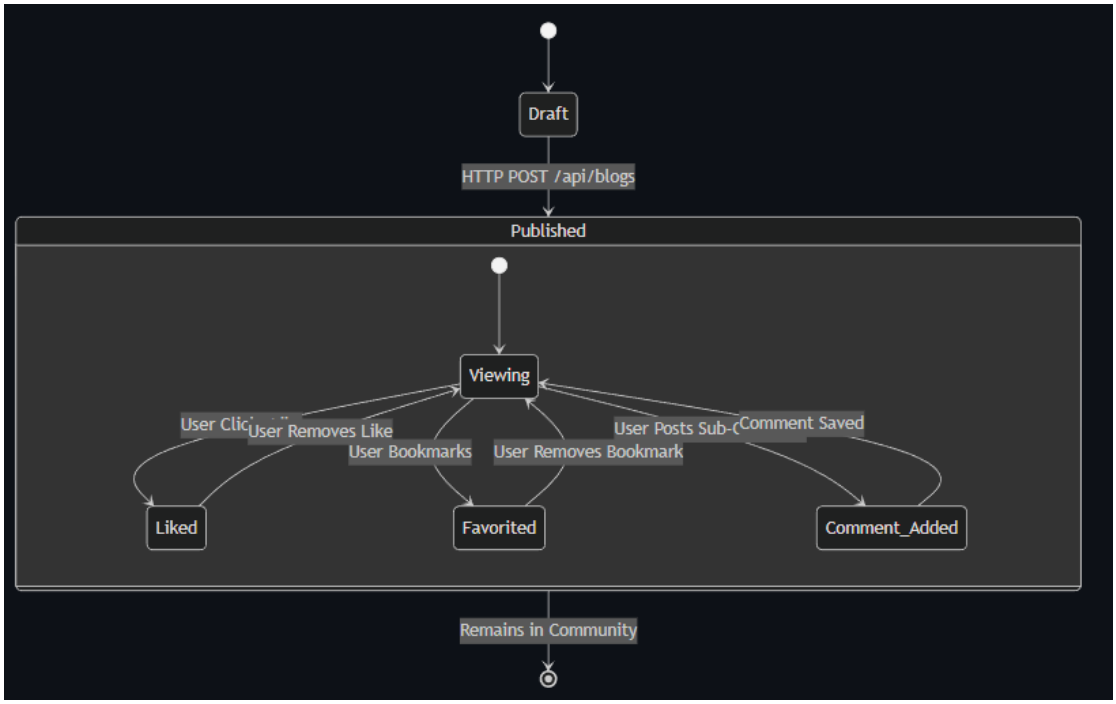
Description:

State change : Logged Out → Normal User → VIP User

In the lifecycle it start form the logged out user and when it inputted the

credentials(email and password) the system will check the user are authentication success or not representing the login success and fail, if success let the user login, if fail show the alert login fail. And also it will check the VIP status of the user to design which UI will the user see(VIP upgrade/VIP dashboard) and , if “isVip = True” the which means it is a VIP user and they will see the VIP dashboard and can using more function(VIP Features), if “isVip = False” the user will see the VIP upgrade to let the user pay for the VIP. At last when the user click the logout button or the token expires the user will be logged out.

2. Blog Post Lifecycle State

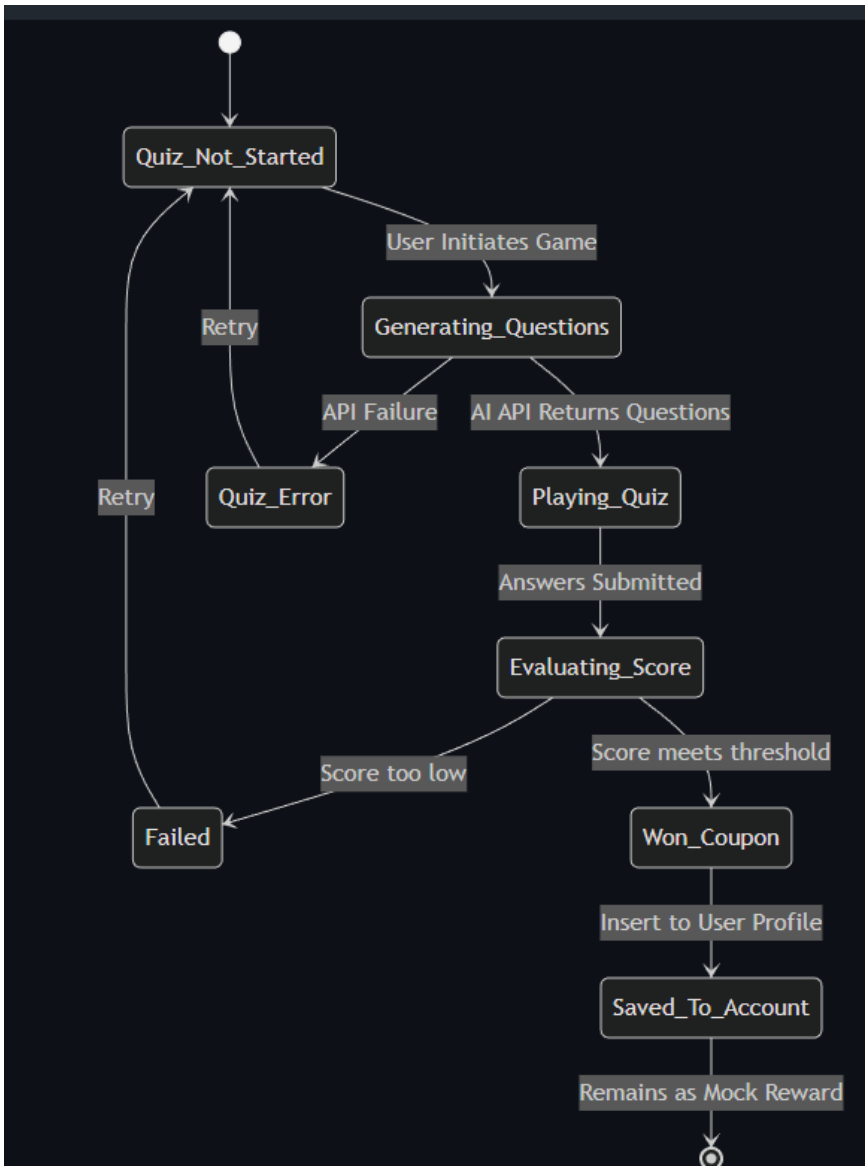


Description:

State change : Draft(blog post create) → Post Published → Interacted(like/dislike and comment)

It start with drafting the blog post and then when the user publish it will go to the published state which can be see by all the user then go to the interacted state when some user are like or dislike the post or comment to the posts.

3. Discount Quiz & Coupon Gamification State

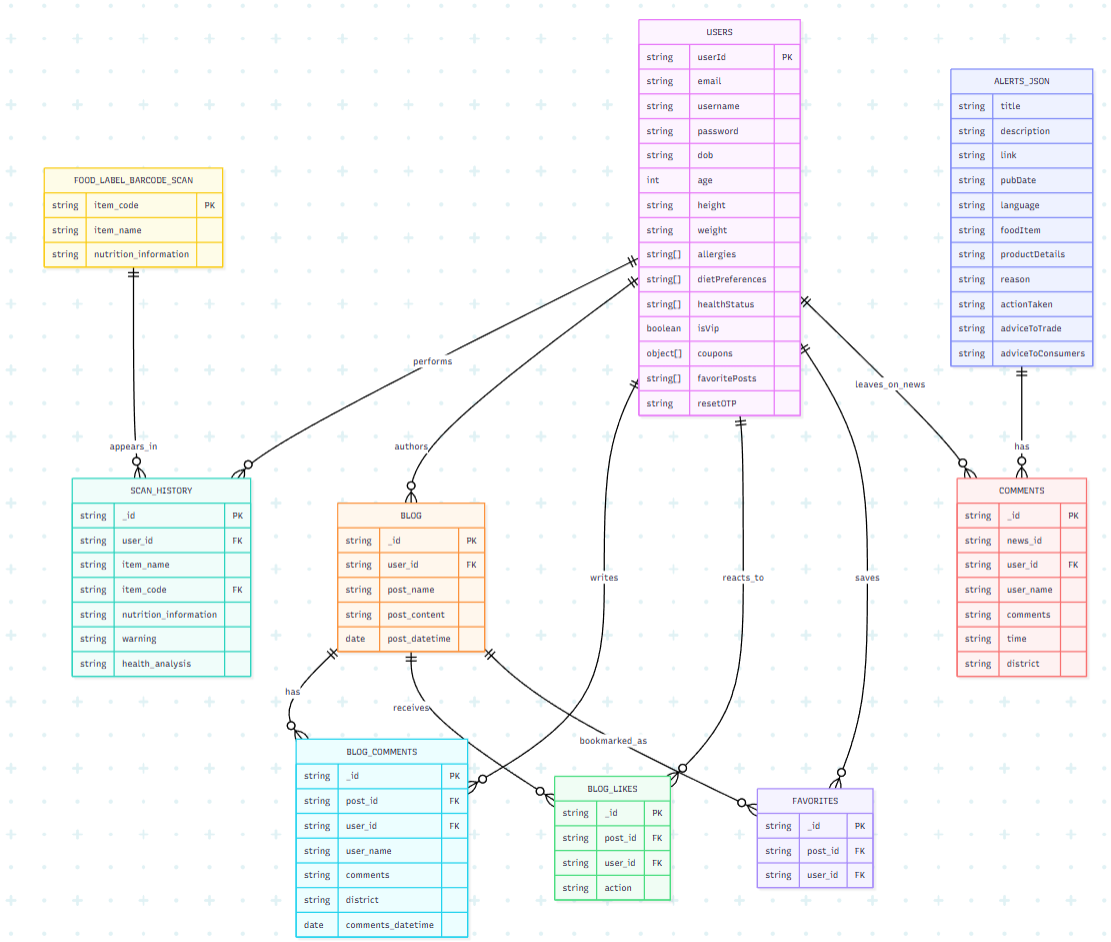


Description:

State change : No Coupon → Gain Coupon(play the game) → Gained The Coupon (win the game and save to the account)

It state form the no coupon in the user account and the user can play the quiz game to gain the coupon and it is the Gain coupon state, then when it is having the score that are meets the threshold it can get the coupon and save it to the user account and it is the gained the coupon state.

4.4 Database Design & Physical Data Model



Description:

The database is designed using MongoDB and a local JSON data store.

Collection	Description
Users	The users collection is mainly storing the account details(user name, email, password etc.), health preferences (allergies/diets/health status), and rewards(coupon). And it is the main collection that interaction with all the collection in app. For activities use this collection handling near all the activities in the system for example the login check, reset password check(email and OTP code check), blog and all the comments(blog comments and alert comment) authors check, likes/dislike action check and also the scan function user data(account details and health preferences) load and check etc.. Overall it is the core collection of the system.
FoodLabelBarCodeScan	The FoodLabelBarCodeScan collection serves as the master product catalog containing barcodes and product data, and this collection is designed to be replace with a real reality product

	database, for easier maintenance and better scalability, because when the scan mode on the app was used the system will first find it in this collection if no found then go to the ScanHistory collection, which means the mechanism in app will search both of the collection so on the scalability if the technician who are using this system can easy to plug down the collection and switch with a new database also if the technician want to change the data in database they can change in a offline database and switch it to the system, no need to change it online which to reduce the error that cause by the human error like technician wrong data input in the database. For activities use when the user using the barcode mode this collection will be use for search and load to return the data.
ScanHistory	The ScanHistory collection is mainly use for storing the user scanning activities history for example the scan results, warnings, and health analysis report, and for activities use this collection is mainly used to support the scan mode(barcode mode) data search and also the history review.
Blog	The Blog collection is mainly use for storing the user ports, and in the collection the all the ports detail like post author, post title, post content etc. will be store in here and it connect with such sub-collection like Blog_comments, Blog_likes collection to form the whole social and interaction features in the system. For activities use when the user are going to the blog page or want to view the blog this collection will be load to the system.
Blog_comments	The Blog_comments collection is a sub-collection to the blog collection, it is mainly storing the comments that user given under the blog posts, other than the comments content it will also store the post time of the comment, comment author, etc., And one main point in here is that the post_id will also be store in here as the key that to let the system to find which post is the comments is belong to. For activities use it is only when the blog the viewing the user it will be use to fetch the corresponding comments.
Blog_likes	The Blog_likes collection is also the sub-collection to the blog collection, it is mainly storing the like/dislike action , also the collection will store the user_id to see who has already given a action to the blog posts avoiding the action collision problem between different user when they using the system, and post_id are used here as the same function to the Blog_comments collection which to being the key to find the corresponding posts. For activities use it is same as the Blog_comments, which when the blog post are being view the system will fetch the like

	or dislike action of that user and display it on the app.
Favorites	The Favorites collection is collection is the sub-collection to the user collection, but it is also interaction with the blog collection and their sub-collection, it is mainly storing post_id and user_id let the user can book mark the posts that they like, by using the user_id to let the system know which favorites list need to be load to the app for user view. For activities use it is only be used when the favorites was clicked in a post or the user want to view the favorites list.
Comments	The Comments collection is mainly use to storing the comments for the food safety alert, the structure of this collection is similar to the Blog_comments collection storing the news_id as the key to let the system can find out the corresponding alert but it was used to the food safety alert parts. For activities use it was used when the alert was view by the user the system will load the corresponding comment to it.
Local Json	Description
alerts.json	The alerts.json is mainly storing the alerts detail that get form the data.gov.hk database and it will be updated every day 3 am in the morning, the data like alert post date, alert content etc. will be store in here and for activities use it is use to the part that handling the alert news, when the user go to the home page or go in to view the detail of the alert, the alerts in the alerts,json will be load to system and after the process of the alert it will be display to the user.

5. Systems Implementation

5.1 Use of Software/Library

5.1.1 Use of Software

- **React Native (react-native):** The core framework used to build the cross-platform mobile app(Android and iOS).
- **Expo (expo):** A framework and platform built around React Native and used for access to device for testing and simplifies the build process.
- **Node.js:** The JavaScript runtime environment used to execute the backend server logic outside of a browser.
- **MongoDB:** The NoSQL database system used to store and manage all application data.

5.1.2 Use of Library

- **Frontend Libraries:**
  1. Expo Router (expo-router): used for react native to handling all the screen navigation.
  2. Expo Camera(expo-camera): used to letting the system can use the camera and crop the photo.
  3. Image Picker (expo-image-picker): used for when the user want to pick the image from the phone and crop the image.
  4. Async Storage (@react-native-async-storage/async-storage): used to allowing the system can locally save the data like such JWT token, language preferences etc.
  5. React Native Render HTML (react-native-render-html): parses and renders the html string content from backend into the proper native mobile views, mainly used on the blog and alert sections.
  6. Expo Vector Icons(@expo/vector-icons):provides a customizable and scalable vector UI icons that used in the bottom navigation bar and the whole interface.
  7. React-native-dotenv: used to hiding the sensitive information like API key and the API endpoint from the source code.
- **Backend Libraries:**
  1. Mongoose (mongoose): An ODM library used to connect to MongoDB

and strictly define the data schemas or structures.

2. Express.js (express): The core backend web framework used to establish RESTful API architecture and handle HTTP routing
3. JSON Web Token (jsonwebtoken): Used for generating and verifying secure JWT to maintain and validate active user login sessions.
4. Bcrypt.js (bcryptjs): A cryptographic library used to securely hash and salt user passwords before storing them in the database
5. Nodemailer (nodemailer): An email automation module used to construct and send Verification OTP code during user registration.
6. Node Cron (node-cron): A task scheduling library used to execute automated background jobs, like the periodic scraper that updates Food Safety Alerts.
7. Day.js (dayjs):
8. XML2JS (xml2js): A parser that converts external XML data formats like government RSS agency feeds into readable JSON formats for the frontend.
9. Axios (axios): A Promise-based HTTP client used heavily to fetch external data for example used for calling the Poe AI API and grabbing government open data in the system.
10.

5.2 System Constraints

- **Hardware Constraints**
    1. Camera Quality: The user mobile device must possess a functioning rear-facing camera and for the accurate of the OCR result the camera are at least 8 Megapixels.
    2. Storage Space: the user mobile device are requested to having at least 50MB to 100MB local storage space for the AsyncStorage to storing the JWT, language preference etc..
    3. Internet Connectivity: The user mobile device are requested to having the capability to connect to the network, which is to having the WIFI functionality, cause the application are mainly relies on the cloud-based service(AI API) to run, otherwise the application will fail.

- **Software / OS Constraint**
  - Operating System Version:
    - Android Devises: For Android devises the system is requested to run on the Android 6.0 (API level 23) or higher vision also it need to be support to the React Native (Expo) runtime environment.
    - iOS Devises: For IOS the devise is requested to run on iOS 13.0 or higher.
  - Usable email: The user email which used to registration must be an active email address, to able to receive a email with OTP validation codes when resetting the password.
- **System Usage & Interaction Constraints**
  - Daily Action Quotas: To prevent server overload and API cost abuse, there will be some limitations on the scan action as shown as follows:
    - Normal user: 5 scan limit per day, and renew on everyday 1200 am.
    - VIP user: No limit
  - Token Expiration: When the user JWT is expiration, for security reasons the system will enforced to logout the user from the system and the user must log back in for continue to use the system.
- **User Permission Constraints**
  - Device Permissions: when the user first use the system there are some device permissions are requested for initial launch:
    - CAMERA_PERMISSION: Required for scanning the physical barcodes or snap ingredient labels.
    - MEDIA_LIBRARY_PERMISSION: Required to upload existing image from the device gallery.
  - Data Access Boundaries: Users can only interact with their own locally bound scanning history and personal favorites via their specific userId, the action like by using the same device to globally synchronized the data is not allow.
- **Language Constraints**

  Chinese/English: the system now has just supporting two languages, Chinese and English, if the user are no knowing any of this two language there are not be able to use it.

- **Environmental Constraints**

  Optimal Lighting: The scan accuracy is highly related on the OCR result, if the user are scanning the label in a over or not enough light environment, the OCR result accuracy will be low.

5.3 Test Plans

Unit Test

Test No.	Description	Expected Result	Actual Result
TC-UT001	Test if the system can correctly convert a complex international timestamp into a simple, readable date layout.	Returning the formatted date string.	Returning "2026-03-28". (21.11ms)
TC-UT002	Test the time-checking feature: Verify that today date is correctly recognized as falling within the last 7 days.	Returning True.	Returning True. (0.56ms)
TC-UT003	Test the time-checking feature: Verify that an old date from 10 days ago is correctly identified as being older than 7 days.	Returning False.	Returning False. (0.27ms)
TC-UT004	Test if the system correctly generates today calendar date as a solid 8-digit number code (used for backend fetching).	Returning an 8-digit text code.	Returning Length 8 String. (0.18ms)
TC-UT005	Test processing a valid food safety alert notification to ensure the system successfully breaks down and extracts the details.	Properly extracting 1 data block.	Error safely caught. (4.48ms)
TC-UT006	Test the system stability when the government alert provider sends an completely empty data folder to ensuring it skips rather than crashes.	Returning an empty list.	Returning Array[0]. (1.11ms)
TC-UT007	Test requesting the government food safety database using the proper Traditional Chinese language requirement.	Receiving the official download webpage link.	Returning URL String. (204.33ms)
TC-UT008	Test requesting the government food safety database using an unsupported language requirement to see its error handling.	Being rejected gracefully (Null).	Returning Null. (52.38ms)
TC-UT009	Test the web detail scraper program when no target website link is provided, to ensure it does not break the background service.	Safely returning a blank block.	Returning Empty Object {}. (0.12ms)
TC-UT010	Test the AI text-cleaning tool: Check if it successfully deletes useless chatter (like “Thinking...”) from the AI food scan response.	Returning pure and clean food text.	Returning Clean Text. (0.01ms)

TC-UT011	Test the standard login flow attempting to log in with a fully registered email address and correct password.	Login approved (Returning True).	Returning True. (241.05ms)
TC-UT012	Test the system unauthorized login protection attempting to log in with a fake account that does not exist in the system.	Login rejected (Returning False).	Returning False. (198.11ms)
TC-UT013	Test the barcode scanning feature when a user scans a product that has not been registered in the database yet.	Finding no record and returning Null.	Returning Null. (302.40ms)

Functional Test

Functional Test Items:

- **User Authentication & Accounts**
  - **User Registration**
  - **User Login / Logout**
  - **Forgot Password**
- **Scanning & Recognition**
  - **Scan Ingredients by Camera / Image**
  - **Search Product by Barcode**
  - **Scan Quota Deduction System**
- **Analysis & Results Display**
  - **View Basic Ingredient Breakdown**
  - **VIP Deep Analysis**
  - **Personalized Allergen Warning Trigger**
- **VIP Subscription**
  - **Trigger VIP Upgrade Prompt**
  - **Process Payment & Unlock Features**
- **Community & Blog**
  - **Create New Blog Post**
  - **Browse Blog Feed**

    - **Add / Remove Favorites**
- **News & Alerts**
    - **Fetch & Display External Food Alerts**
    - **View Alert Details**
- **User Engagement & Personalization**
    - **Play Nutrition Quiz Game**
    - **Calculate & Display Quiz Score**
    - **Switch System Language**
    - **View Scan History**
    - **Edit Profile / Settings**

User Authentication & Accounts

1.1 User Registration

Items	Descriptions
FT ID	TC-FT001
Objective	To test whether the system can correctly register a new user account.
Input	Valid email, username, and password.
Expected Output	1. If email already exists, an error message will show asking the user to login. 2. If inputs are valid, the system creates the account. 3. After account creation, the user will be automatically navigated to the Home page. 4. A success message is shown to acknowledge the registration is completed.
Pass / Fail Criteria	1. If the system recognizes an existing email and shows an error, the test passed. 2. If the system registers valid input and navigates to Home, the test passed. 3. If valid input fails to navigate to Home, the test failed.

1.2 User Login / Logout

Items	Descriptions
FT ID	TC-FT002
Objective	To test whether a user can log in with correct credentials and wipe session data upon logout.

Input	Registered email and password.
Expected Output	1. If wrong password or email is entered, an error warning will show. 2. If inputs are correct, the user is navigated to the main dashboard. 3. When the user clicks "Logout", the session is cleared, and they return to the Login page.
Pass / Fail Criteria	1. If the system denies wrong credentials with a prompt, the test passed. 2. If the system accepts correct credentials and logs in, the test passed. 3. If logout clears cache and returns to the login screen softly, the test passed. 4. If the user can still bypass back to the dashboard after logging out, the test failed.

1.3 Forgot Password

Items	Descriptions
FT ID	TC-FT003
Objective	To test whether the system can trigger a password reset flow via email.
Input	Registered email address.
Expected Output	1. If the email is not in the database, an "Email not found" error is shown. 2. If the email is valid, the system dispatches a reset link/code. 3. A message prompts asking the user to check their email inbox.
Pass / Fail Criteria	1. If the system successfully dispatches a reset email and shows a hint, the test passed. 2. If the system fails to identify an unrecorded structured email address, the test failed.

Scanning & Recognition

2.1 Scan Ingredients by Camera / Image

Items	Descriptions
FT ID	TC-FT004
Objective	To test whether the system can correctly extract text from an image and identify food additives.
Input	Image containing clear ingredient text.
Expected Output	1. If the system (AI) determines the image is not clear enough, an error message "Unable to read" will show. 2. If readable, the system extracts the text and uses regex to clean redundant AI formatting tags.

	3. The system automatically matches extracted ingredients against the internal database. 4. After scanning completes, the user is navigated to the "Product Details" page.
Pass / Fail Criteria	1. If the system extracts text and highlights hazardous additives properly, the test passed. 2. If an unreadable image causes a safe UI rejection without crashing, the test passed. 3. If the system extracts text but fails to auto-navigate, the test failed.

2.2 Search Product by Barcode

Items	Descriptions
FT ID	TC-FT005
Objective	To test whether the system can retrieve product data via barcode scanner queries.
Input	Barcode retrieved from scanner (e.g., "4891028164395").
Expected Output	1. If the barcode doesn't exist, a "Product Not Found" message shows, prompting manual entry. 2. If recognized, product data is fetched from the server/DB. 3. The user is navigated to a Product Details page populated with accurate properties.
Pass / Fail Criteria	1. If the system retrieves and shows product details from a valid barcode, the test passed. 2. If an unrecorded barcode suggests adding manually successfully, the test passed. 3. If the system clicks the scan and has absolutely zero response, the test failed.

2.3 Scan Quota Deduction System

Items	Descriptions
FT ID	TC-FT006
Objective	To test if free-tier users have their daily scan quota deducted accurately without loopholes
Input	Trigger scan actions repetitively.
Expected Output	1. Quota allowance number decreases by 1 after every successful request. 2. When the quota reaches 0, the system blocks scanning operations and renders an "Empty Quota" alert.
Pass / Fail Criteria	1. If the scan allowance deducts perfectly sequentially, the test passed. 2. If the system locks down functionality successfully upon reaching a flat zero, the test passed.

	3. If users can bypass the limit somehow after zero counts, the test failed.

Analysis & Results Display

3.1 View Basic Ingredient Breakdown

Items	Descriptions
FT ID	TC-FT007
Objective	To test whether the product detail page renders parsed hazard safety ratings via color codes.
Input	Redirect landing sequence into the populated Details Layout.
Expected Output	1. List of ingredient outputs is exhibited broadly centered. 2. Additive text boxes must be color-coded distinguishing risks (i.e. Red = Hazard, Green = Safe).
Pass / Fail Criteria	1. If the UI presents the list perfectly adapting color tones per additive risk tag, the test passed. 2. If the UI loads plain text completely failing to inject warning colors, the test failed.

3.2 VIP Deep Analysis

Items	Descriptions
FT ID	TC-FT008
Objective	To test whether VIP-enabled users can unfold the high-tier NOVA classifications interface.
Input	Pressing the "Deep Analysis" view option (isVip = True).
Expected Output	1. Server authorizes parameter clearance without objection. 2. Canvas expands pulling further detailed graphs regarding E-Number properties and NOVA limits.
Pass / Fail Criteria	1. If an authorized user is awarded access showcasing correct charts cleanly, the test passed. 2. If standard payloads error out blocking an actual premium account, the test failed.

3.3 Personalized Allergen Warning Trigger

Items	Descriptions
FT ID	TC-FT009
Objective	To evaluate whether cross-allergic interactions are pinpointed properly cross-referencing user profile configurations.
Input	Scan containing "Casein" structure while User profile lists "Milk" restrictions.
Expected Output	1. Internal algorithm overlaps Casein as an intersection to

	dairy properties. 2. Prompts a highly intrusive red critical warning denying safe consumption instantly post-scan.
Pass / Fail Criteria	1. If matching implicit dairy allergies results in forced UI warnings effectively tracked, the test passed. 2. If identical wording is checked only missing cross-allergic bonds entirely, the test failed.

VIP Subscription

4.1 Trigger VIP Upgrade Prompt

Items	Descriptions
FT ID	TC-FT010
Objective	To verify constrained functionalities protect content against unauthorized free-tier accounts safely.
Input	Tap "Deep Analysis" lock (isVip = False).
Expected Output	1. Controller refuses execution logic detecting false context param. 2. Halts routing and launches an overlay pop-up suggesting VIP subscription merits.
Pass / Fail Criteria	1. If bounded attempts securely trap progression raising correct payment dialogues, the test passed. 2. If free users slip beneath blocks accessing protected graphs, the test failed.

4.2 Process Payment & Unlock Features

Items	Descriptions
FT ID	TC-FT011
Objective	To assure front-line elements drop bounds instantaneously reacting towards gateway payment success strings.
Input	Type fictional digits processing "Purchase" in Checkout.
Expected Output	1. Virtual receipt pings DB writing True towards user's VIP Boolean structure. 2. The client notices changes wiping global padlocks entirely allowing unbarred motion.
Pass / Fail Criteria	1. Providing successful mocks forces UI constraint drops gracefully changing structure, the test passed. 2. Mock payment claims success yet interface lock mechanisms linger un-updated, the test failed.

Community & Blog

5.1 Create New Blog Post

Items	Descriptions
FT ID	TC-FT012

Objective	To test whether a valid payload composed from inputs gets inserted and propagated to front panels.
Input	Constructing text in Body & Title segments then clicking Upload.
Expected Output	1. System validates denying utterly hollow content forms directly. 2. Permitted queries bridge over towards database merging seamlessly sliding towards primary board view. 3. Content rises to the highest absolute order pushing existing articles downwards.
Pass / Fail Criteria	1. Record written smoothly rendering in immediate live view arrays reliably, the test passed. 2. Payload posts yet necessitates software re-launch finding its listing position, the test failed.

5.2 Browse Blog Feed

Items	Descriptions
FT ID	TC-FT013
Objective	To ensure query fetching pulls multiple external nodes scrolling unhinderedly matching time arrays.
Input	Vertically swipe down screen center spanning the community zone.
Expected Output	1. Visual loader indicates request process reaching APIs pulling newer iterations. 2. Dropdowns populate dynamically appending further segments without lagging out page physics.
Pass / Fail Criteria	1. Array parsing works correctly loading infinite sequence seamlessly downwards visually, the test passed.

5.3 Add / Remove Favorites

Items	Descriptions
FT ID	TC-FT014
Objective	To assess component toggling parameters sync accurately alongside secondary listing collections.
Input	Hit the 'Heart / Favorite' icon tagged to any arbitrary published feed.
Expected Output	1. The prior vacant outline instantly floods solid red coloring. 2. Backed record creates a linked instance showing vividly inside individual isolated 'My Favorite' views.
Pass / Fail	1. Interlocking action stores identical records across split

Criteria	screens while permitting identical removal, the test passed.

News & Alerts

6.1 Fetch & Display External Food Alerts

Items	Descriptions
FT ID	TC-FT015
Objective	To gauge XML extraction routines pulling contemporary Government reports successfully.
Input	Clicking "Alert & News" icon via bottom main slider tab.
Expected Output	1. Microservice engages Gov APIs deciphering RSS blocks translating them contextually. 2. Displays neatly organized row-items dating exclusively over the seven prior continuous days. 3. Antiquated materials filter aside maintaining relevancy priorities.
Pass / Fail Criteria	1. Timeline limits enforce exactly fetching functional reports rendering successfully, the test passed. 2. Prolonged requests trigger complete white-screening failures without safety loading indicators, the test failed.

6.2 View Alert Details

Items	Descriptions
FT ID	TC-FT016
Objective	To scrutinize sub-detail crawler programs extracting exact contexts without collapsing beneath invalid links.
Input	Tap arbitrary generated row instances proceeding deeper inside.
Expected Output	1. Crawler investigates targeted distinct Web HTML lifting exact photos alongside text. 2. Supposing the base domain crashes providing HTTP 404, layout suppresses the error generating a gentle visual void lacking crashes.
Pass / Fail Criteria	1. Complex text bodies unearth securely portraying context flawlessly, the test passed. 2. Dead/unresponsive URL targets circumvent fatal crashes rendering stable placeholders independently, the test passed.

User Engagement & Personalization

7.1 Play Nutrition Quiz Game

Items	Descriptions
FT ID	TC-FT017
Objective	To gauge direct-touch element responsiveness analyzing instant boolean calculations.
Input	Pick multi-choice answer A,B,C,D manually upon gaming UI.
Expected Output	1. Positive Boolean yields instant bright green surface shading. 2. Erroneous selections turn red subsequently flashing the parallel correct line-item green inherently.
Pass / Fail Criteria	1. Surface layer modifies coloring parameters precisely without uncalibrated delays following presses, the test passed.

7.2 Calculate & Display Quiz Score

Items	Descriptions
FT ID	TC-FT018
Objective	To decide mathematical accumulative validity scaling towards finale endpoints reliably.
Input	Run fully through all queried levels pushing terminal end phases.
Expected Output	1. Layout transitions generating the final scoring animation popup correctly. 2. Values precisely translate right-answer totals logically matched mirroring input tracks exactly.
Pass / Fail Criteria	1. Scoring mechanisms produce bulletproof tallies omitting glitch drops entirely smoothly, the test passed.

7.3 Switch System Language

Items	Descriptions
FT ID	TC-FT019
Objective	To observe state rehydration altering comprehensive interface dictionary trees globally concurrently.
Input	Interact altering standard 'Language/Localization' dropdown within App parameters.
Expected Output	1. Main contextual state triggers refreshing internal library values. 2. Instantaneous language metamorphosis spans entire nested fragments abstaining mandatory hard re-boots.
Pass / Fail Criteria	1. Absolute interface overhaul executes natively switching syntax dictionaries actively mid-session fluently, the test

	passed.

7.4 View Scan History

Items	Descriptions
FT ID	TC-FT020
Objective	To assert historical tracking matrices append records pulling chronologically reverse indices cleanly.
Input	Jump explicitly into isolated "History Directory" screen sector.
Expected Output	1. Routine pulls arrays mapping previously queried products linked natively inside. 2. Formatting forces structural presentation starting from immediate current descending towards longest past instances systematically.
Pass / Fail Criteria	1. Output precisely arranges prior requests retaining descending order logic appropriately spaced, the test passed.

7.5 Edit Profile / Settings

Items	Descriptions
FT ID	TC-FT021
Objective	To review write/read loop closures linking modifying menus and homepage avatar boards cohesively.
Input	Rewrite basic text form ('Username') pushing submission logic.
Expected Output	1. Value merges solidly atop older DB parameters establishing dominance. 2. Returning standard flow reflects exact user alterations pasted actively replacing past designations completely.
Pass / Fail Criteria	1. Live modifications map directly echoing revisions instantly following successful server commit cycles conclusively, the test passed.

The black box test detail in A1.

5.4 Test Strategies

In the test there will be using two types of tests method, the white box test and also the black box test for the unit test and the function test.

For the unit test will be using the white box test on it, to test is the system are stable and dependable. And white box test is well fit on the unit test because it can make sure there are no logic problem on the code, by testing every little unit of the code to make sure the app system stable and dependable.

For the function test will be using the black test on it, to test is the function work well, and the black box test is well fit on it, because it can by viewing the output to see is there are any wrong output and error on the function of the app.

6. Discussions, Contributions and Conclusion

6.1 Discussions and Conclusion

First of all, I want to thank you all the people that help me with the project, including my professor Prof. ZHANG, Eric Lu, the instructor Mr. CHENG, Kenny Kam Kong and all my school mate and the my friend and family , all of them are given a huge help on the system analysis, Design, implementation and my life when I am doing the project.

Next, I want to talk about my achievement, I am doing the nearly 100% of my plan to the project, on the core function I have successfully using the hottest technology now in the world – AI, to help solving the problem that with the computer vision, by using it I my system can clearly and accurately read out the food additives and nutritional information and analysis it to a report, achieving one-stop service resolved the problem of fragmented functionality in other applications and greatly help on the people that who need some kind of this application, letting them can more easily to know more about what food additives or nutritional they have taken, and is it good, bad or harmful to them, with that the user can get rid of to suffering by select healthy and suitable food product when they were shopping.

Furthermore, I have also developing bunch of side function on my system, on one side those side function make the system function more abounding, leading the system are more attractive to the user, and on the other side the side function make the system are highlights the close alignment between the system and the theme, and greatly enhances the commercial value of my application, for example the quiz game, it is a interface that the system can co-op with the shop or supermarket, and for the blog and alert news if on the future with good operation on it, the system can being a one-stop platform for food safety information exchange and dissemination, and on that time advertising revenue and social value are immeasurable. Even less need to talk about the revenue that the VIP mode brings.

Although there are so many achievement on my system but still there are some limitation I want to talk about which we can developing it better on the future, first on the database, now the system are using a user growth model database plus provided food info. database, which need user to scan more product to let the database being more huge and it led to the problem that it may take some time to growth the database on the user growth model database and for the provided food info. database because this kind of database normally created by the company itself and not open to the public so on this stage there some limitation for the system to use a comprehensive database to support the user needs, but I have also making a solution on it, which I have decided that part to a pluggable database, on the future if the company who using the system want to change the database they can easily plug in a new comprehensive database.

And the next limitation is the, payment and coupon, now on the system the payment ways are still on mock mode, because of the personal are hard to apply for the payment method for the company, but it is not a big problems on the future when some company want to use this system, and for the coupon like what I have mention above the coupon is used to when the system are co-op with the shop outside and it was using for getting discount to shopping by using the coupon, but on this stage it is also the mock coupon since I and not co-op with any shop.

Lastly, is the admit, for now the system having develop a admin system to handle all the function on the system, the company need to manage the system in code or in the database, and it greatly increase the management complexity of the system, but it is cause by the develop time, by developing the system and handle my school work it has no time for me to done it, so it is also one of the future development to the system.

In conclusion, although there are still some limitation area on my system, but it is not some big problems and can be fix on the future, and I have developed a one-stop platform that uses advanced technology to serve a healthy lifestyle, bringing real convenience to users.

6.2 Contributions

Contributor	Section	Consultor/Reference
WONG HO FAI	**System analysis**	**Prof. ZHANG, Eric Lu**
	System design	**Prof. ZHANG, Eric Lu**
	System testing	**Mr. CHENG, Kenny Kam Kong**

6.3 Future Development

- Mock payment system and coupon go online
    - using a real company to apply for the payment and use on the payment system.
    - Co-op with some shop to get some re coupon for discount.
- Admin UI development
    - Develop a admin user interface for managing the system.
- Comprehensive database plug in
    - using a real company or develop a comprehensive database plug to the system.
- More system language add
    - Now the system are only supported Chinese and English
    - Adding more system language to the system(e.g. Japanese , Spanish)

APPENDICES

A1 Black-box Test Cases

1. User Authentication & Accounts

Test code	Input	Expected Result	Actual Result	Result
TC-FT001	Enter valid email, username, and password on the Registration page.	Account should be successfully created and navigate to Home.	Account secured and automatically navigated to home page.	Pass
TC-FT002	Enter registered email and corresponding password on the Login page.	Authentication passes logging out removes sessions.	Validated accurately login and logout swap works smoothly.	Pass
TC-FT003	Enter the registered email on the Forgot Password page.	System should dispatch a password reset code/link.	Reset request dispatched, UI confirms successful sent.	Pass

2. Scanning & Recognition

Test code	Input	Expected Result	Actual Result	Result
TC-FT004	Capture an ingredients list photo natively and select "Scan".	Text correctly parsed via OCR additives identified.	Noise skipped properly, detailed ingredient list generated.	Pass
TC-FT005	Call the system's barcode reader to scan a product code.	System correctly fetches matching product from DB/API.	Matched corresponding product metadata to UI successfully.	Pass
TC-FT006	Trigger multiple mock scans for a non-VIP account.	Quota deducts incrementally and prevents over-usage.	Free limits applied correctly, locking scanning when zeroed.	Pass

3. Analysis & Results Display

Test code	Input	Expected Result	Actual Result	Result
TC-FT007	Render the scanning result details layout.	Should show parsed additive list with color-coded safety tiers.	Additive risks safely parsed out and highlighted red correctly.	Pass
TC-FT008	Provide VIP-user input to access the "Deep Analysis" view.	App provides advanced NOVA standards and exact breakdowns.	Sub-chart rendered perfectly showcasing precise NOVA stats.	Pass
TC-FT009	Set profile allergy to "Milk", scan product containing "Casein".	App alerts user of hidden/derived allergen danger.	Deep checker detected hidden link and pushed alert popup.	Pass

4. VIP Subscription

Test code	Input	Expected Result	Actual Result	Result
TC-FT010	Trigger a VIP-restricted button from a Free tier login.	Blocks navigation and prompts VIP promo upgrade screen.	System intercepted restricted tab & opened the VIP overlay.	Pass
TC-FT011	Proceed via the mocked gateway clicking "Purchase".	UI drops locks, changing account structure to VIP.	Fictional payment ran, instantly unlocking all UI views.	Pass

5. Community & Blog

Test code	Input	Expected Result	Actual Result	Result
TC-FT012	In 'Create Blog', provide Title & Body and hit Publish.	Post saves to server and immediately lists atop	Saved to database, updated public	Pass

		community.	community list stream.	
TC-FT013	Pull to refresh the community blog feed overview.	Fetches external/other users' generated contents.	Rendered public list correctly enabling seamless scroll.	Pass
TC-FT014	Toggle the "Heart" favorite icon adjacent to a post.	Fills heart icon indicating item belongs in Saved Favorites.	Icon toggled, item cloned successfully into User Favorites.	Pass

6. News & Alerts

Test code	Input	Expected Result	Actual Result	Result
TC-FT015	Shift tab selection to main external Alert/News board.	Should parse chronological warnings from local Govt DB.	Rendered latest alerts safely, stripping malformed fields.	Pass
TC-FT016	Choose specific warning payload accessing its Details page.	Recalls complete textual description without app crashing.	Detailed extraction executed gracefully, 404s bypassed cleanly.	Pass

7. User Engagement & Personalization

Test code	Input	Expected Result	Actual Result	Result
TC-FT017	Play Quiz minigame, tapping an arbitrary multiple-choice response.	Color states dynamically adapt (Green=Win, Red=Lose).	Interacted correctly delivering an immediate visual answer state.	Pass
TC-FT018	Exhaust all game questions achieving the summary dashboard.	Accurately calculates total correct numbers mapping user	Final scoreboard populated cleanly scoring	Pass

		output.	accurate math logic.	
TC-FT019	Flip language module preferences internally within Settings.	Complete DOM/routing tags translate uniformly.	Active session rebuilt components showing target language translation.	Pass
TC-FT020	Access dedicated historical scanning record tracker tab.	Poses chronologically reversed order arrays matching logs.	Output perfectly sorted retaining items queried globally.	Pass
TC-FT021	Update display username string via the Profile modifier.	New naming parameter pushes cleanly mapping back to dashboard.	Profile written correctly immediately echoing the fresh text.	Pass

A2 Executive Summary

- **Overview**

  This project presents an AI-powered mobile application designed to bridge the gap between complex food labeling and consumer health. By integrating Computer Vision and Natural Language Processing, the system provides an automated way to analyze food additives and nutritional facts, empowering users to make informed dietary choices.

- **Problems**

  Many consumers suffering to understand food ingredient lists due to technical jargon, small print, and a lack of transparency regarding chemical additives. Existing solutions often lack personalization or require manual data entry, which is time-consuming and prone to error.

- **Technical Solution**

  The system adopts a Client-Server architecture using React Native (Expo) for the mobile frontend and Node.js/Express.js for the backend, supported by a MongoDB database.

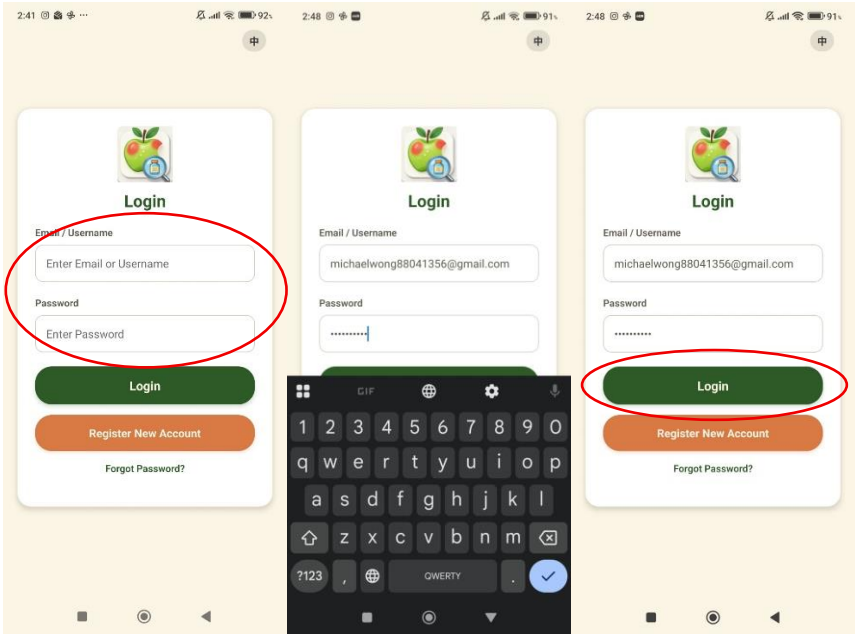
  - **Core Innovation:** Leveraging the Gemini-1.5-Pro API, the app performs high-accuracy OCR to extract text from food labels and provides a "one-stop" analysis report. And it scrapes real-time data from data.gov.hk to provide up-to-date food safety alerts.

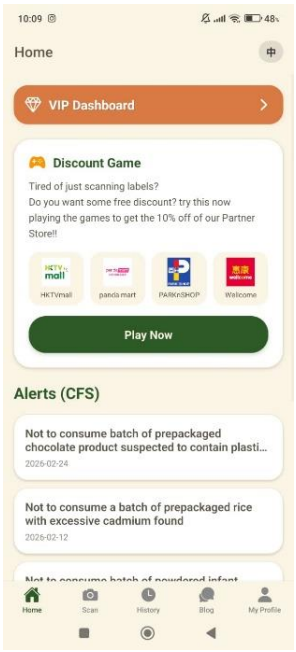
- **Key Features & Commercial Viability**
  - The system's commercial viability is rooted in its one-stop functional ecosystem, which transforms a technical tool into a high-value consumer service. By integrating Gemini-1.5-Pro AI for high-accuracy OCR label analysis, the application provides personalized health alerts tailored to individual allergies and dietary goals, creating strong user retention. Its business model is diversified through a VIP subscription tier for advanced nutritional tracking and a gamified Quiz Mode designed for O2O partnerships with supermarkets to drive foot traffic via digital coupons. Furthermore, the pluggable database architecture ensures long-term scalability, allowing the system to seamlessly integrate with professional third-party data providers, while a projected 31.9% ROI underscores its financial sustainability and market readiness.
  - **Conclusion**

  With a projected return on investment of 31.9% and a strong focus on social value, this application serves as a practical, scalable, and technologically advanced solution for modern food safety management.

A3 User's Manual

Login:

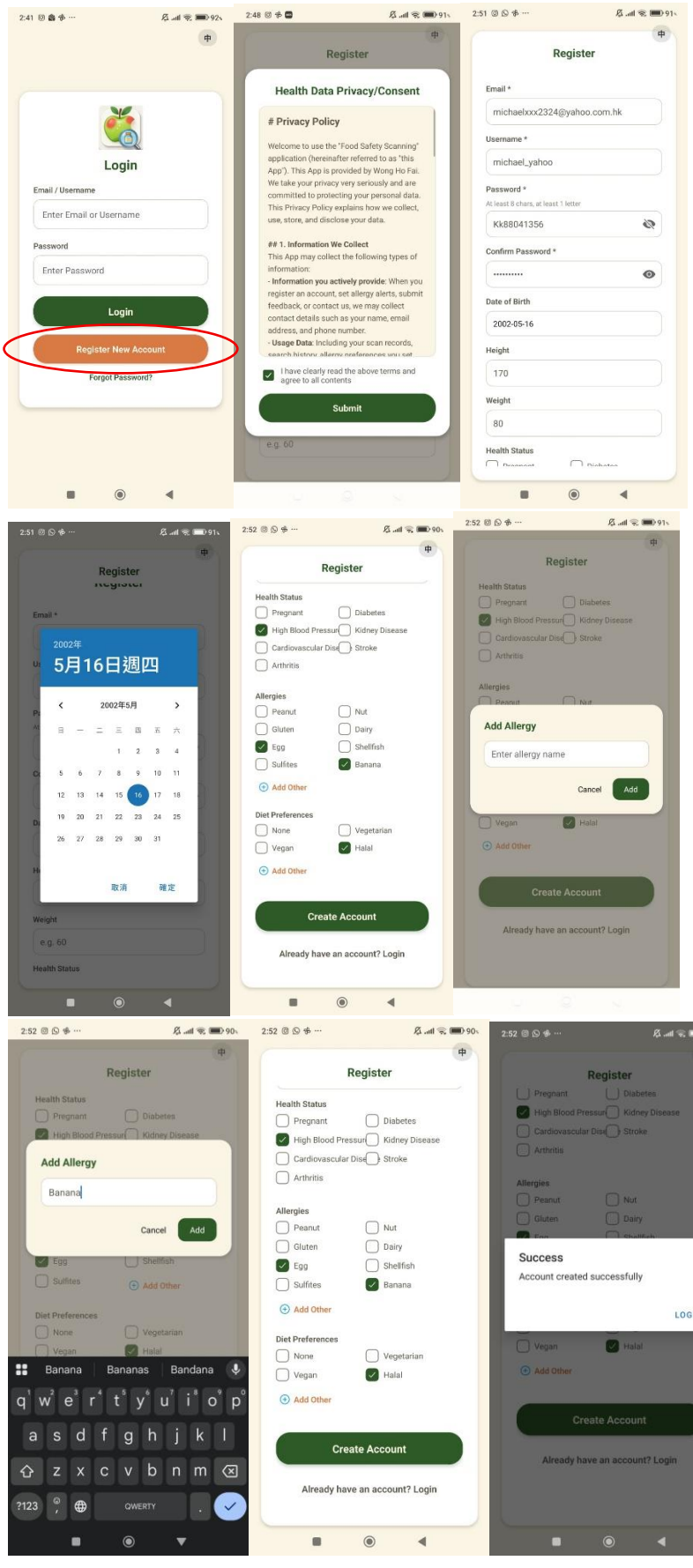




1. Type the email and password in the box

2. click login button

Register:


Login
Email / Username
Enter Email or Username
Password
Enter Password
Login
Register New Account
Forgot Password?
Register
Health Data Privacy/Consent
Privacy Policy
I have clearly read the above terms and agree to all contents
Submit
Email *
michaelxxx2324@yahoo.com.hk
Username *
michael_yahoo
Password *
Kk88041356
Confirm Password *
Date of Birth
2002-05-16
Height
170
Weight
80
Health Status
2002年
5月16日週四
2002年5月
取消
確定
Health Status
Pregnant
Diabetes
High Blood Pressur
Kidney Disease
Cardiovascular Dise
Stroke
Arthritis
Allergies
Peanut
Nut
Gluten
Dairy
Egg
Shellfish
Sulfites
Banana
Add Other
Diet Preferences
None
Vegetarian
Vegan
Halal
Create Account
Already have an account? Login
Add Allergy
Enter allergy name
Cancel
Add
Banana
Success
Account created successfully
LOGIN

1. The user need to click the “register new account”

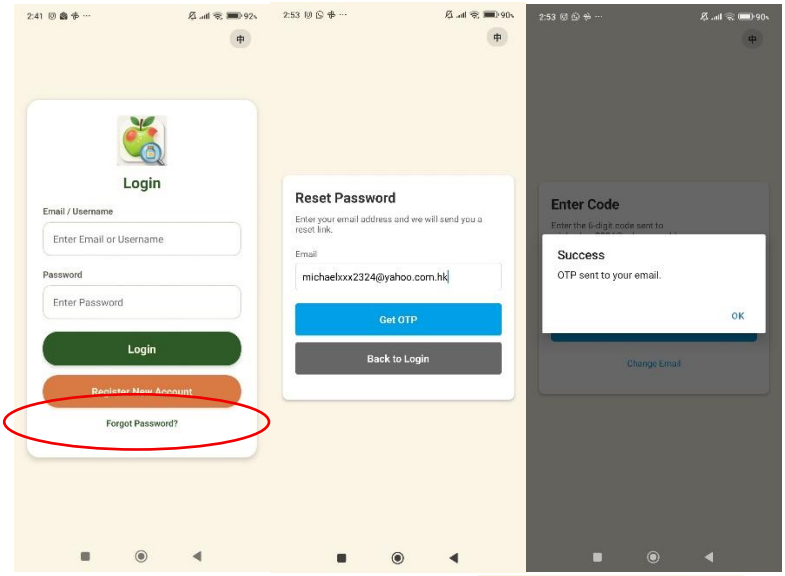
2. Read and submit the agreement

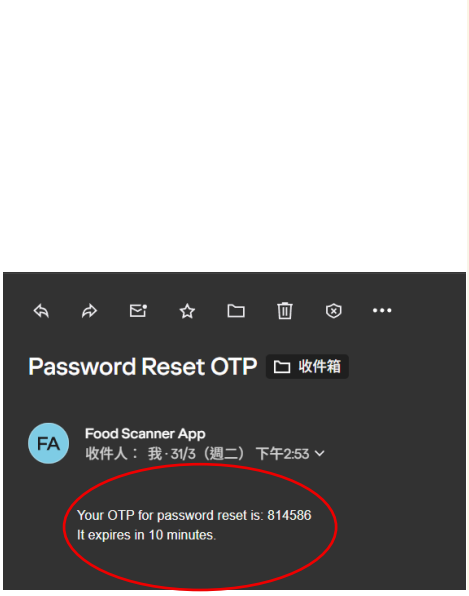
3. Fill up with the info.(name, email, pick the date of birth etc.)

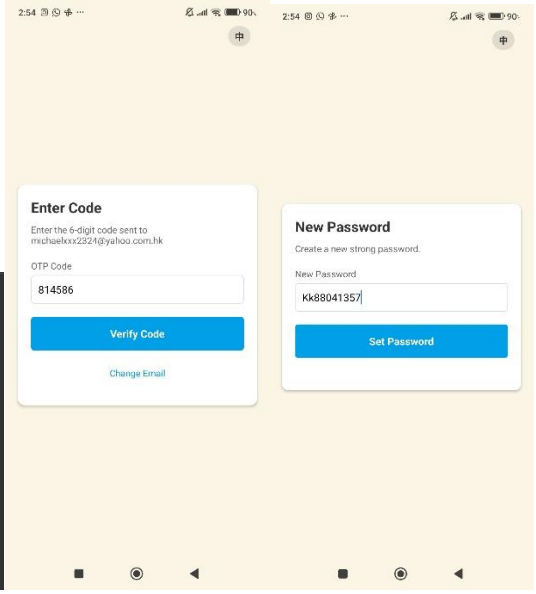
4. Select the health status, allergies and diet preferences(on this user can also manual entry some option that not set default in the app)

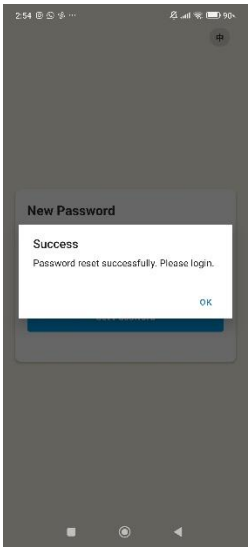
5. click create account to submit the data and click login back to the login page.

Forgot password:









1. click the “forgot password” button.

2. entry the email address and click send.

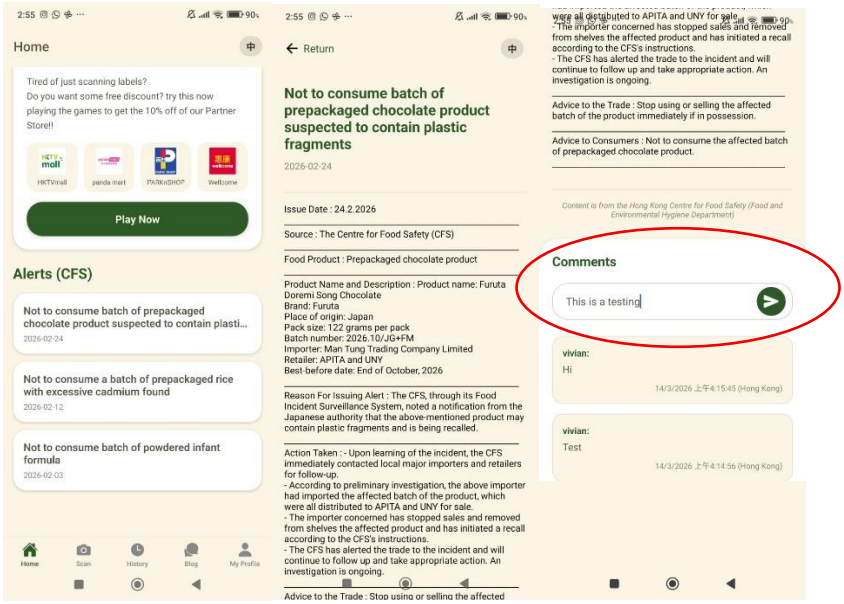
3. wait for the OTP code.

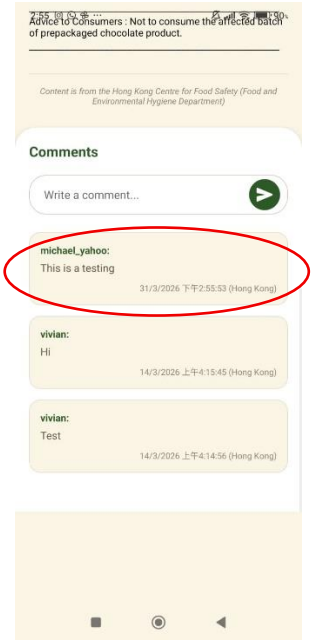
4. back the app and entry the OTP code and click “verity code”.

5. entry the new password and click "set password”.

6. click ok to back to the login page.

Alert news view and comment:





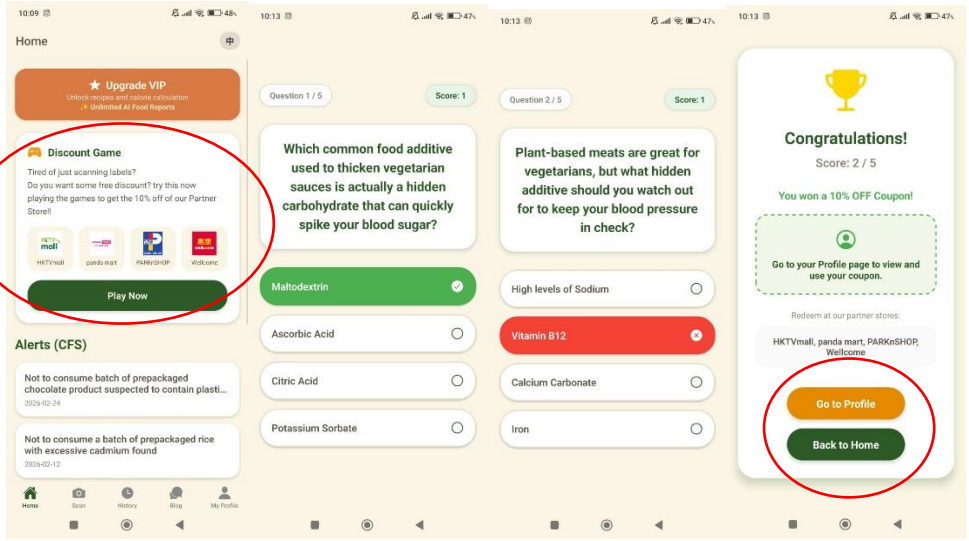
1. click the alert that you want to view.

2. roll down to the comment section to comment.

3. click on the input box and type the comment.

4. click the send arrow.

Quiz game:



1. click on the “play now” on the discount game section.

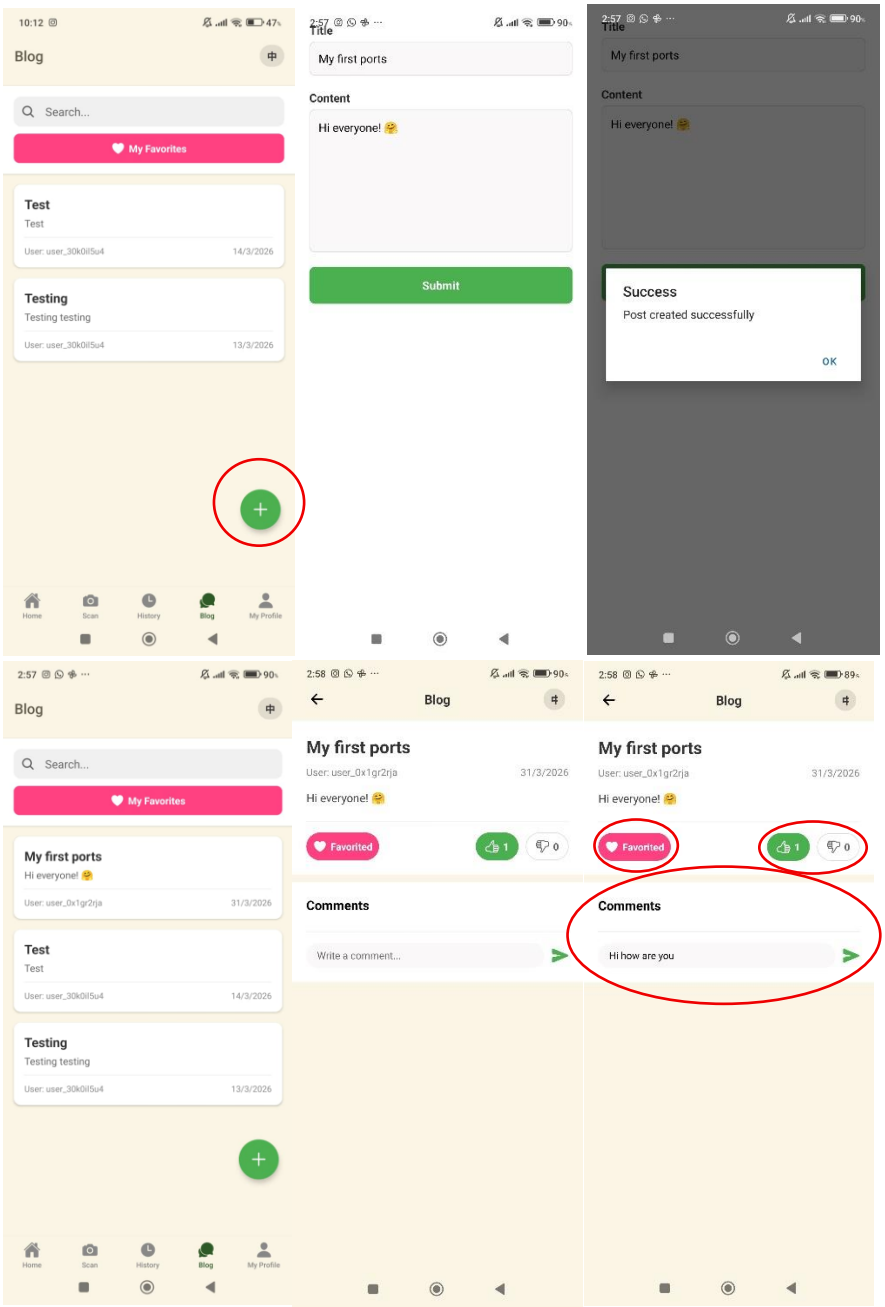
2. answer the question(green is correct and red are wrong).

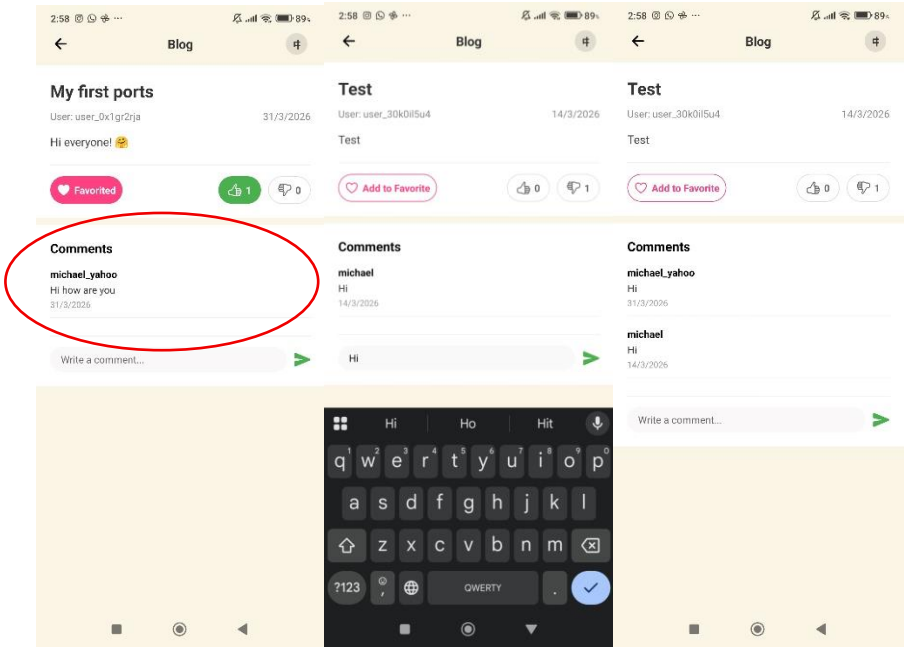
3. after all 5 question are done view the score are get the coupon.

4. choose to back to the home page or to go to profile page to view the

coupon.

Blog view, create, comment, add to favorites, like/dislike:





1. user can view all the blog posts in the blog page.

2. user can click on the “+” button to create.

3. type in the title and cintent and click on submit button.

4. click ok to back to the blog page.

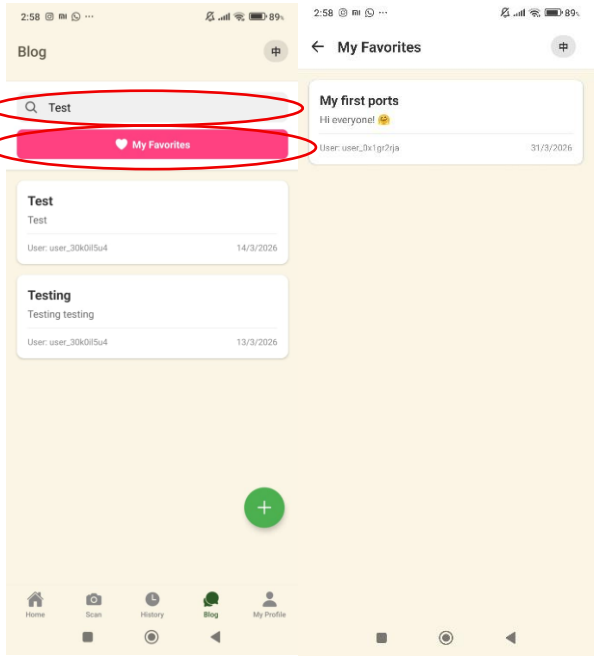
5. for comment, add to favorites and like and dislike action user can select any blog ports that they like to do it.

6.when get into the detail page user can add the blog posts to favorites also can do the like and dislike action.

7. for comment user need to go to the comment section and type on the input box and click the send arrow.

8. user can also comment on the other user blog posts too.

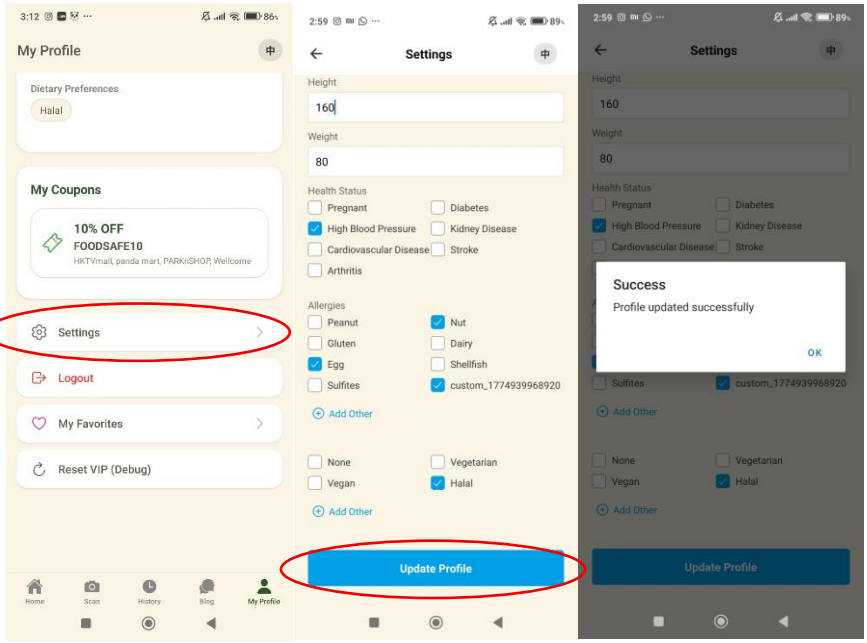
Search and favorites show case:



1. user can type the blog title to search for the blog post.

2. when the post are added to favorites, user can click the “my favorites”: button to view it on the favorites list page.

Profile setting change:



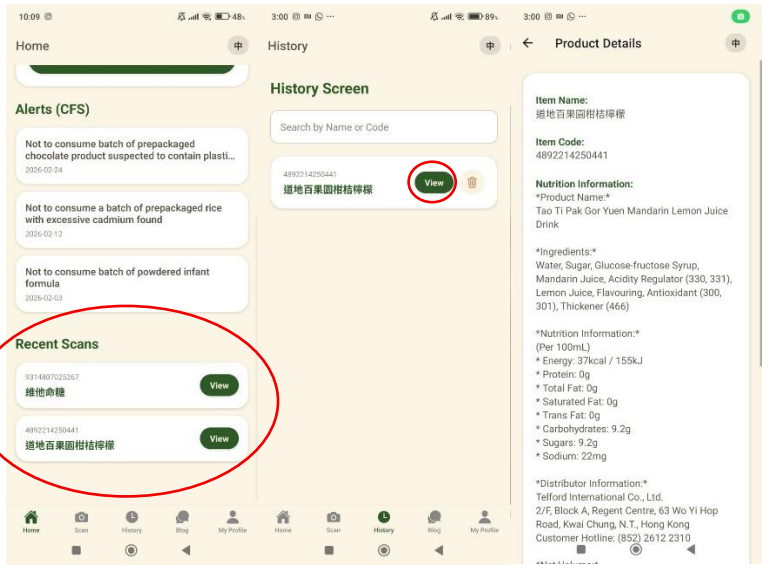


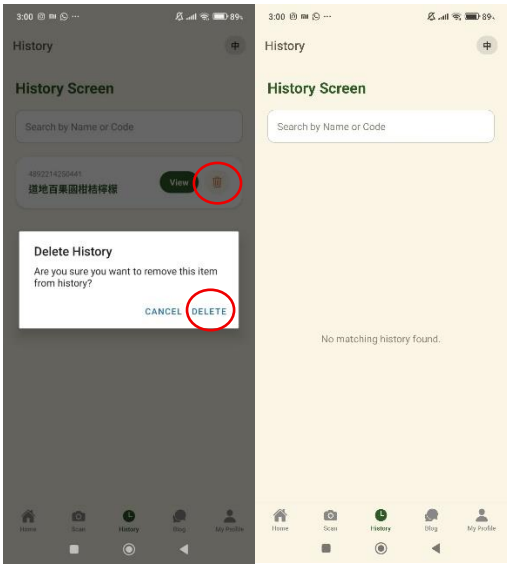
1. user can go to the profile page to click the setting button.

2. change any thing user wants and click the update button.

3. click ok to back to the profile page.

History view and delete:



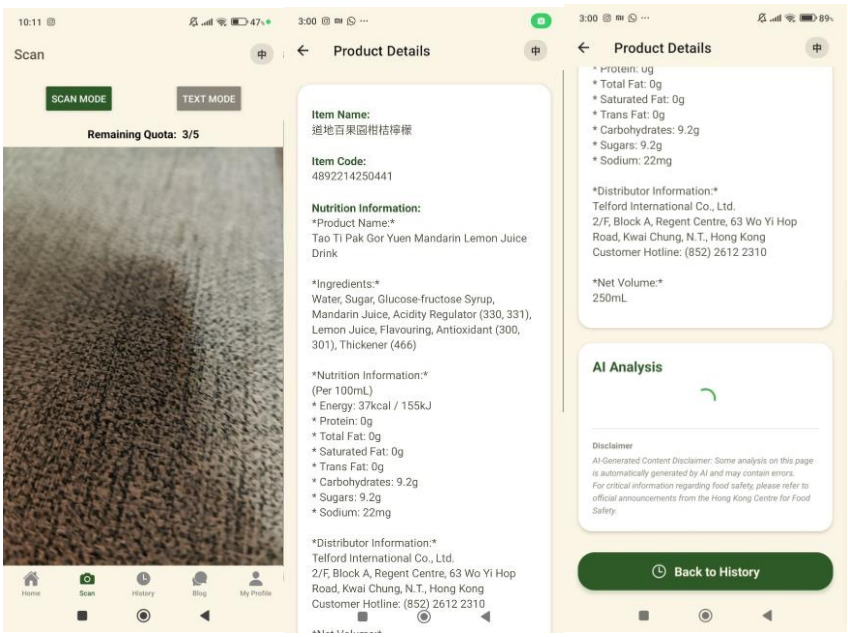


1. the first way to view the history is user can view the recent 3 scan history in the home page.

2. next is to view all the history in the history page

3. by click the “rubbish bin icon” can delete the history.

Scan (bar code):

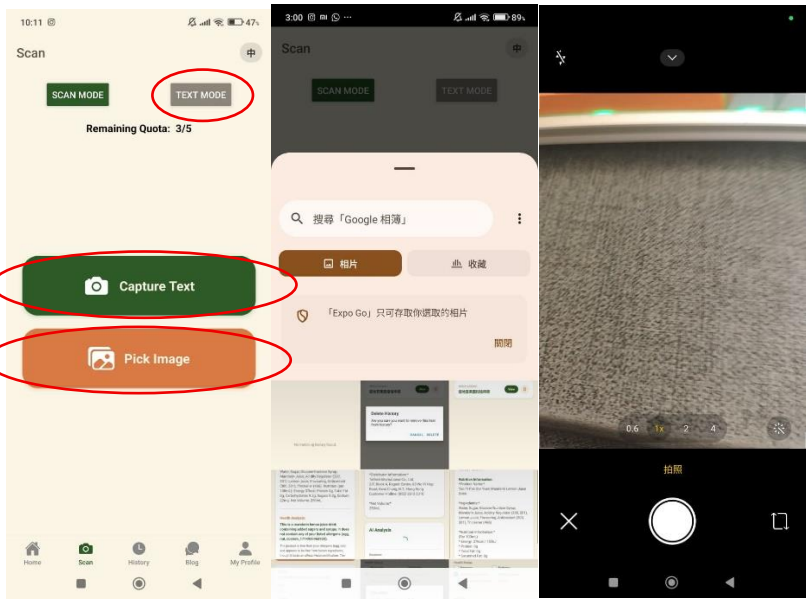


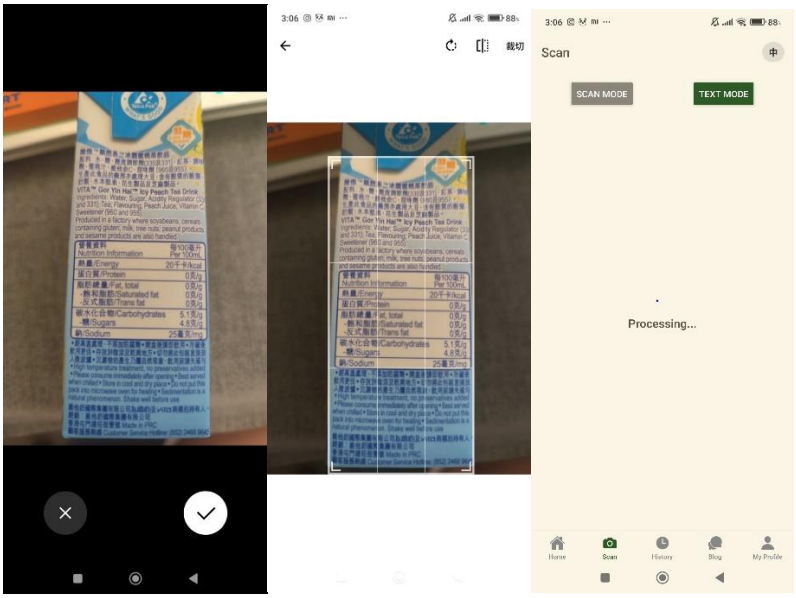
1. user can click the scan mode to scan for the bar code.

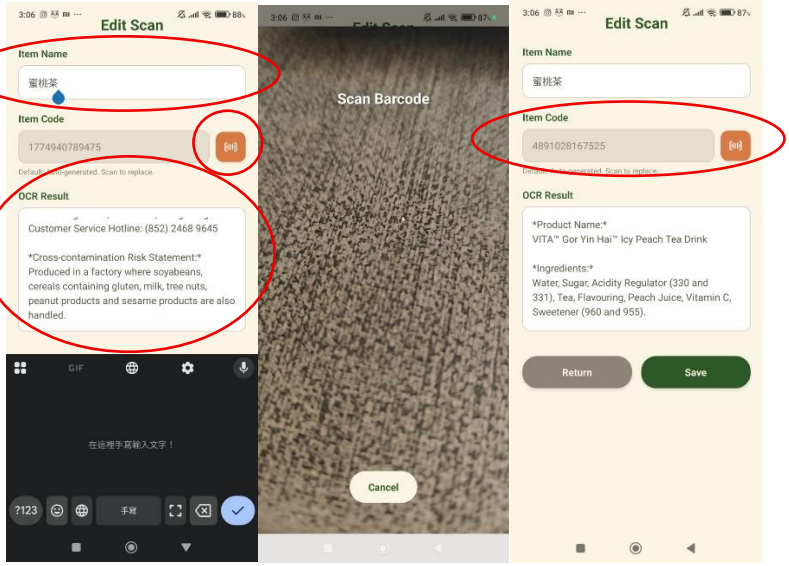
2.then the system will go to the detail page and wait for the report.

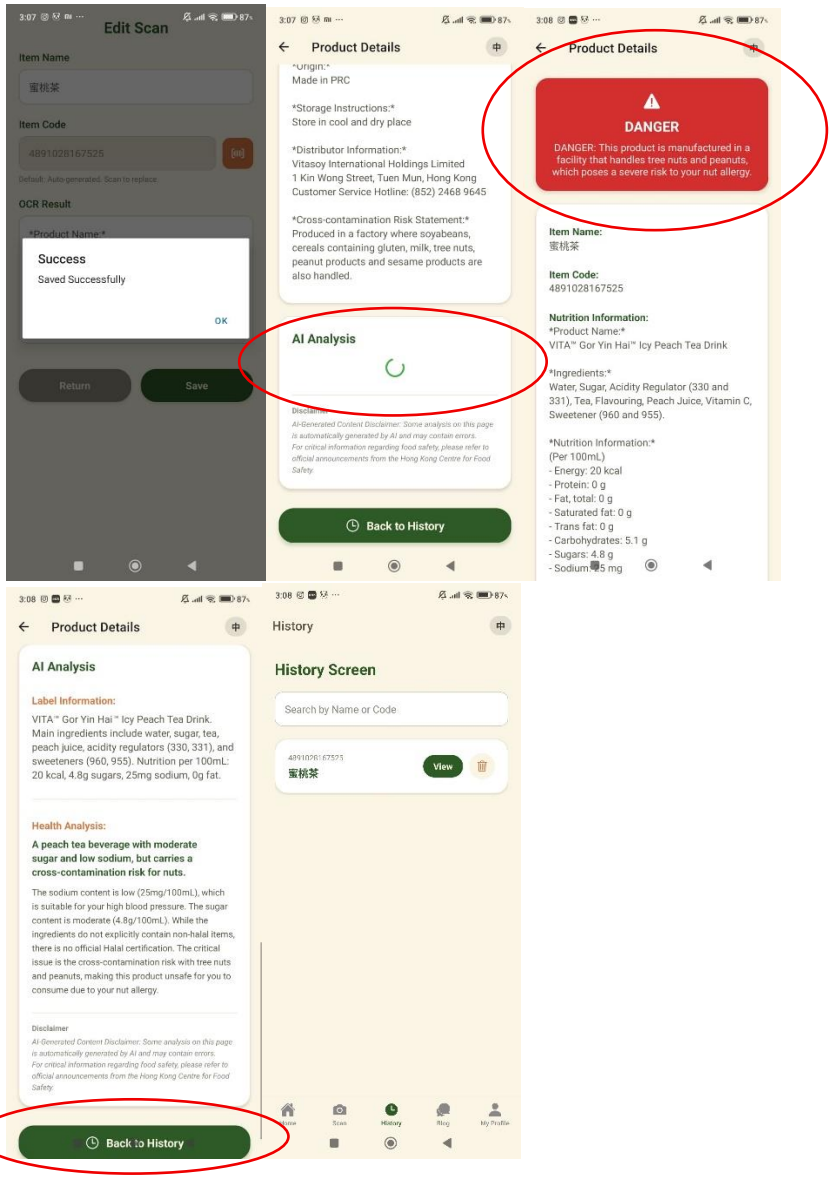
3. once the report ok, it will show on the page.

Scan (bar code):









1. when user using the text mode, the user can select upload photo to the system or take photo.

2. if take photo user need to take a clear photo of the food label and crop the important part for the OCR scan.

3.wait for the OCR process

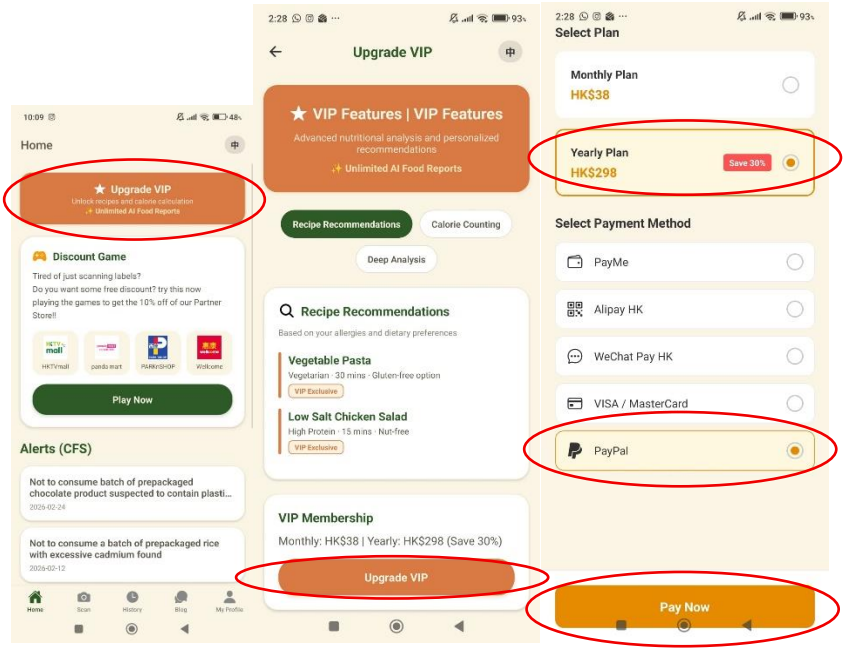
4.input the name and scan the bar code of the product(if no bar code scanned the system will get a default code number to the product) also check and edit the OCR result.

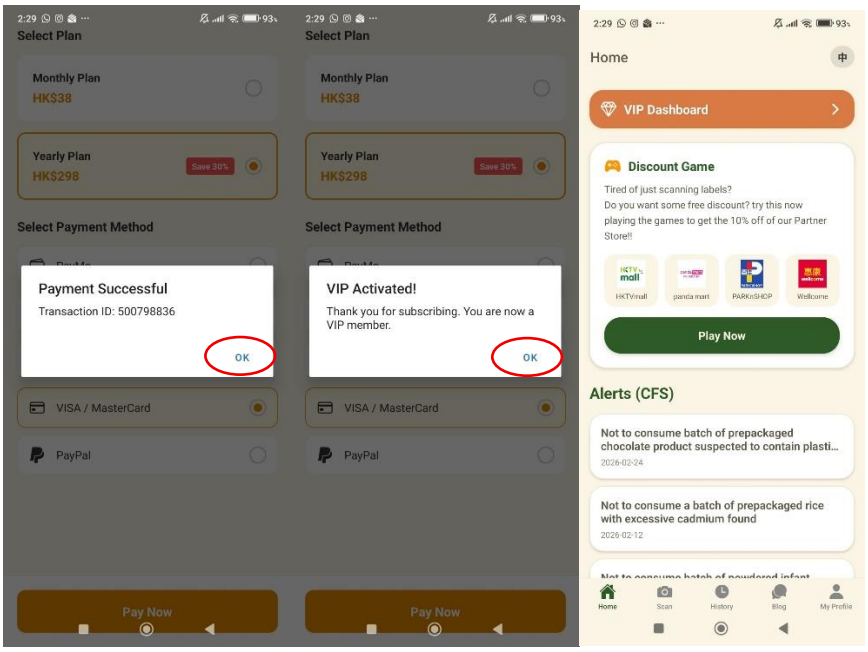
5.wait for the report

6.view the report(the warning sign is in 3 level of color, green = good and safe, yellow = long-term consumption may pose risks, red = dangerous, do not consumption.)

7. click the “back to history” button to finish the scan process.

VIP upgrade and payment page:





1. user can find the upgrade button on the home page and the profile page

2. when click in the page user can view the VIP mode detail.

3. click upgrade VIP button to go for the payment.

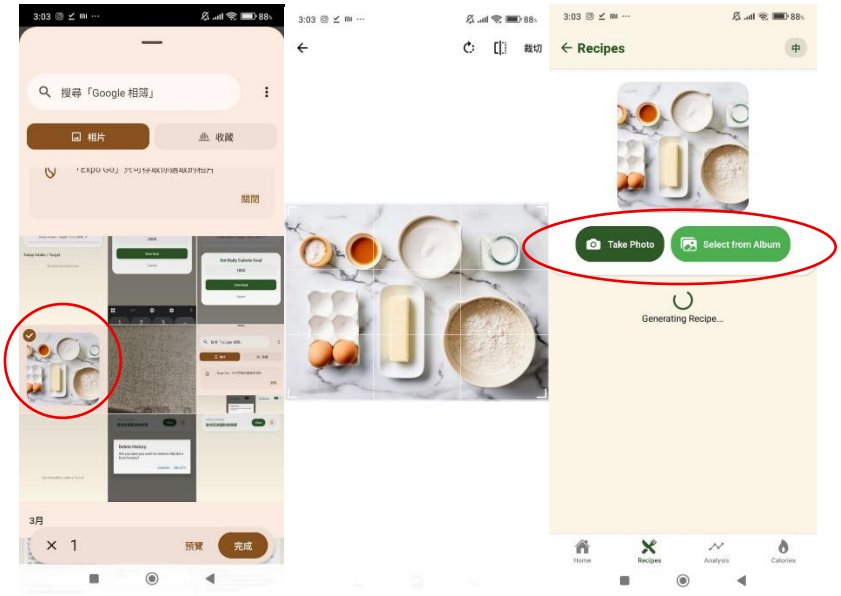
4. select the plan and payment and click the pay now button.

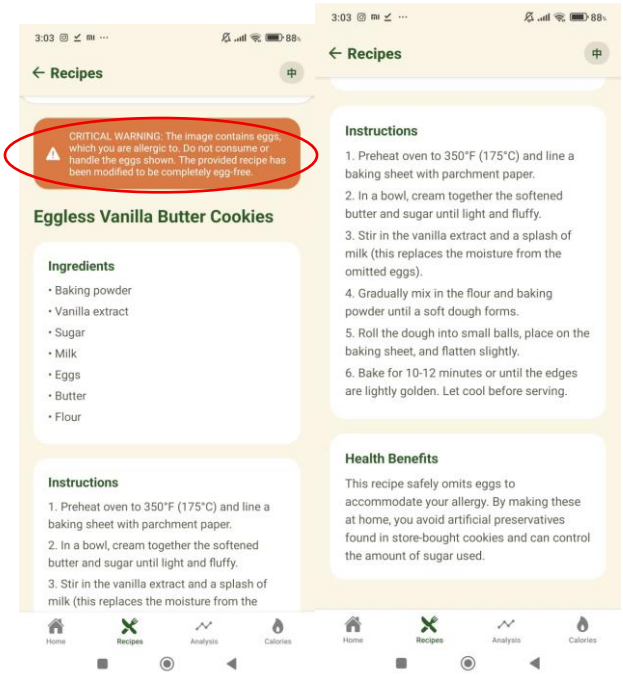
5. click ok to all the pop up message.

6. back to the home page with VIP.

VIP feature(recipes, calories, deep analysis):

Recipes:





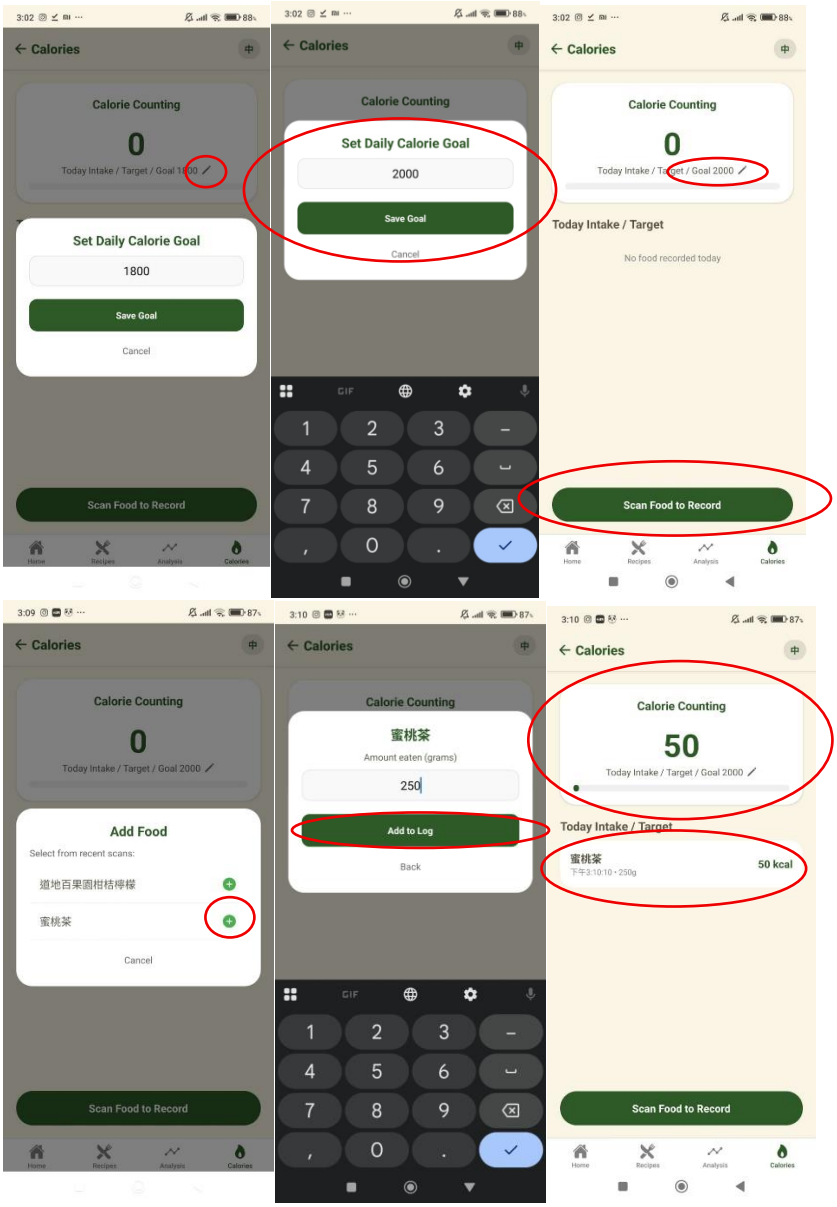
1. user can choose to take photo or upload photo of the ingredients.

2. if upload user can crop the ingredients photo and upload to the system.

3.wait for the process of the recipes.

4. view the recipes(there are also will some warning if the ingredients photo are contain some allergies you have, but no level classify)

Calories:



1. the user can set the calorie they want, by default is 1800.

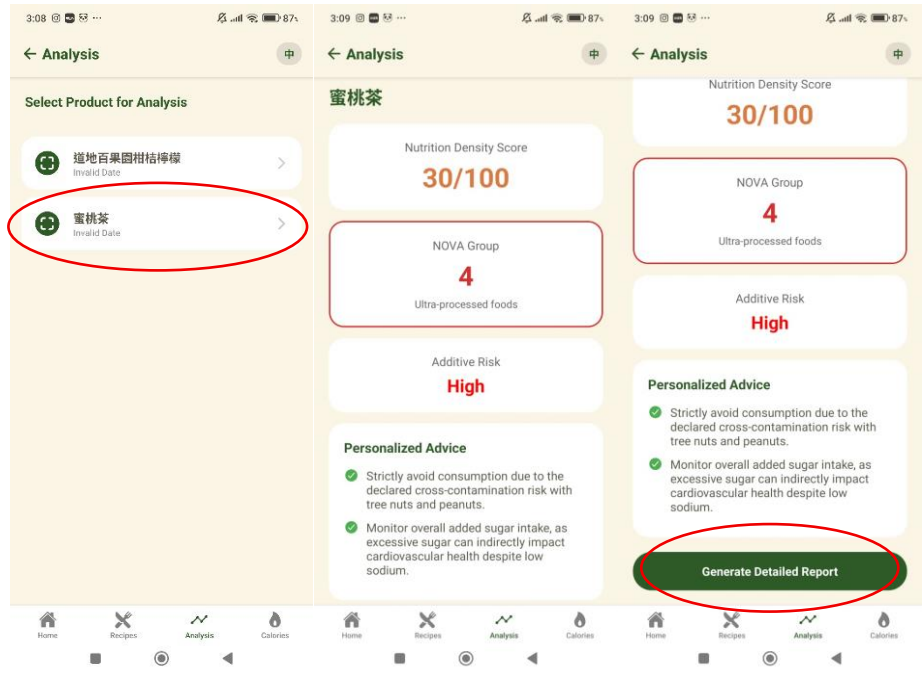
2. click the “scan food to record” button to add food product.

2. then user can select some scanned food product.

3. input how many food product you have taken in grams and click the “add to log” button.

4. the system will count for the calorie.

Deep Analysis:





1. user can select a scanned food product to having the deep analysis.

2. after selecting, user need to wait for the report

3. when the report is ok, the user can view the report(NOVA group,nutrition score,additive risk etc.)

4. click the “generate detail report” for the detail info.(the reason that why the food product will be analysis in this ways)

Language change:



1. User can click the language switch button on every page top right corner of the app.

2. The system will change from one language to the other(Chinese to English / English to Chinese).

A4 Technical Manual

1. System Architecture

The system uses a Client-Server architecture. Font end is using the React Native (Expo) to built the cross-platform mobile app, and the backend is using the Node.js and Express.js to built the RESTful API server.

2. Core APIs

There are two main API that the system are using one is the web scraper that connect to the data.gov.hk for taking the alert data for the app, and the other one is the AI API, most of the function like OCR scanning, analysis report, VIP function, etc. is handling by the AI.

3. Database Management

The system are using the MongoDB as the main database to store the data, and using the JSON file to handle the alert part of the app.

A5 System Setup Guide

Per-install

1. Install Node.js (v16.x or newer).
2. Install Expo Go application on your physical Android/iOS device.
3. Install MongoDB compass and MongoDB Community Server

Step to running the system:

1. npm install
2. setup the .env file(AI API key, API IP and the AI model)

Font end:

1. Go to the food_additives_scanning_app folder(cd food_additives_scanning_app)
2. Run the code(npx expo start)

Backend:

1. Go to the simple-backend folder(cd simple-backend)
2. Run the code(npm start)

Database:

1. Setup a local database and connect it in MongoDB compass